

Practical Workbook

IT Fundamentals and Applications

EF-101



Name: _____

Seat #: _____

Batch: _____

Semester: _____

Year: _____

Practical Workbook

Information Technology Fundamentals and Applications

EF-101



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LAB SESSION 01

Learning basics of DOS Environment

THEORY

MS-DOS is a text-based system of Microsoft Operating System, The users interact with the computer by typing text-based commands rather than a Graphical User Interface. These commands allow users to perform various tasks, such as copying, deleting, or moving files, and managing programs.

Why are MS-DOS Commands used?

- **File Management:** MS-DOS commands enable users to navigate through directories and create, delete, or copy files, enabling efficient file management.
- **System Configuration:** Users can configure system settings, manage drives, and analyze hardware issues using MS DOS commands, providing a robust toolkit for system customization.
- **Program Execution:** MS-DOS for commands facilitate the execution of programs and scripts, offering a streamlined approach to launching applications without the need for graphical interfaces.

DOS: A DOS, or disk operating system, is an operating system that runs from a disk drive. The term can also refer to a particular family of disk operating systems, most commonly MS-DOS, an acronym for Microsoft DOS. It has four functional units.

- **Command Interpreter** The command interpreter (command.com) is an interactive interface used to execute DOS commands.
- **System Files** The system files define the hardware to the software. When the computer is booted these system files provide a unified set of routines for controlling the computer operation.
- **Basic Input Output System (BIOS)** The system files or hidden files interact with special read-only memory on the motherboard. The special ROM is called Basic Input Output System (BIOS).
- **DOS Utilities** The DOS utilities carry out useful tasks such as preparing disks, comparing files, finding free space on a disk and printing.

PROCEDURE

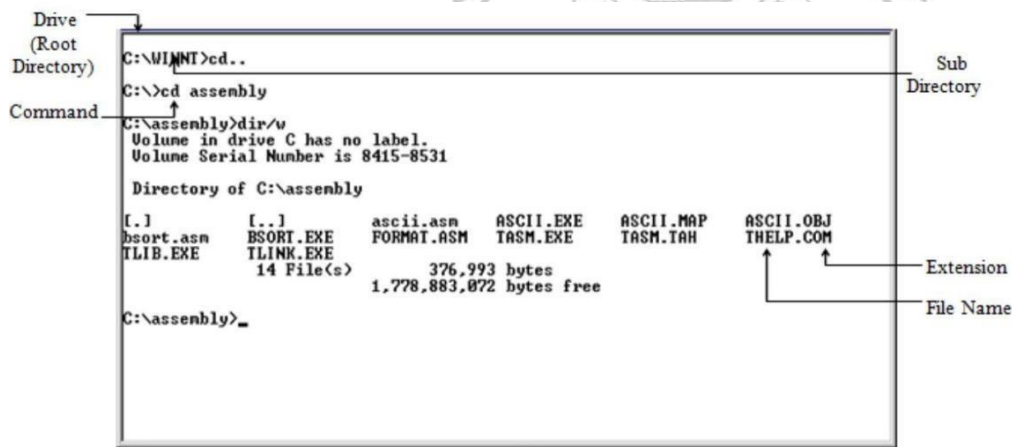


Fig. 1.1. DOS Environment

DOS commands are of two types namely internal commands and external commands.

- Internal commands are those which are built into command.com.
- External commands are those that must be located from a file loaded by command.com before it can be executed.

A brief description of important DOS commands is given below:

Changing drive

Type drive letter of the drive to which you want to switch to, on the command prompt followed by ':'

Example: c:>d: This will change the current drive from C to D. Result:
d:>

Wild Cards

DOS recognizes two wild cards:

- The asterisk (*) represents one or more characters that a group of files has in common.
- The question mark (?) represents a single character that a group of files has in common.

Working with Directories

- **dir** to view the contents of a directory
Prompt on screen: drive:>
Syntax: dir

Example: c:>dir This will list all the files and subdirectories on drive C, which is being prompted on screen. If another drive needs to be accessed, first change the drive, as previously explained and then enter the **dir** command.

Use wild cards to display selected lists. For example:

- dir *.* Displays all files and subdirectories on the drive
- dir ?????*.com Displays all files having names up to five characters and extension 'com'

dir command can be modified using these wild cards and other switches as follows so that only one screen of selected information is displayed at a time.

- **Cd** to change from one directory to another
Prompt on screen: drive:>
Syntax: cddrive:\path

Example: c:>cd dos This will change the current directory from the root directory to 'dos' directory on the prompted drive C.

Result: c:\dos>

- **md** to create a new directory
Prompt on screen: drive:>
Syntax: mddrive:\path\dirname

Example: c:>md neduet This will create a new directory, named 'neduet' on the drive being prompted, in this case drive C.

- **cd..** to switch back one level up in the directory structure
Prompt on screen: drive:>
Syntax: cd..

- **cd** to switch back to the root drive directory
Prompt on screen: drive:>
Syntax: cd\

Example: c:\dos\subdos>cd\

This will switch back to the root drive, C

Result: c:\>

- **rd** to remove a directory
Prompt on screen: drive:>
Syntax: rddrive:\path\dirname

Example: c:\>rd neduet This will delete the directory, named '**neduet**' from the prompted drive C.

- **tree** to view directory listing in a hierarchical structure.
Prompt on screen: drive:>
Syntax: treedrive:\path

Example: tree d: This will display the hierarchical directory structure of drive D.

Use /f to display the names of the folders as well as the files in each folder, in a hierarchical manner.

Working with Files

- **copy** to copy a file from one directory or drive to another
Prompt on screen: drive:>
Syntax: copy drive:\source path\filename drive:\destination path\filename
Example: copy c:\neduet\cfile.exe c:\windows
This will copy the file, named '**cfile.exe**' from '**neduet**' directory to '**windows**' directory with the same name for the new file.

- **ren** to rename a file Prompt on screen: drive:>
Syntax: rendrive:\path\old filename drive:\path\new filename
Example: c:\>ren cfile.exe edit.exe
This will rename the file '**cfile.exe**' to '**edit.exe**' in the prompted drive C.

- **del** to delete a file Prompt on screen: drive:>
Syntax: del drive:\path\filename
Example: c:\>del cfile.exe
This will delete the file, '**cfile.exe**' present in the prompted drive C.

Wild cards can also be used with any of the above commands to work with group of files.

- **Type** to view a file on the screen. Only text based files can be viewed, as DOS supports only such files.
Prompt on screen: drive:>
Syntax: typedrive:\path\filename
Files longer than one screen scroll off to the top. To avoid this, use |more suffix following the command.
Example: c:\>type cfile.txt |more

Disk Management

- **format** to format a disk Prompt on screen: drive:> Syntax: formatdrive: Example: c:\>format Z: This will format Flash memory stick in *drive Z*.
- **chkdsk** to get a report on statistics of the disk Prompt on screen: drive:> Syntax: chkdskdrive: Example: c:\>chkdsk c: This will generate a report on statistics of *drive C*.
- **vol** to view a disk's volume label and it's serial number Prompt on screen: drive:> Syntax: vol drive: Example: c:\>vol d: This will display volume label of *drive d*.
- **ver** to find current version of DOS installed on your system Prompt on screen: drive:> Syntax: ver Example: c:\>ver This will display the current version of DOS on your system.

ver

There are many versions of DOS and ver command is used to find the exact version.

cls

Used to clear the screen.

label

Displays the volume label of the disk. Also allows the user to add, modify, or delete the volume label of a disk.

vol

Displays the volume label of the disk.

dir

Displays a list of a directory's files and subdirectories. Supports multiple parameters.

md

Creates a directory or subdirectory. The name of the new directory is provided right after the command.

Cd

Displays the name of the current directory or changes the current directory. To change directory, provide the name of the directory right after the command.

rd

Deletes a directory.

tree

Displays the directory structure of a path or of the disk in a drive graphically.

copy con

Makes a new file in the current directory. A file extension must be provided along with the file name. Once the file is created, an editor opens. To exit the editor, press Ctrl+Z, then Enter.

ren

Renames files or directories.

del

Deletes one or more files. If a directory name is provided, then all files in the directory are deleted.

type

Displays the contents of a text file.

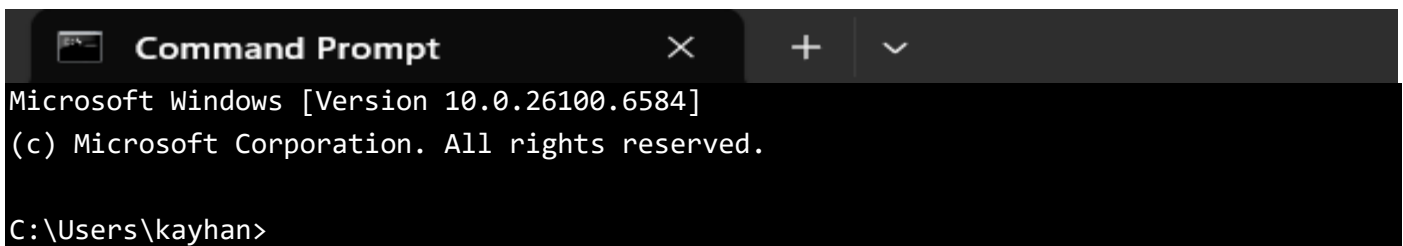
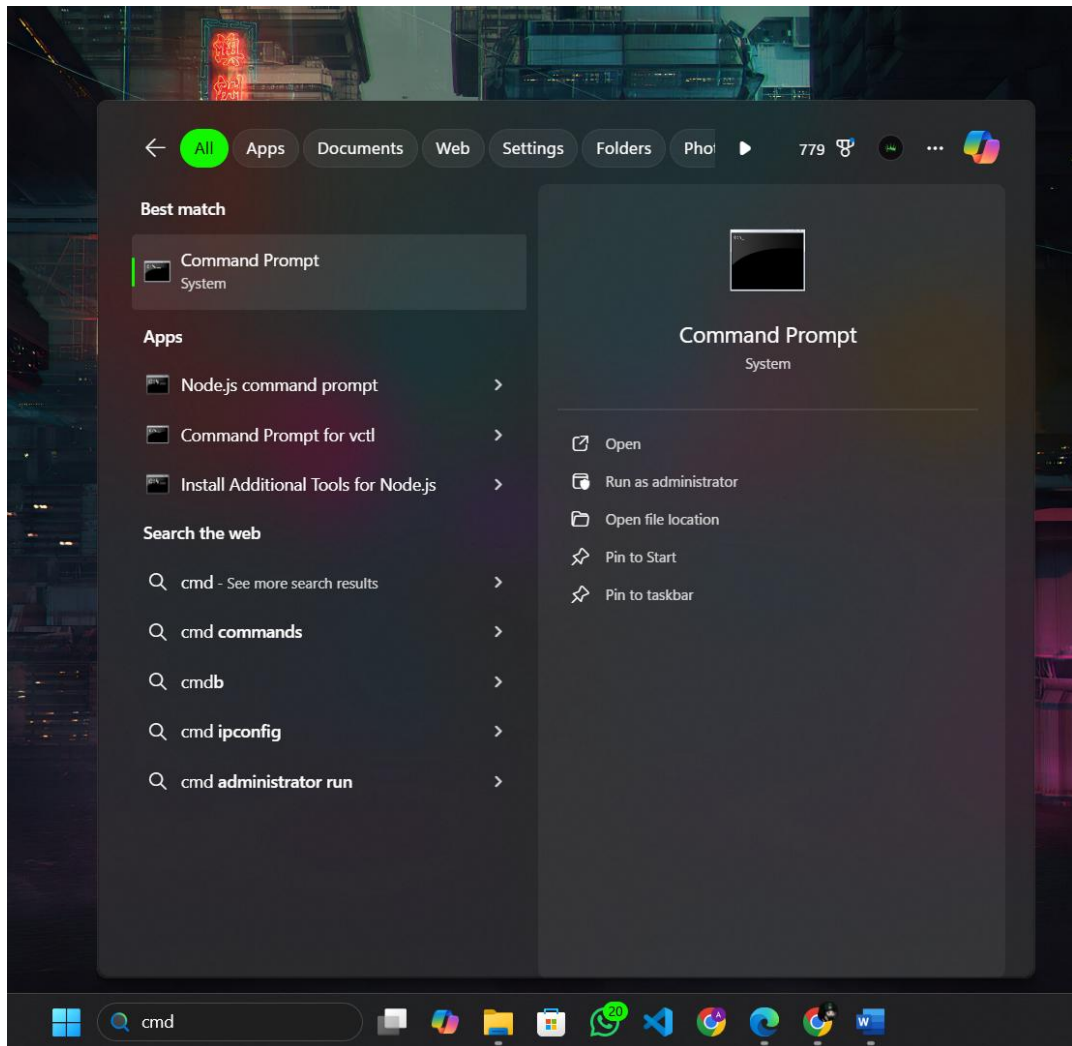
copy

Copies one or more files from one location to another.

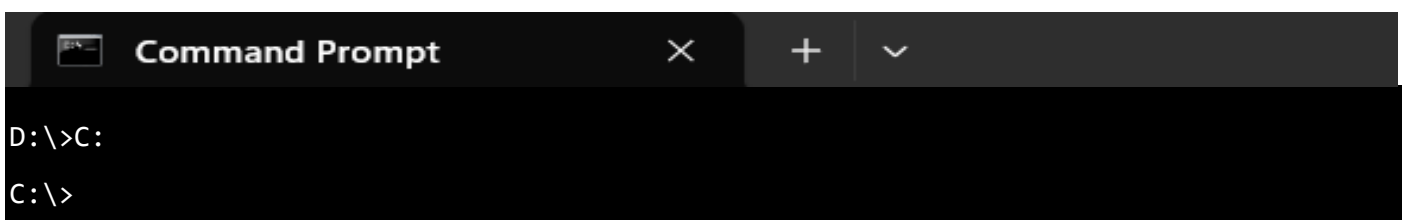
EXERCISE:

Perform the following operations on DOS and present your output. Attach relevant screenshots.

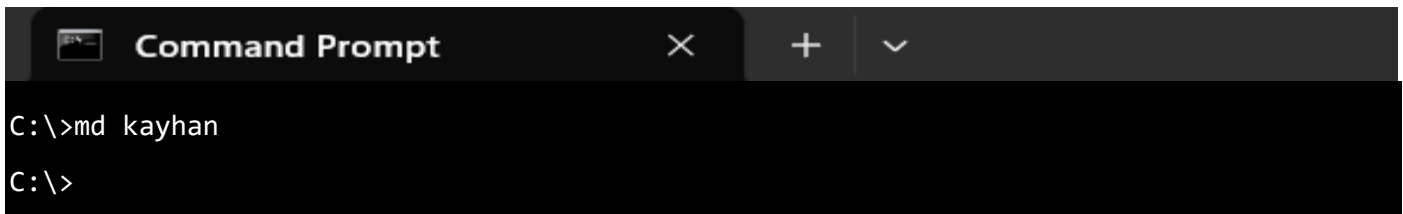
1. Open an MS-DOS window by typing *cmd* in the Start Menu.



2. Change the current drive to C:\>

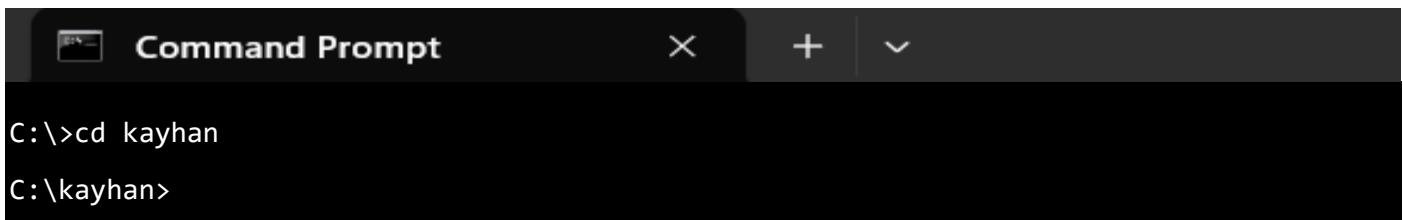


3. Create a new directory with your name.



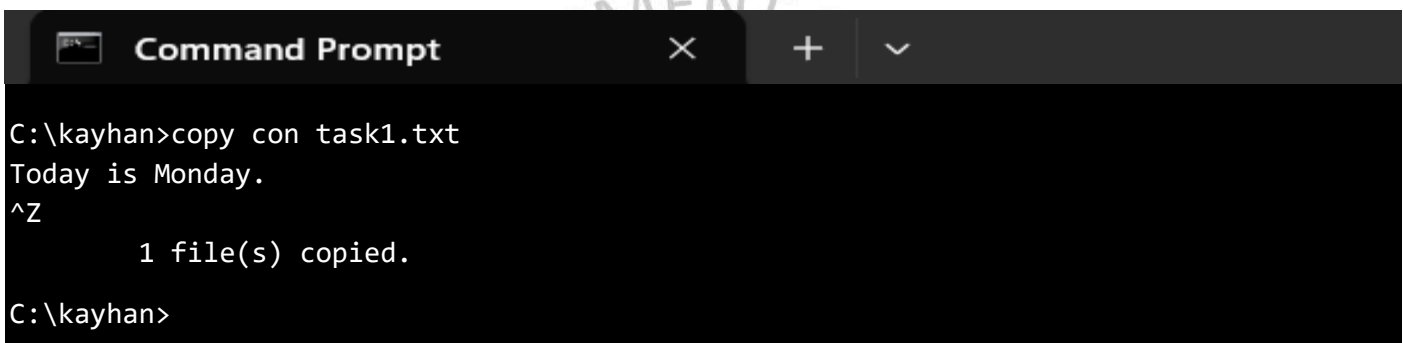
```
Command Prompt
C:\>md kayhan
C:\>
```

4. Change to the newly created directory in.



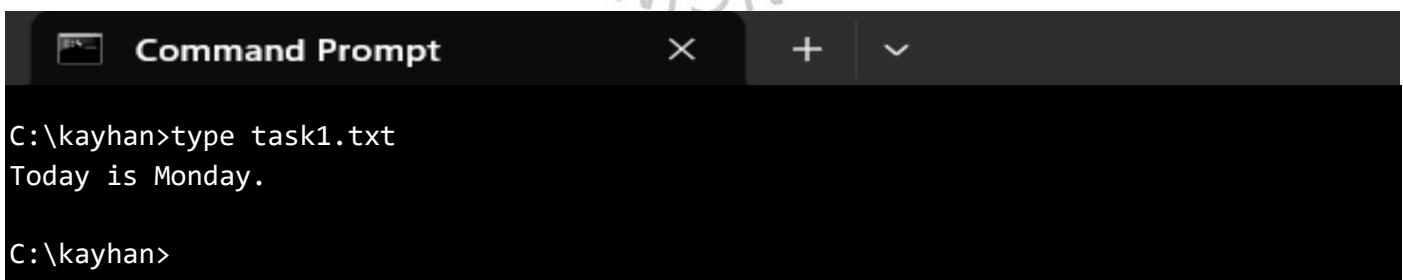
```
Command Prompt
C:\>cd kayhan
C:\kayhan>
```

5. Create a new text file and fill it with text.



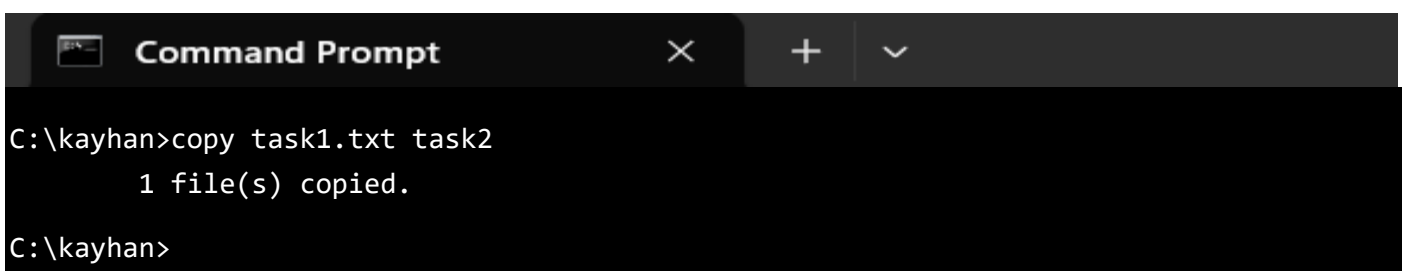
```
Command Prompt
C:\kayhan>copy con task1.txt
Today is Monday.
^Z
        1 file(s) copied.
C:\kayhan>
```

6. Display the contents of the file created in



```
Command Prompt
C:\kayhan>type task1.txt
Today is Monday.
C:\kayhan>
```

7. Make a new copy of the file created in



```
Command Prompt
C:\kayhan>copy task1.txt task2
        1 file(s) copied.
C:\kayhan>
```

8. List all the files in the current directory.

```
Command Prompt
C:\kayhan>dir

Volume in drive C is OS
Volume Serial Number is A061-C9F9

Directory of C:\kayhan

19/09/2025  08:00 pm    <DIR>          .
19/09/2025  07:30 pm                18 task1.txt
19/09/2025  07:20 pm                18 task2
                2 File(s)                36 bytes
                1 Dir(s) 272,681,673,920 bytes free

C:\kayhan>
```

9. Delete the file created in.

```
Command Prompt
C:\kayhan>del task1.txt
C:\kayhan>
```

10. Exit the directory and delete it.

```
Command Prompt
C:\kayhan>cd..
C:\>rd kayhan
C:\>
```

11. Write command to display all files having extension 'docx'.

dir*.docx/s

12. Write command to display all files of all types starting with letter 's'.

dir /s s*

13. Write a command to copy the file *assignment* of type *docx* from the directory 'ITC' of drive D to drive F.

copy d:\ITC\assignment.docx F:\

14. Write a command to delete the directory as well as the files of the directory 'world' on drive E.

Rd/s E:\world

15. Write command to copy all the files beginning with 'm' and whose file names has a 'txt' extension from drive A to the '\document' directory on drive C.

copy A:*m.txt C: \document

16. Write set of commands to create a directory 'practical' on drive F, and then move into it. Now list all files present in it, then go back to root drive F.

F:
md practical
cd practical
dir
cd



LAB SESSION 02

Getting Started with MS Word

THEORY

Introducing Microsoft Word

Microsoft Word is a document and word processing application that enables a user to create, edit, and format different types of documents, such as letters and reports. Word provides a lot of document creation tools that have been refined to be easy to use. It also enables a user to add effects to picture from within the document and allow multiple authors to collaborate efficiently on a document.

PROCEDURE

Understanding the Microsoft Word Interface

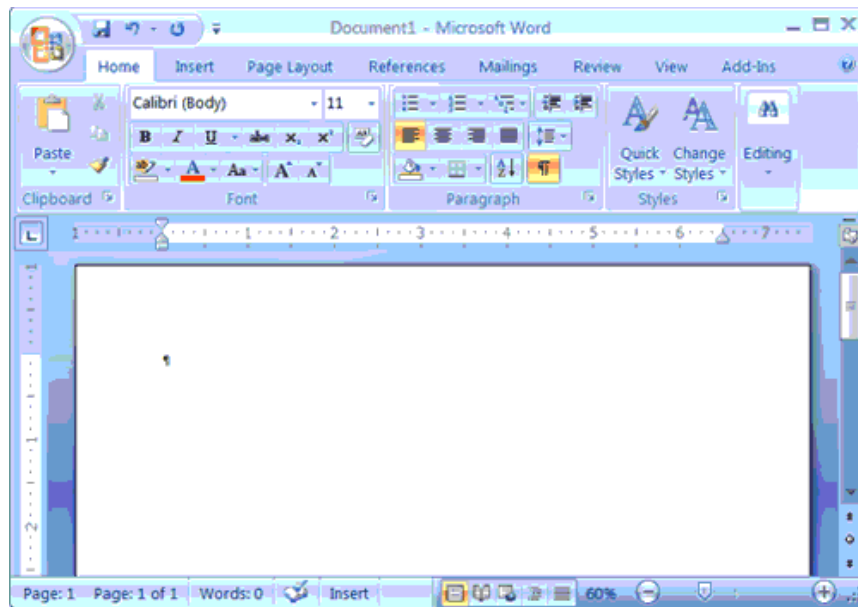


Fig 2.1 MS Word interface

Create Sample Data and Select Text

If you type =rand() in your Word document and then press Enter, Word creates three paragraphs. You can use these paragraphs to practice what you learn. Throughout these lessons, you will be asked to select text. The following exercise teaches you how to create data and how to select data. You can select by using the arrow keys or by clicking and dragging. When using the arrow keys, use the up arrow to move up, the down arrow to move down, the left arrow to move left, and the right arrow to move right. When using the mouse, press the left mouse button and then drag in the direction you want to move.

Understanding the Ribbon Interface

The Ribbon interface is a common feature in all applications included in the Microsoft Office Suite. It is an area above the document workspace. It provides easy and quick access to all the commands by organizing them in a set of main tabs. The File tab is the only non-contextual tab which provides access to the Backstage view; all other tabs are main tabs.

These main tabs are divided into groups that represent different command groups and can be customized to add buttons for frequently repetitive tasks. Also, additional contextual tabs appear as the user works with different elements. For example, when the user is editing a table, the Design and Layout tabs are displayed under a new set of tabs called Table Tools.

The Quick Access Toolbar

Next to the Microsoft Office button is the Quick Access toolbar. The Quick Access toolbar provides you with access to commands you frequently use. By default Save, Undo, and Redo appear on the Quick Access toolbar. You can use Save to save your file, Undo to rollback an action you have taken, and Redo to reapply an action you have rolled back.

Setting Page Orientation

Page Orientation is related to how the user wants to display the document, when it will be printed. Page orientation is of two types, portrait and landscape. Portrait orientation is suitable for documents that are printed vertically and landscape orientation is suitable for documents that are printed horizontally.

Working with Lists

Numbered and bulleted lists allow the user to emphasize a list of items and make them stand out from the rest of the document. In other words, bulleted lists work best when you want to separate a list of items from other text. With Microsoft Word 2010, users can display a list of items as ordered or unordered lists. Users can also create multi-level lists with a maximum of nine levels.

Working with Tables

Tables enable the user to present the data in a grid of rows and columns.

Microsoft Word allows user to quickly add a table and format it in one of the different available styles.

Working with Graphics

Microsoft Word allows user to add different types of graphics, such as pictures, clip arts, and charts into their document. The options to insert graphics in a document are located in the Illustrations group of the Insert tab.

Borders, Lines, and Shading

For a finished look, the user can add borders and shading to the documents. Just select the paragraph(s) to apply a border or shading to, then choose Insert, Table and click on Insert Table. In the title bar, there appears a Table Tools box. Click on this box. Click on Border and then on Border and Shading. Then click the desired line style; Click the Line Weight button, then click the line weight to be used; Click the drop-down arrow on the Borders button, then click the type of border to be applied; Click the drop-down arrow on the Shading Color button to display the palette of gray percents and colors, then click the percent of gray or color button.

Drop Cap

A drop cap is large capital letters of the first word that is set into a paragraph to add visual interest. Just select the first letter, word, or section of the paragraph to be formatted with a drop cap; Choose Insert, Drop Cap; Select Dropped or In Margin; In the Drop Cap Options section, select the Font from the drop-down list, and increase or decrease the Lines to Drop if the default of three lines is not to be used; If the distance of the text from the drop cap is to be changed, use the increment buttons on the Distance from Text option, then choose OK. To remove drop caps, click the drop caps text, choose Drop Cap, click the None option in the Position section of the dialog box, then choose OK.

Inserting Special Characters

To insert special characters, place the insertion point at the point in the paragraph where a symbol or special character is to be placed; Choose Insert, Symbol to display the Symbol dialog box. Click the Symbols tab to select a symbol, or click the Special Characters tab to select a special character; On the Symbols tab, select the font set that contains the symbol to be inserted in the document in the Font drop-down list box. To view a symbol in the displayed table, click the symbol. The symbol is then displayed in an enlarged and highlighted view; To insert a symbol or special character, click the item to be inserted, then click the Insert button. Click the Close button to close the dialog box and return to the document.

Zooming the Document Display

Zooms customize the on screen size of the document in a way the user wants to view it. However, zoom setting does not affect the actual formatting of the document. To adjust zooming option, click View, Zoom; In the drop-down list, select the zoom percentage to be used. The document display changes to the zoom percentage selected; If the required zoom percentage is not in the list, type the percent in the text box part of the Zoom button and press Enter.

Creating Custom Colors/Gradients/Textures

To create custom colors, right click on the graphic element to be colored. Along with the shortcut menu appears the Drawing toolbar. Then click the drop-down arrow of the Fill Color or Line Color button and select More Fill Colors to open the Colors dialog box; Click the Custom tab at the top of the Colors dialog box. Set desired color. For gradient settings, in the drop-down arrow beside the Fill Color button, select Gradient at the bottom of the color palette instead of More Fill Colors. Click the Gradient tab and make desired settings. Texture tab can also be selected to apply designs to the objects.

Inserting Clip Art/AutoShapes/Pictures/WordArt

To insert a Clip Art/ AutoShape, Picture or WordArt, position the insertion point in the document where the image is to be appeared. Choose Insert and then the desired option. Related dialog box appears, then make required selections. The inserted object can be customized in a number of ways. The user can change colors of the image, edit, reposition and resize them.

There are two ways to rotate or flip the inserted object. One is to right click on the object and then choose Format Picture/Shape. Then click on the 3-D Rotation button. From here, the desired rotation in the x, y and z axes may be chosen. The other gives access to that same tool and to the commands Rotate Left, Rotate Right, Flip Horizontal, and Flip Vertical. To do this, click the object to be adjusted choose Page Layout, click on Rotate and then choose the desired option.

These commands automatically execute. If Free Rotate is chosen (if we bring the cursor on the small green circle that appears above any object when we select it, this gives the free rotate option) continue with the following steps; Move the pointer over the rotate handle and, when it assumes the shape of the rotate tool icon, hold down the right mouse button; The pointer changes to four circling arrows. Drag right or left. While dragging, a dotted outline indicates where the object would be if the mouse button is released; until the object is deselected it can be continuously rotated, even if the mouse button has been released. Place the pointer over a circle handle, hold down the mouse button and continue to rotate. Release the mouse button when the object is rotated to the desired position, and click outside the object to deselect it.

Creating Headers and Footers

Headers and footers contain information repeated at the top or bottom of the pages in a document. To set Header and Footer, with the document open, choose Insert, then click on Hheader or Footer. Choose the desired format from the drop down list. The Header pane is displayed in the document; Type and format the information for the header; To include the Page Number, Number of Pages, or current Date or Time, click the corresponding button on the Header and Footer toolbar; To create a footer, click the Go to Footer Button on the toolbar and type and format the footer just as was done for the header. Click Close Header & Footer button to return to the document.

Document Map

The Document Map is a very functional way to move quickly through long or online documents. To use Document Map, choose View, Navigation Pane. Use the mouse to click the heading or text in the Document Map pane to move to that section of the document; Click the arrow on the left of any heading to expand or collapse the headings; To adjust the size of the Navigation pane, move the mouse pointer onto the right edge of the pane so the pointer changes to a resizing pointer, then click and drag the edge to the left or right. To use the keyboard, press F6 to move to the Navigation pane. Arrow keys and the Enter key will move to the desired location in the document.

Drawing Tables

Drawing a table allows the user to place the rows and columns where he wants them. To create a table, click the Tables and Borders button to bring up the Tables and Borders toolbar and change the mouse pointer to a pencil; Drag the mouse pointer from one corner of the new table to the opposite corner to create the rectangle outline for the table; If a line is to be removed, click the Eraser button on the Tables and Borders toolbar and drag across the line.

If the user needs the data from a table, he may (rarely) want to convert the table to text with some sort of separator between the data for the former columns. To do this, select the entire table by positioning the mouse directly at the left top of the table until the pointer changes to a four directional arrow and then click it. A Table toolbar appears at the top of the document window. In the Layout tab, choose Convert to Text. Choose to separate the text with Paragraph Marks, Tabs, Commas, or type a new character in the Other text box; then choose OK. Similarly the user may want to convert the lines of text into a table. To do this, select the rows of tabbed text to be converted; Choose Insert, click on Table and then choose Insert Table.

The cells of the table can be edited and formatted, like moving and copying cells, splitting and merging cells, adjusting row and column spacing/width, inserting and deleting rows and columns, by selecting appropriate options from the menu that appears on right clicking the table or the selected cells.

Creating Bookmarks

A bookmark is a named marker for a block of text, an entire table or a graphic, a cell or range of cells in a table, or simply a position in a document. First enter a bookmark where it is wanted, then the user can move to it or cite it as a reference in a field or formula. To create a book mark, click in the document at the location where the bookmark is to be inserted, or select the text or graphic to be named; Choose Insert, Bookmark to display the Bookmark dialog box; enter

the new bookmark name in the Bookmark Name text box. The user can also select an existing bookmark name from the list, and Word moves the bookmark from its existing location to the place selected; Click the Add button to add the bookmark to the bookmark list and close the dialog box

To delete a book mark, choose Insert, Bookubookmark; Select the name of the bookmark to be deleted from the Bookmark Name list box; Click the Delete button. The selected bookmark name is removed from the list box and the document; Click the Close button to close the dialog box and return to the document.

Hyperlinks

The user can create a link that jumps to a document, also called a page, on his computer and on a network or intranet. Two types of hyperlinks can be created: one that jumps to a bookmark in the same document and another that jumps to a URL.

EXERCISE

Perform the following operations on a single document in MS Word. Print and attach your final result. *No screenshots allowed.*

1. In the header, put your name, today's date, and course name using Arial, font size 11. Align the text on the right side of the document.
2. In the footer, put the lab number on the left side and the page number on the right side of the document using Arial, font size 11.
3. Inside the document, type the name of the task (heading) using Times New Roman, font size 16.
4. Center the heading, bold and underline it, and change its color to red.
5. Skip a line and type "Task Completed" in Calibri, font size 11 without any formatting.
6. Add a horizontal line and start Task 02.

Task-1

“Task Completed”



1. Create a two page document with four headings. Each heading has two sub headings. The first sub heading of heading 4 also contain two sub headings.
2. Change the template font and page setup settings of the first paragraph following the standard and format specified.

Font	Times New Roman, 12pt, Bold, Italic
Page Columns	Number of Columns: 2, Width: 1:3, Spacing: 0.2
Paragraph Formatting	Indentation: 0.5" (Both Left and Right), 1" (Both Top and Bottom), Alignment: Justify
Page Margins	Left and Right: 0.75", Top and Bottom: 0.75"
Page Size	A4
Page Background	Watermark: NED Logo without Ayah

Task-2

Foundations of AI: History of Artificial Intelligence: AI research

began in the 1950s with pioneers like Alan Turing and John McCarthy. Early systems focused on symbolic reasoning and problem solving. The field grew rapidly with computing power, data, and algorithms. Key AI Concepts and Terminologies: Core ideas include machine learning, neural networks, and natural language processing. Algorithms allow computers to "learn" from data and improve over time. AI also includes concepts like training, inference, and automation.

AI Technologies:

- **Machine Learning:**

Machine learning enables systems to recognize patterns and make predictions. It relies on supervised, unsupervised, and reinforcement learning methods. Applications range from spam filtering to stock market forecasting.

- **Deep Learning:**

Deep learning uses multi-layered neural networks inspired by the human brain. It powers breakthroughs in computer vision, speech recognition, and NLP. Training deep models requires large datasets and high computing power.

Applications of AI:

- **AI in Healthcare:**

AI assists in disease diagnosis through image recognition and predictive models. It supports personalized medicine by analyzing genetic and patient data. Virtual assistants help doctors and improve patient care efficiency.

- **AI in Business and Industry:**

Businesses use AI for customer service chatbots and recommendation systems. Predictive analytics optimizes supply chains and decision-making. Automation increases productivity and reduces human error.

Future of AI:

- **Ethical and Social Implications:**

AI raises questions about fairness, accountability, and transparency.

Unchecked algorithms can reinforce discrimination and inequality.

Governments and organizations are working on AI regulations.

- **Bias in AI Systems:**

AI can inherit biases from training data, leading to unfair outcomes.

Biased facial recognition has shown errors in identifying minorities.

Addressing bias requires diverse datasets and ethical guidelines.

- **Privacy and Security Concerns:**

AI collects and processes massive amounts of personal data.

Without safeguards, this can lead to surveillance and misuse.

Strong encryption and privacy-preserving AI are vital solutions.

- **Emerging Trends in AI:**

Generative AI creates images, text, and music with human-like quality.

AI integration with robotics is advancing automation further.

Future AI will likely focus on explainability and human collaboration.

Task 3

Create a CV selecting one of the given templates.

(Next page)

MUHAMMAD KAYHAN

SOFTWARE ENGINEERING STUDENT

CONTACT

- 📞 00000000000
- ✉️ abc@example.com
- 📍 Pakistan

EDUCATION

Bachelor of Software Engineering (In Progress)

NED University of Engineering & Technology

Intermediate

DJ Sindh Government Science College

Matriculation

Karachi Public School

SKILLS

Programming Languages:

Python (Beginner–Intermediate)

Web Development:

HTML, CSS, JavaScript (Front-End)

Tools & Technologies:

VS Code, Git/GitHub

Other Skills:

Problem-solving, Teamwork,
Time Management

ACHIEVEMENTS

Lorem ipsum dolor
sit amet,
consectetur
adipiscing elit.
Sed do eiusmod

PROFESSIONAL OVERVIEW

Motivated and detail-oriented Software Engineering student at NED University with strong academic background and a growing skill set in **front-end development** and **Python programming**. Passionate about problem-solving, continuous learning, and building efficient software solutions. Seeking opportunities to apply technical skills in practical projects and internships.

WORK EXPERIENCE

(No formal work experience yet — open to internships and entry-level opportunities.)

Task 4

Insert any complex equation that atleast contains an integration or summation or limit value.

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n} \int_0^1 e^{x^2} dx$$





NED University of Engineering & Technology

Department of Software Engineering EF-101 IT Fundamentals And Applications

Criteria	Level of Attainment					
	(5)	(4)	(3)	(2)	(1)	(0)
Problem Identification	Able to correctly identify the problem(s) and all required data (input and output).	Able to correctly identify the problem(s) but not all the required data (input and output).	Able to partially identify both the problem(s) and the required data (input and output).	Able to partially identify the problem(s) but cannot identify required data (input and output).	Not able to correctly identify the problem(s) and all required data (input and output).	Not tried to identify the problem(s) and all required data (input and output).
Problem Solving Skills	Uses an effective strategy to solve the problem(s) and clearly identifies the steps.	Uses an effective strategy to solve the problem(s) but cannot clearly identify the steps.	Uses an adequate strategy to solve the problem(s) but misses some details.	Uses an inadequate strategy to solve the problem(s) and misses most details.	Uses an ineffective strategy to solve the problem(s) and completely misses the steps.	Didn't use any strategy to solve the problem(s) and completely misses the steps.
Efficiency in Tool Usage	Efficient implementation and excellent tool usage skills.	Efficient implementation and good tool usage skills.	Adequate implementation and fair tool usage skills.	Inadequate implementation and tool usage skills.	Incomplete implementation and ineffective tool usage skills.	No implementation and tool usage skills. .
Deliverable(s)	Generates correct and complete output.	Generates correct but partially complete output.	Generates partially correct and partially complete output.	Generates partially correct and incomplete output.	Cannot generate the desired output.	No output generated.
Depth of Knowledge	Thorough grasp of the concepts covered in the lab and can apply it in real-life situations.	Good grasp of the concepts covered in the lab but cannot effectively apply it in real-life situations.	Fair grasp of the concepts covered in the lab and can vaguely identify a real-life application.	Limited grasp of the concepts covered in the lab and cannot identify any real-life application.	Does not grasp the concepts covered in the lab and cannot identify any real-life application.	Didn't try to grasp the concepts covered in the lab and cannot identify any real-life application.
Weighted CLO Score						
Remarks and Signature						

LAB SESSION 03

Getting Started with MS Excel

THEORY

Introduction

Excel is a spreadsheet program that allows you to store, organize, and analyze information. While you may believe Excel is only used by certain people to process complicated data, anyone can learn how to take advantage of the program's powerful features. Whether you're keeping a budget, organizing a training log, or creating an invoice, Excel makes it easy to work with different types of data.

Getting to know Excel

If you've previously used Excel 2010 or Excel 2013, then Excel 2016 should feel familiar. If you are new to Excel or have more experience with older versions, you should first take some time to become familiar with the Excel interface.

The Excel interface

When you open Excel 2016 for the first time, the Excel Start Screen will appear. From here, you'll be able to create a new workbook, choose a template, and access your recently edited workbooks.

- From the Excel Start Screen, locate and select Blank workbook to access the Excel interface.

PROCEDURE

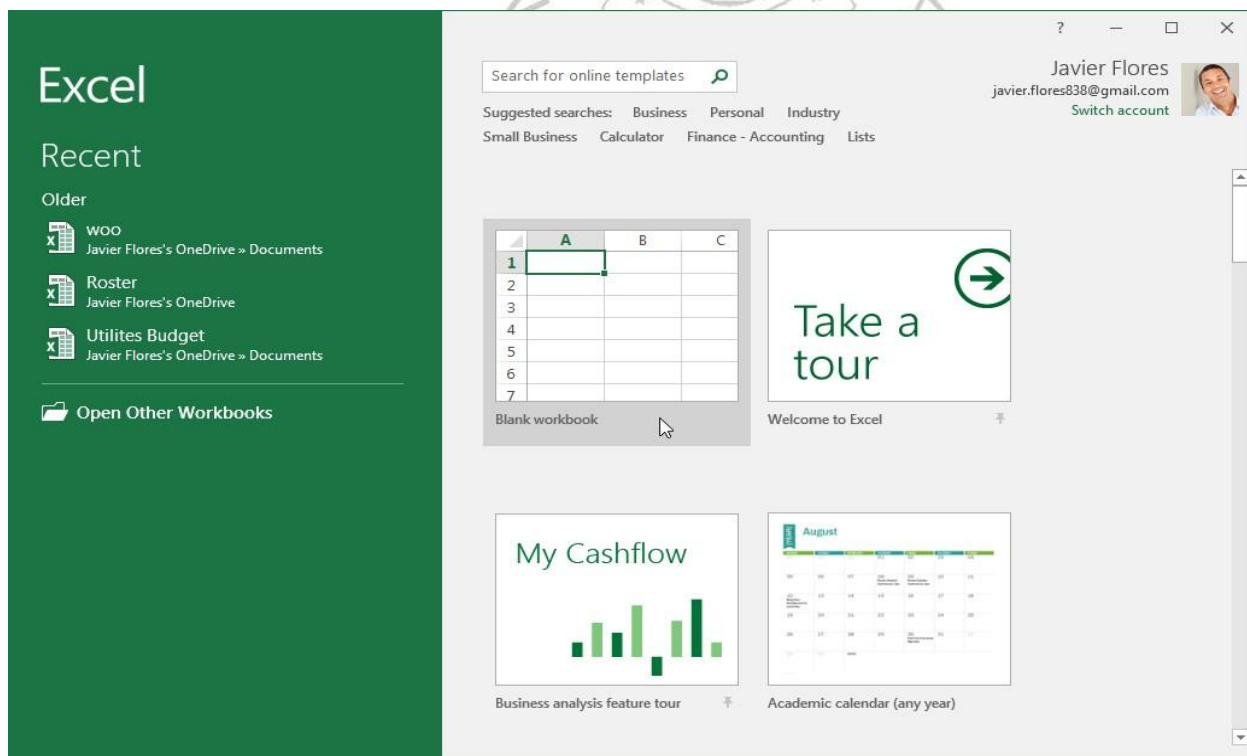
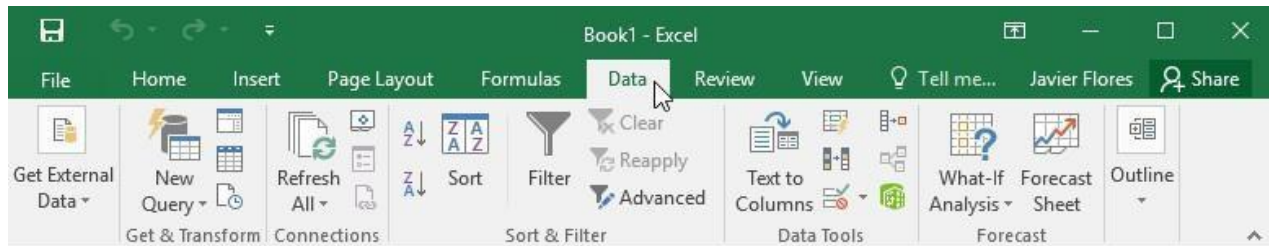


Fig 3.1 Excel Environment

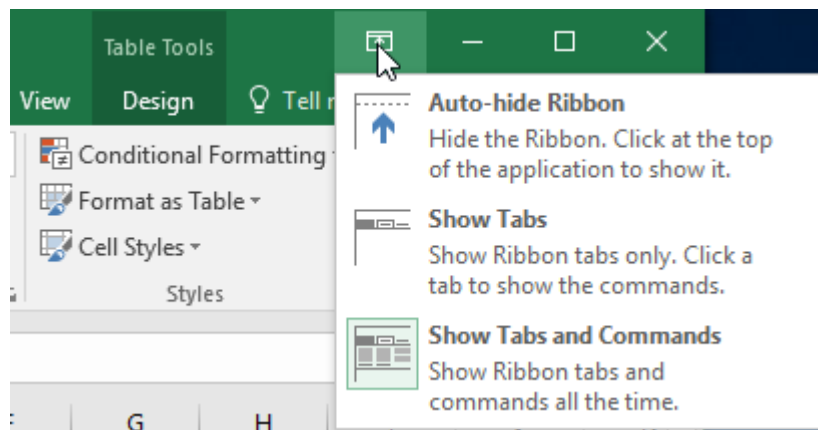
The Ribbon

Excel 2016 uses a tabbed Ribbon system instead of traditional menus. The Ribbon contains multiple tabs, each with several groups of commands. You will use these tabs to perform the most common tasks in Excel.

- Each tab will have one or more groups.
- Some groups will have an arrow you can click for more options.
- Click a tab to see more commands.

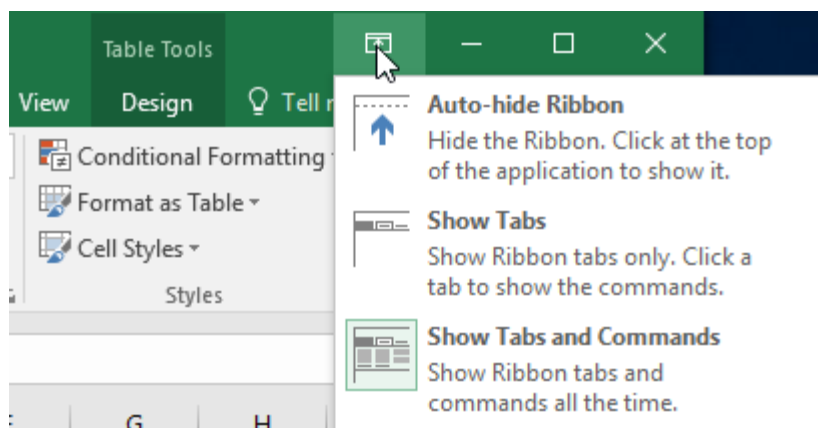


- You can adjust how the Ribbon is displayed with the Ribbon Display Options.



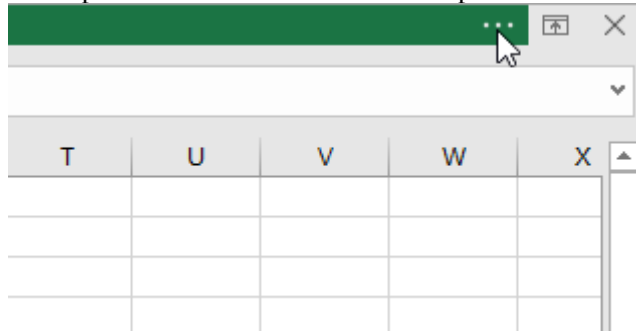
Certain programs, such as Adobe Acrobat Reader, may install additional tabs to the Ribbon. These tabs are called add-ins. To change the Ribbon Display Options:

The Ribbon is designed to respond to your current task, but you can choose to minimize it if you find that it takes up too much screen space. Click the Ribbon Display Options arrow in the upper-right corner of the Ribbon to display the drop-down menu.

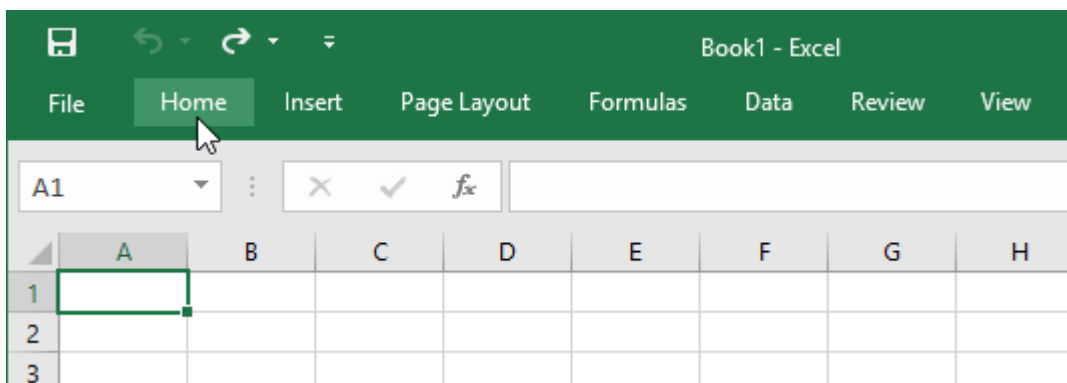


There are three modes in the Ribbon Display Options menu:

- Auto-hide Ribbon: Auto-hide displays your workbook in full-screen mode and completely hides the Ribbon. To show the Ribbon, click the Expand Ribbon command at the top of screen.



- Show Tabs: This option hides all command groups when they're not in use, but tabs will remain visible. To show the Ribbon, simply click a tab.



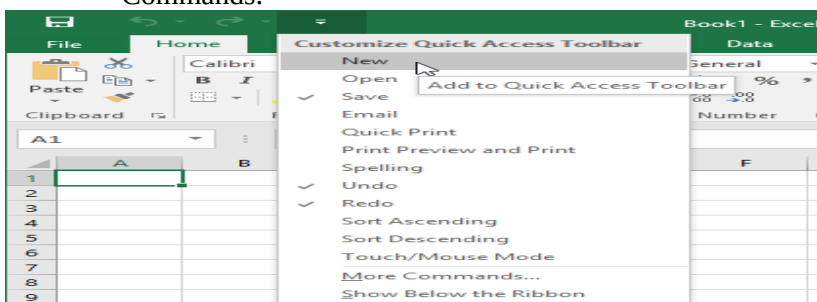
- Show Tabs and Commands: This option maximizes the Ribbon. All of the tabs and commands will be visible. This option is selected by default when you open Excel for the first time.

The Quick Access Toolbar

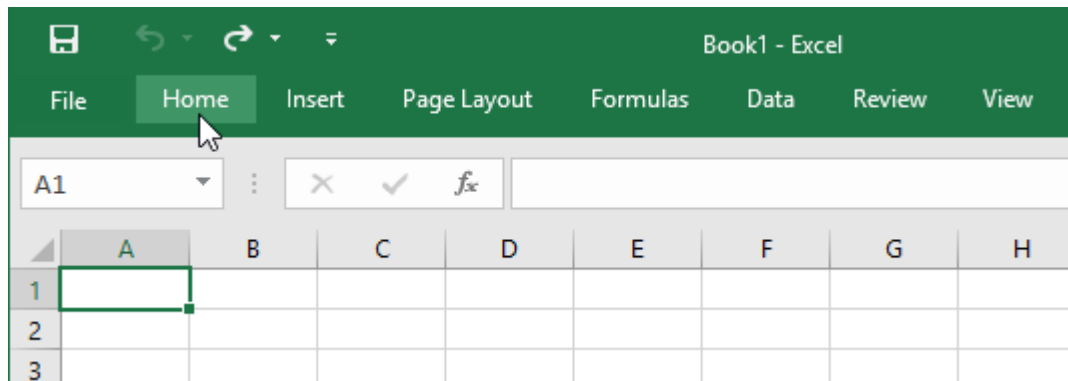
Located just above the Ribbon, the Quick Access Toolbar lets you access common commands no matter which tab is selected. By default, it includes the Save, Undo, and Repeat commands. You can add other commands depending on your preference.

To add commands to the Quick Access Toolbar:

1. Click the drop-down arrow to the right of the Quick Access Toolbar.
2. Select the command you want to add from the drop-down menu. To choose from more commands, select More Commands.



Show Tabs: This option hides all command groups when they're not in use, but tabs will remain visible. To show the Ribbon, simply click a tab.



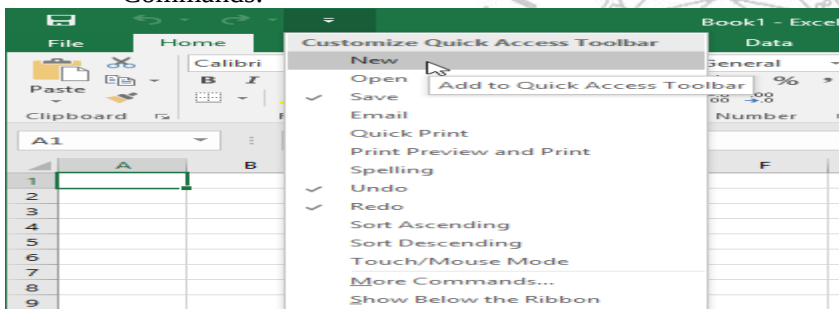
- **Show Tabs and Commands:** This option maximizes the Ribbon. All of the tabs and commands will be visible. This option is selected by default when you open Excel for the first time.

The Quick Access Toolbar

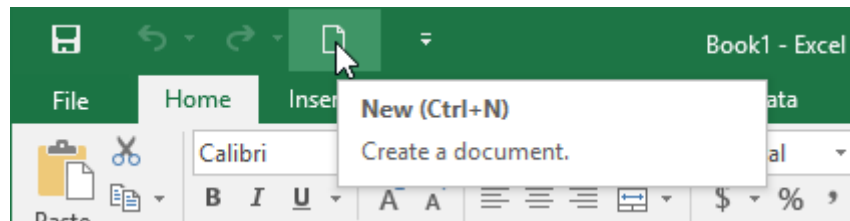
Located just above the Ribbon, the Quick Access Toolbar lets you access common commands no matter which tab is selected. By default, it includes the Save, Undo, and Repeat commands. You can add other commands depending on your preference.

To add commands to the Quick Access Toolbar:

3. Click the drop-down arrow to the right of the Quick Access Toolbar.
4. Select the command you want to add from the drop-down menu. To choose from more commands, select More Commands.

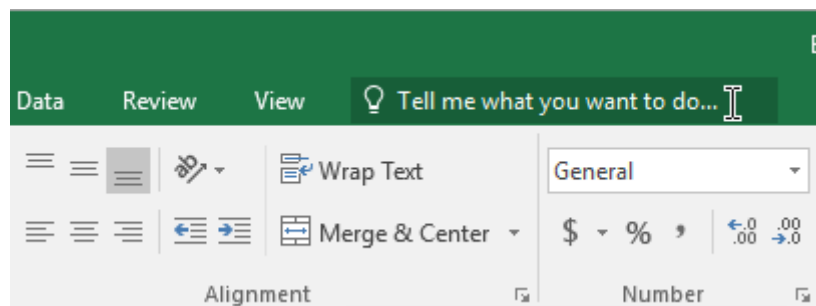


The command will be added to the Quick Access Toolbar.

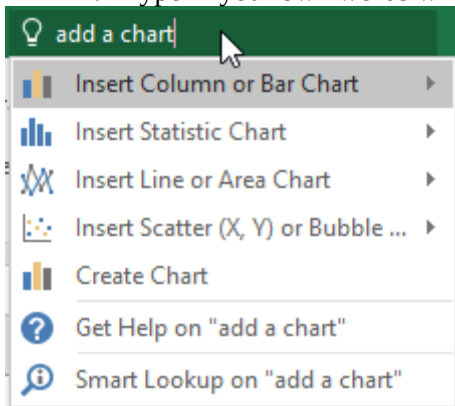


How to use Tell me:

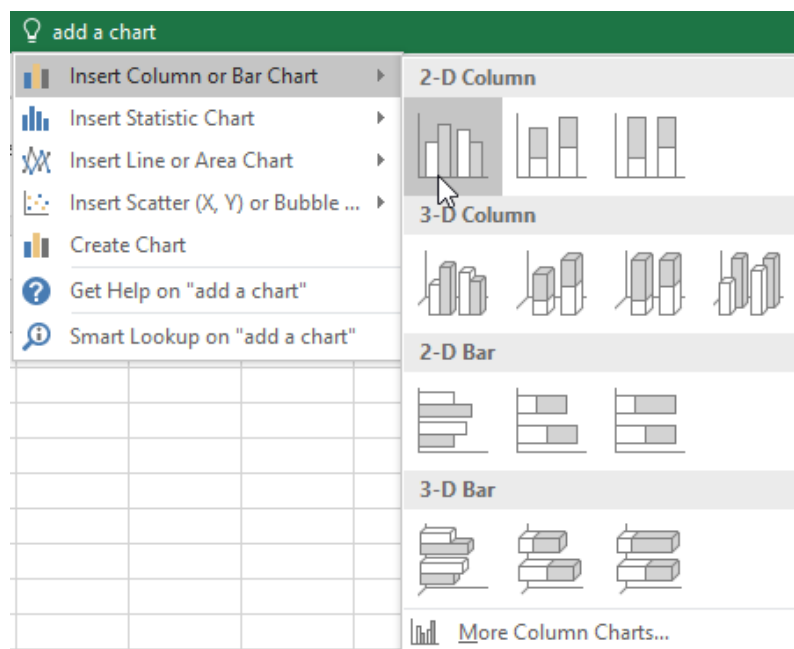
The Tell me box works like a search bar to help you quickly find tools or commands you want to use.

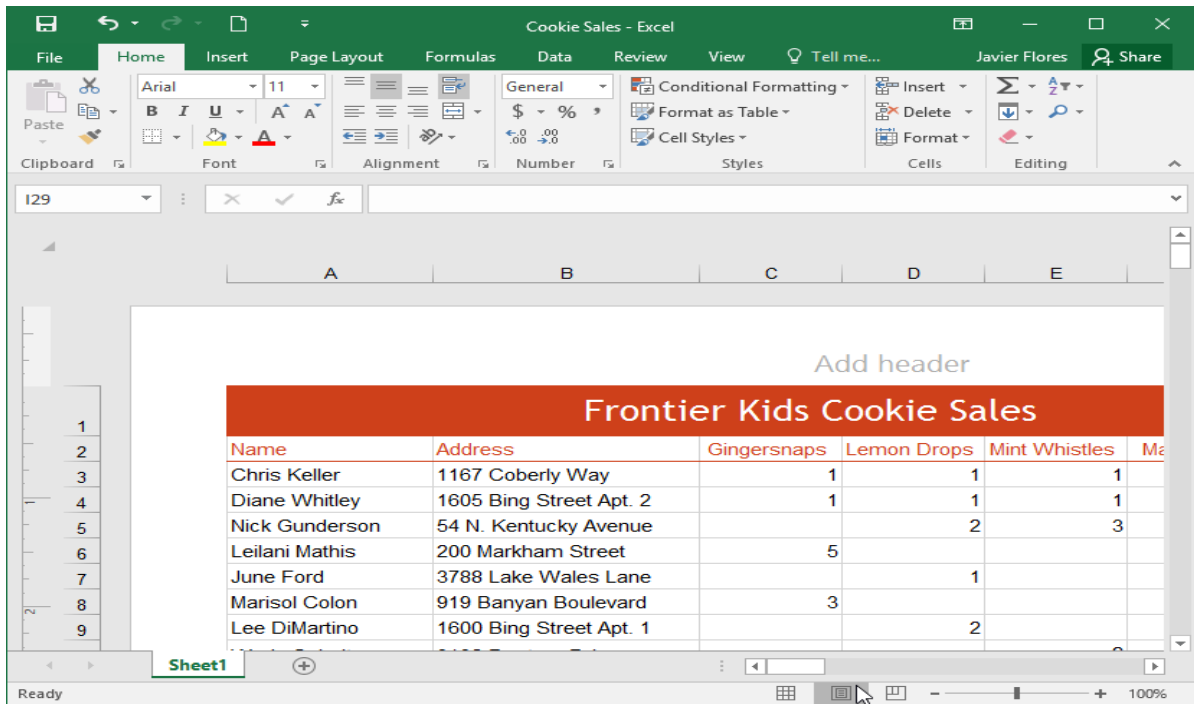


1. Type in your own words what you want to do.



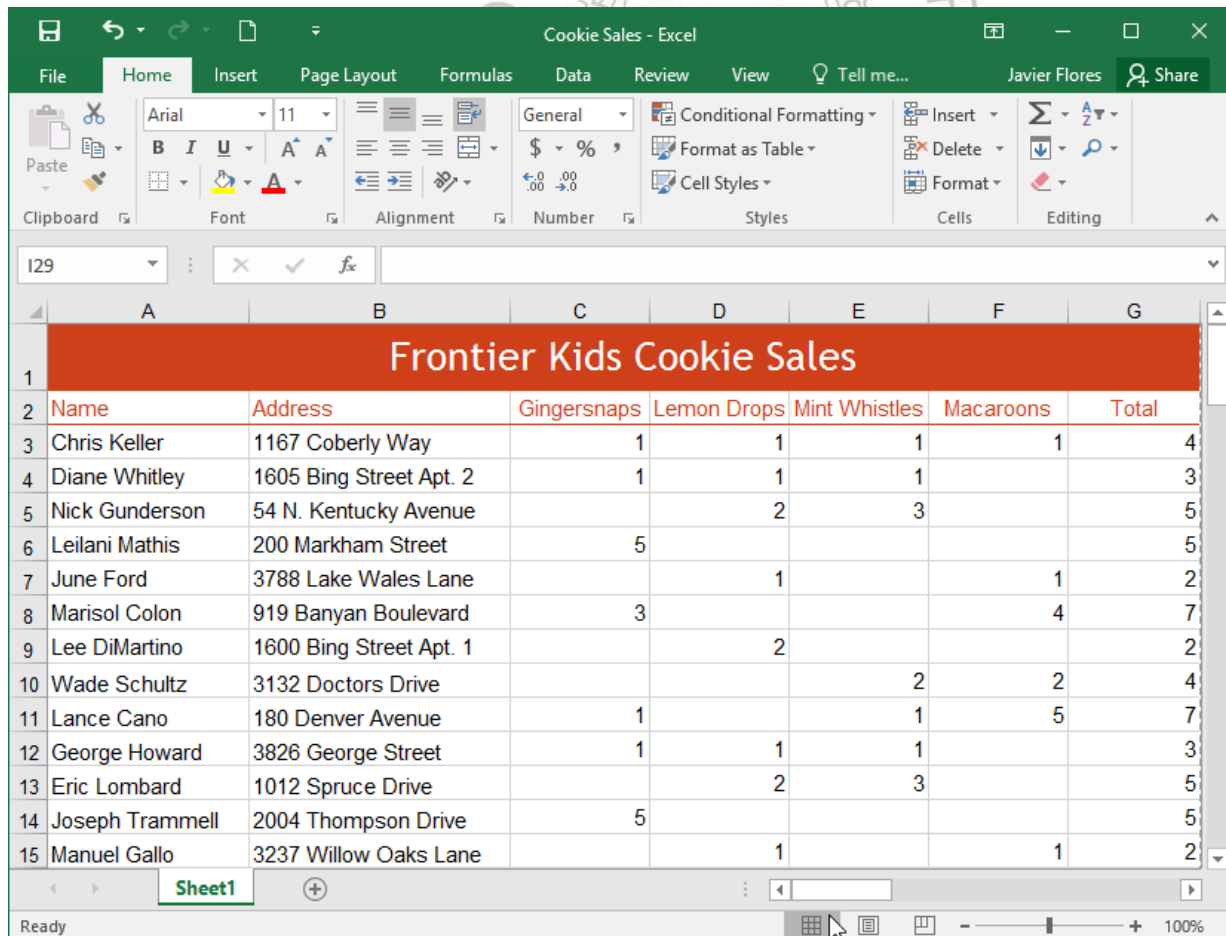
2. The results will give you a few relevant options. To use one, click it like you would a command on the Ribbon. Worksheet views





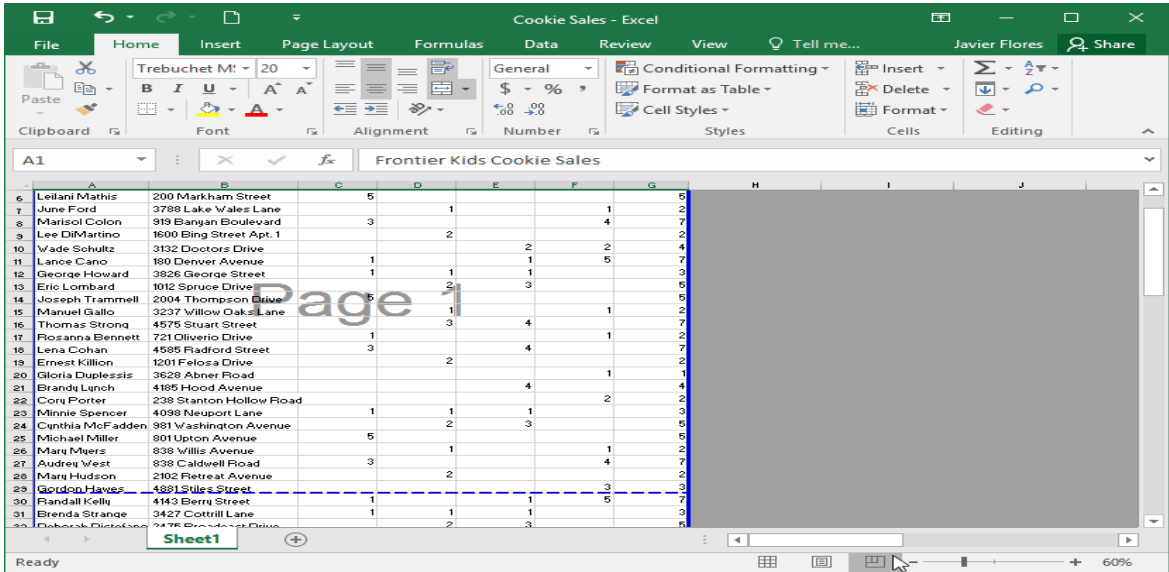
Excel 2016 has a variety of viewing options that change how your workbook is displayed. These views can be useful for various tasks, especially if you're planning to print the spreadsheet. To change worksheet views, locate the commands in the bottom-right corner of the Excel window and select Normal view, Page Layout view, or Page Break view.

- Normal view is the default view for all worksheets in Excel.



- Page Layout view displays how your worksheets will appear when printed. You can also add headers and footers in this view.

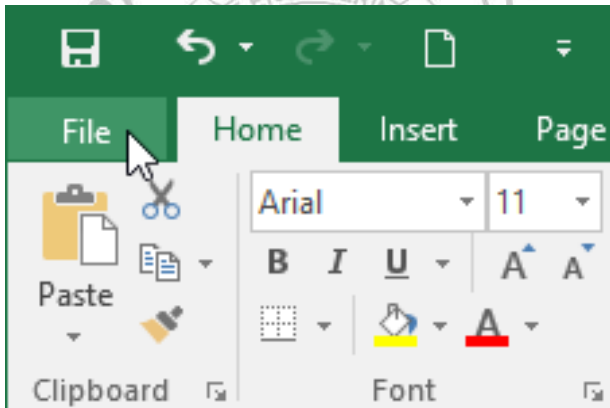
Page Break view allows you to change the location of page breaks, which is especially helpful when printing a lot of data from Excel.



Backstage view

Backstage view gives you various options for saving, opening a file, printing, and sharing your workbooks. To access Backstage view:

1. Click the File tab on the Ribbon. Backstage view will appear.

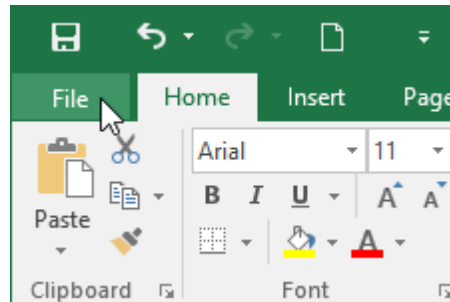


Creating and Opening Workbooks

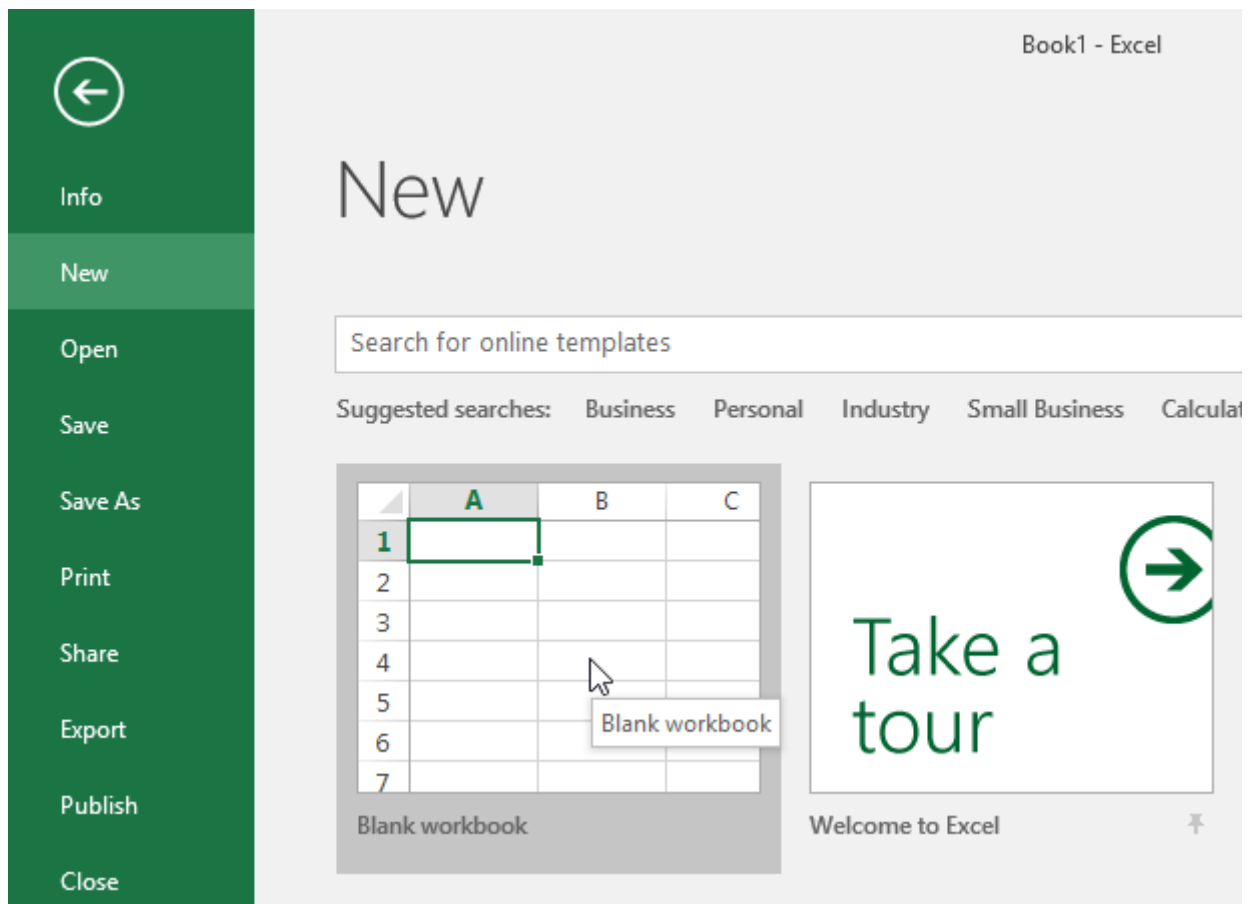
Excel files are called workbooks. Whenever you start a new project in Excel, you'll need to create a new workbook. There are several ways to start working with a workbook in Excel. You can choose to create a new workbook—either with a blank workbook or a predesigned template—or open an existing workbook.

To create a new blank workbook:

1. Select the File Tab. Backstage view will appear.



2. Select New, then click Blank workbook.

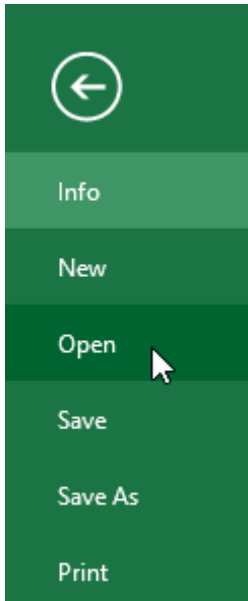


3. A new blank workbook will appear.

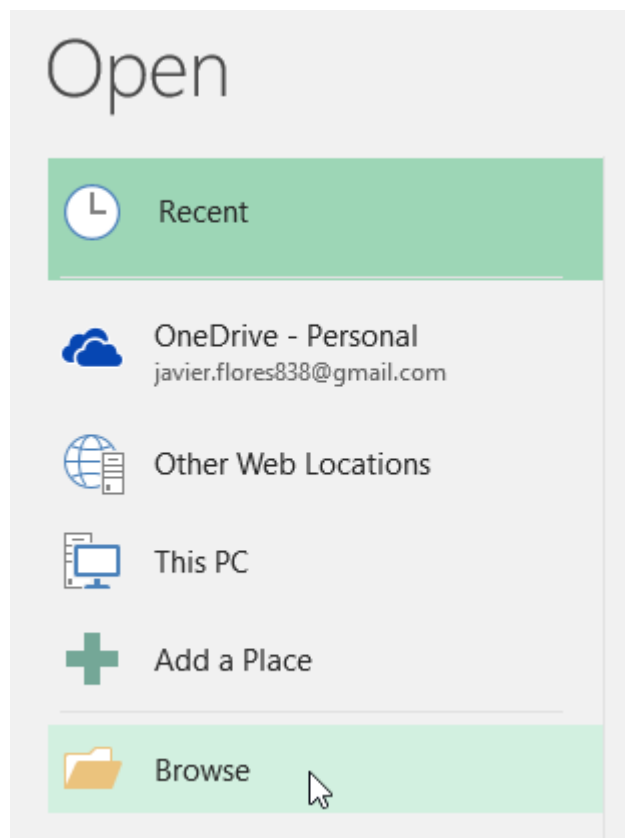
To open an existing workbook:

In addition to creating new workbooks, you'll often need to open a workbook that was previously saved.

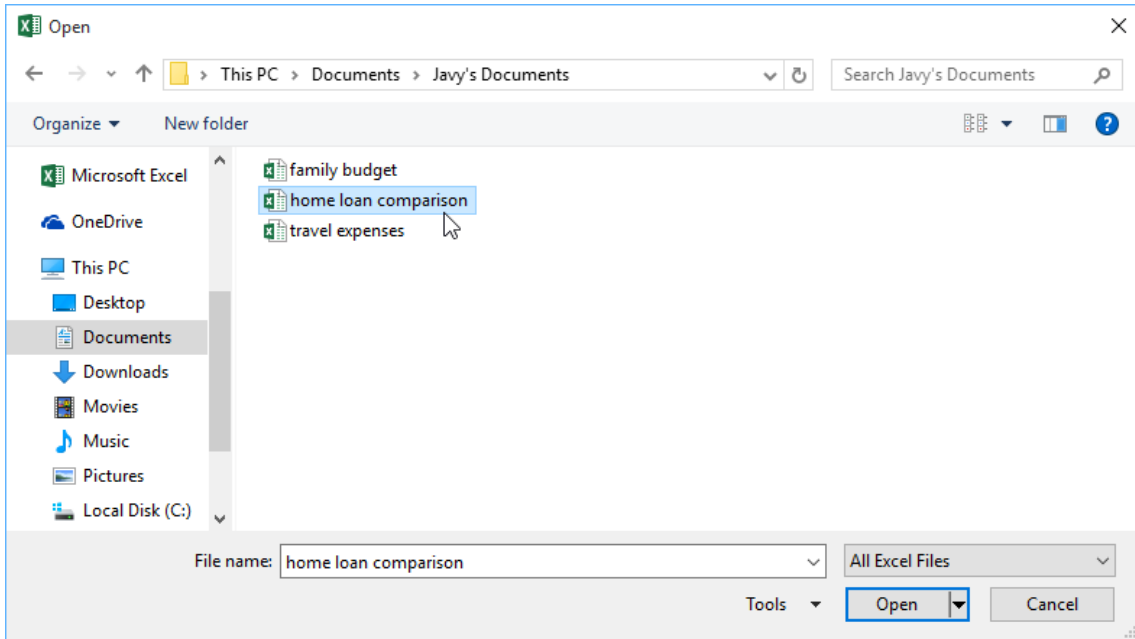
1. Navigate to Backstage view, then click Open.



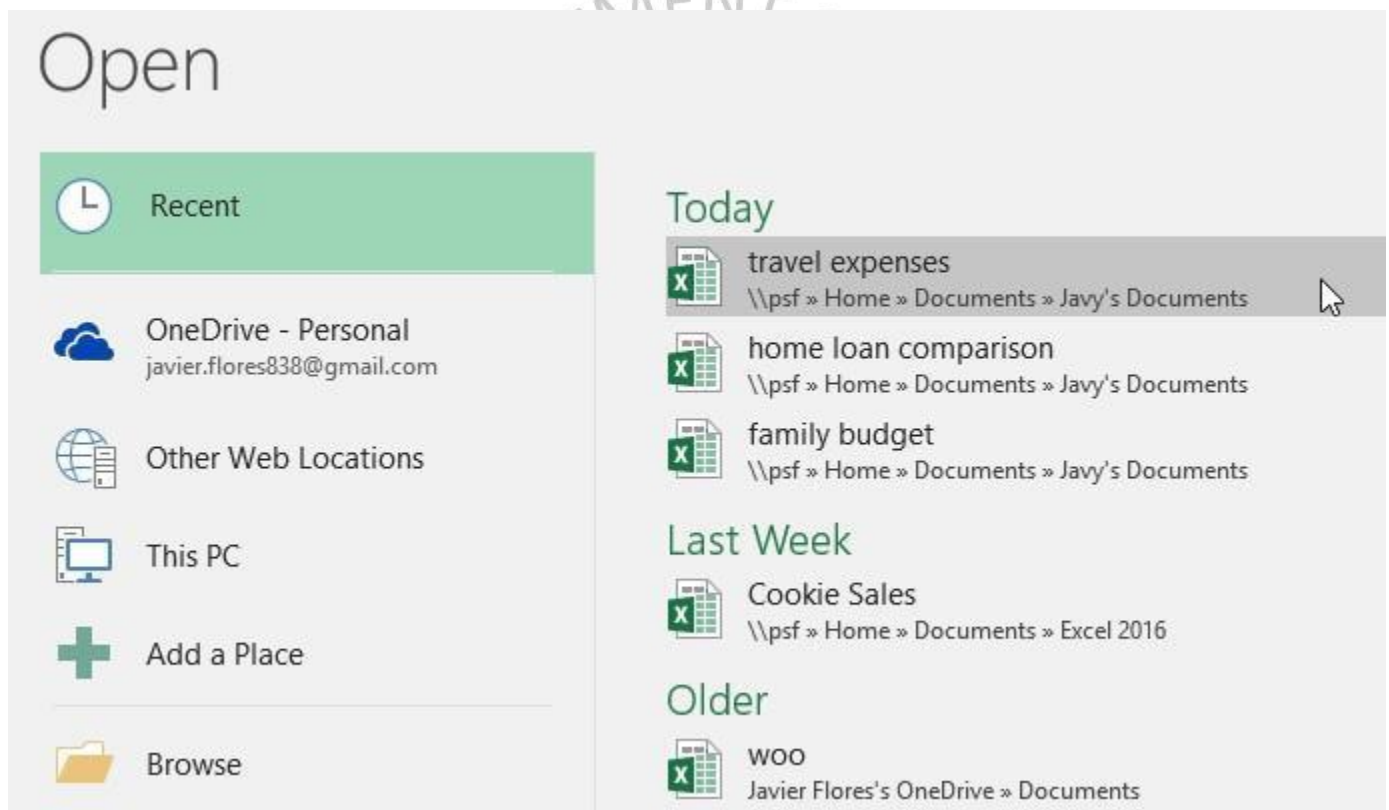
2. Select Computer, then click Browse. Alternatively, you can choose OneDrive to open files stored on your OneDrive.



3. The Open dialog box will appear. Locate and select your workbook, then click Open.



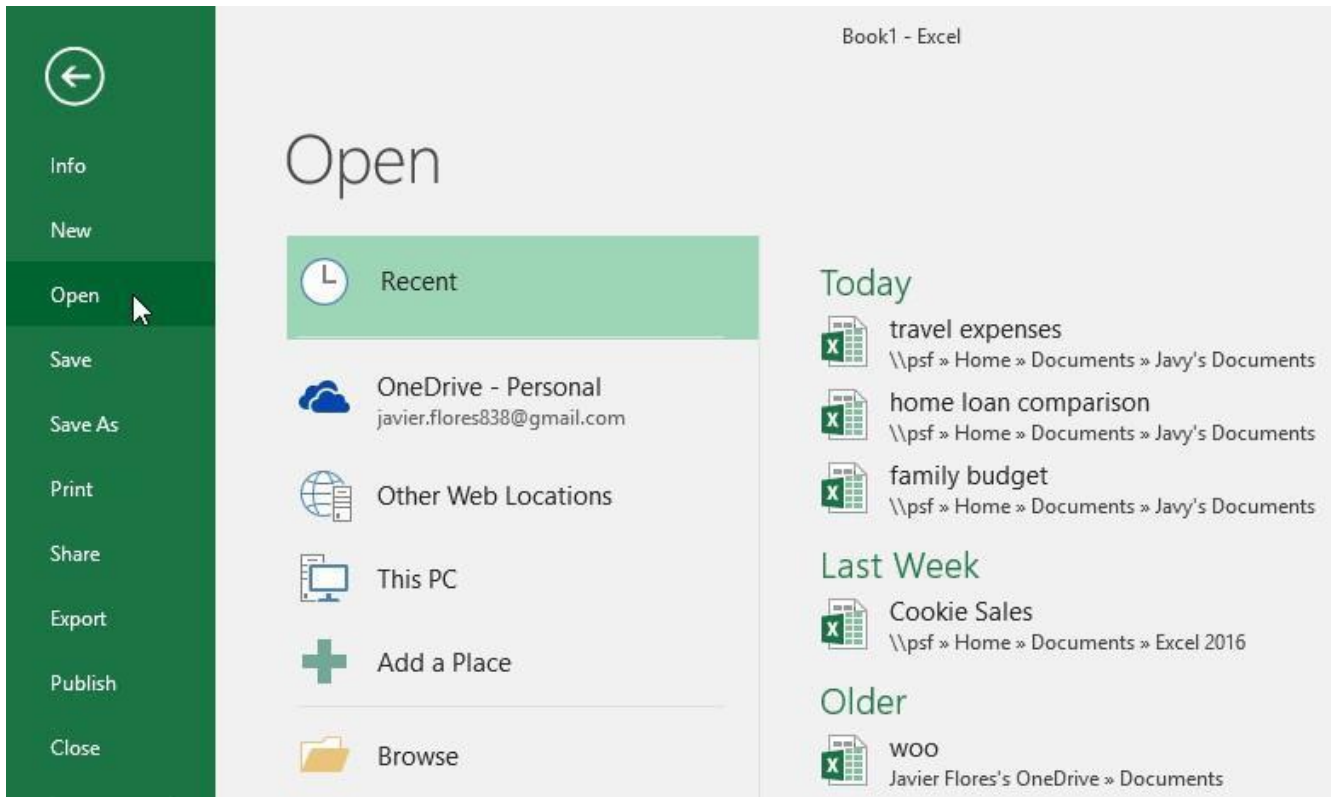
- If you've opened the desired workbook recently, you can browse your Recent Workbooks rather than search for the file.



To pin a workbook:

If you frequently work with the same workbook, you can pin it to Backstage view for faster access.

1. Navigate to Backstage view, then click Open. Your recently edited workbooks will appear.



2. Hover the mouse over the workbook you want to pin. A pushpin icon will appear next to the workbook. Click the pushpin icon.



3. The workbook will stay in Recent Workbooks. To unpin a workbook, simply click the pushpin icon again.

Pinned

	travel expenses \\psf » Home » Documents » Javy's Documents	11/9/2015 4:33 PM	↑
---	--	-------------------	---

Today

	home loan comparison \\psf » Home » Documents » Javy's Documents	11/9/2015 4:33 PM	
	family budget \\psf » Home » Documents » Javy's Documents	11/9/2015 4:33 PM	

Last Week

	Cookie Sales \\psf » Home » Documents » Excel 2016	11/6/2015 9:45 AM	
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Older

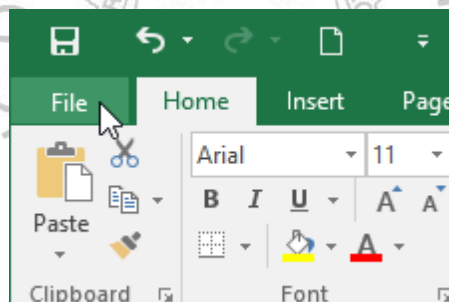
	WOO Javier Flores's OneDrive » Documents	8/3/2015 3:25 PM	
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Using templates

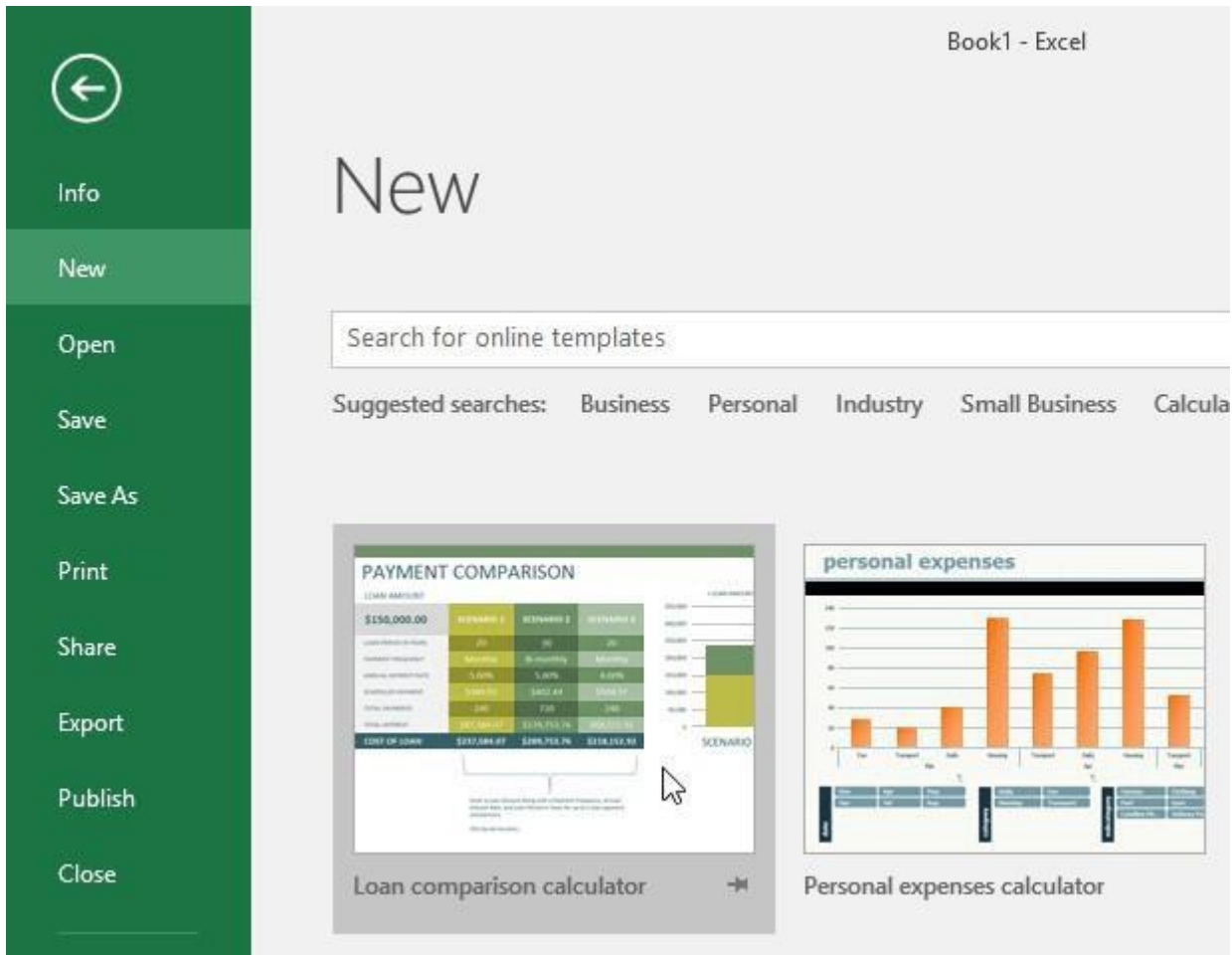
A template is a predesigned spreadsheet you can use to create a new workbook quickly. Templates often include custom formatting and predefined formulas, so they can save you a lot of time and effort when starting a new project.

To create a new workbook from a template:

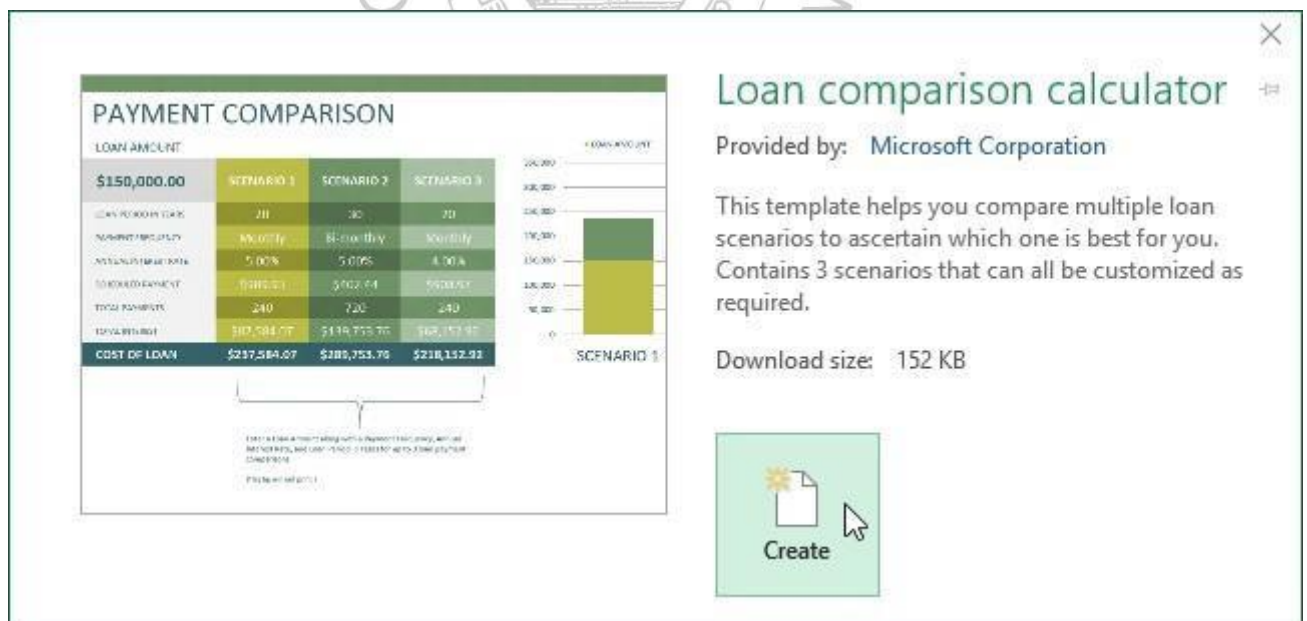
1. Click the File tab to access Backstage view.



2. Select New. Several templates will appear below the Blank workbook option.
3. Select a template to review it.

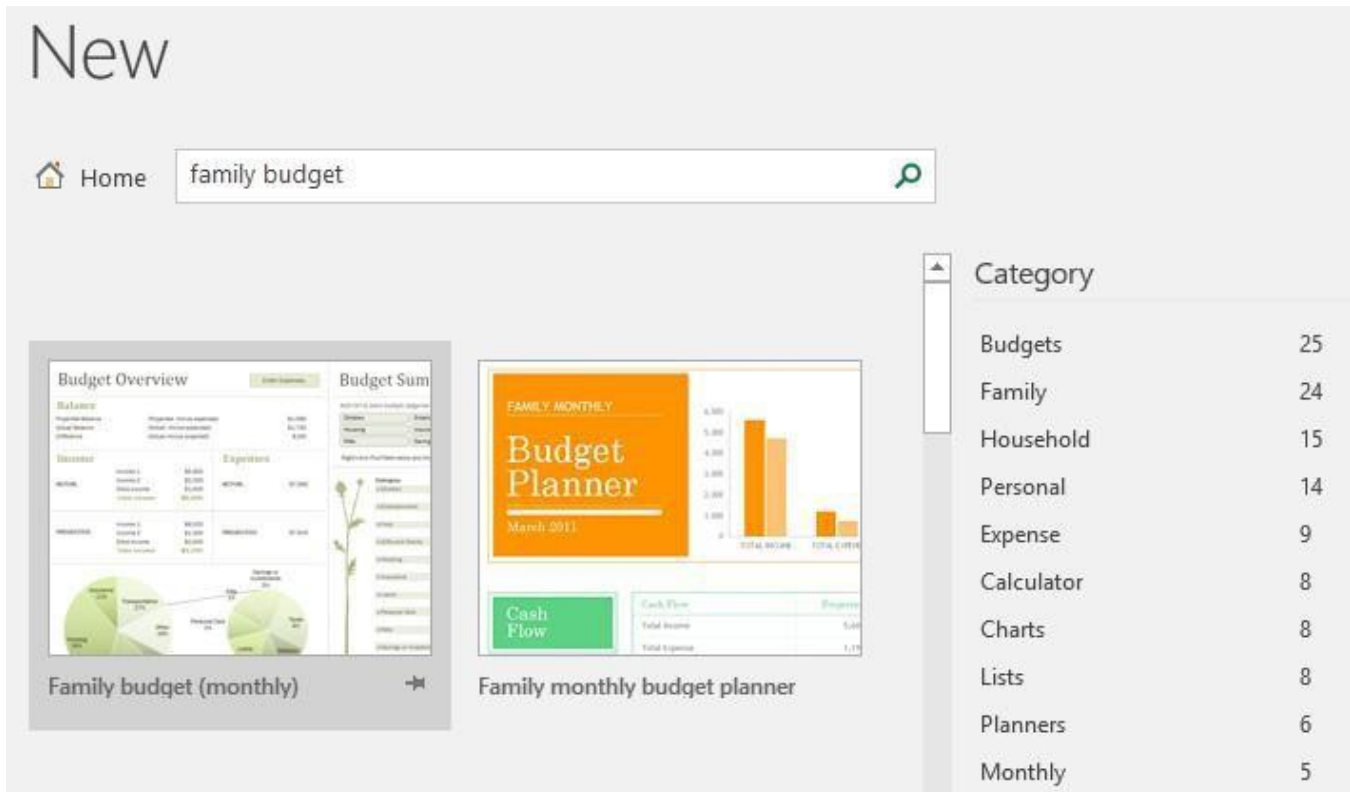


4. A preview of the template will appear, along with additional information on how the template can be used.
5. Click Create to use the selected template.



6. A new workbook will appear with the selected template.

You can also browse templates by category or use the search bar to find something more specific.



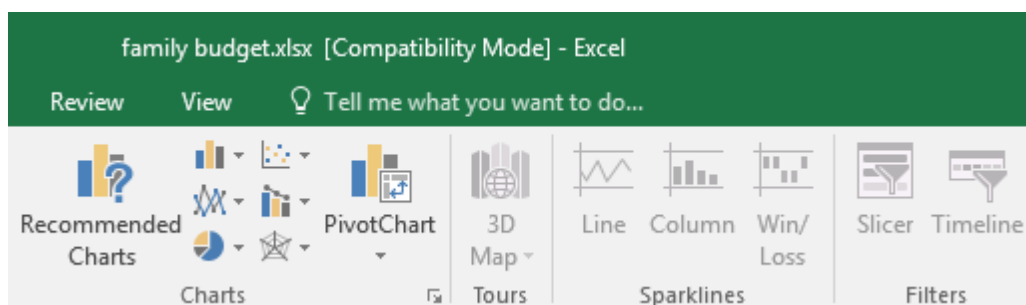
- It's important to note that not all templates are created by Microsoft. Many are created by third-party providers and even individual users, so some templates may work better than others.

Compatibility Mode

Sometimes you may need to work with workbooks that were created in earlier versions of Microsoft Excel, such as Excel 2003 or Excel 2000. When you open these types of workbooks, they will appear in Compatibility Mode.

Compatibility Mode disables certain features, so you'll only be able to access commands found in the program that was used to create the workbook. For example, if you open a workbook created in Excel 2003, you can only use tabs and commands found in Excel 2003.

In the image below, you can see that the workbook is in Compatibility Mode, which is indicated at the top of the window to the right of the file name. This will disable some Excel 2016 features, and they will be grayed out on the Ribbon.

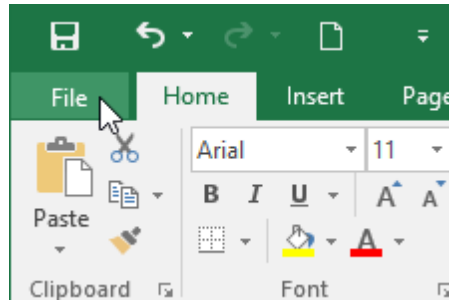


In order to exit Compatibility Mode, you'll need to convert the workbook to the current version type. However, if you're collaborating with others who only have access to an earlier version of Excel, it's best to leave the workbook in Compatibility Mode so the format will not change.

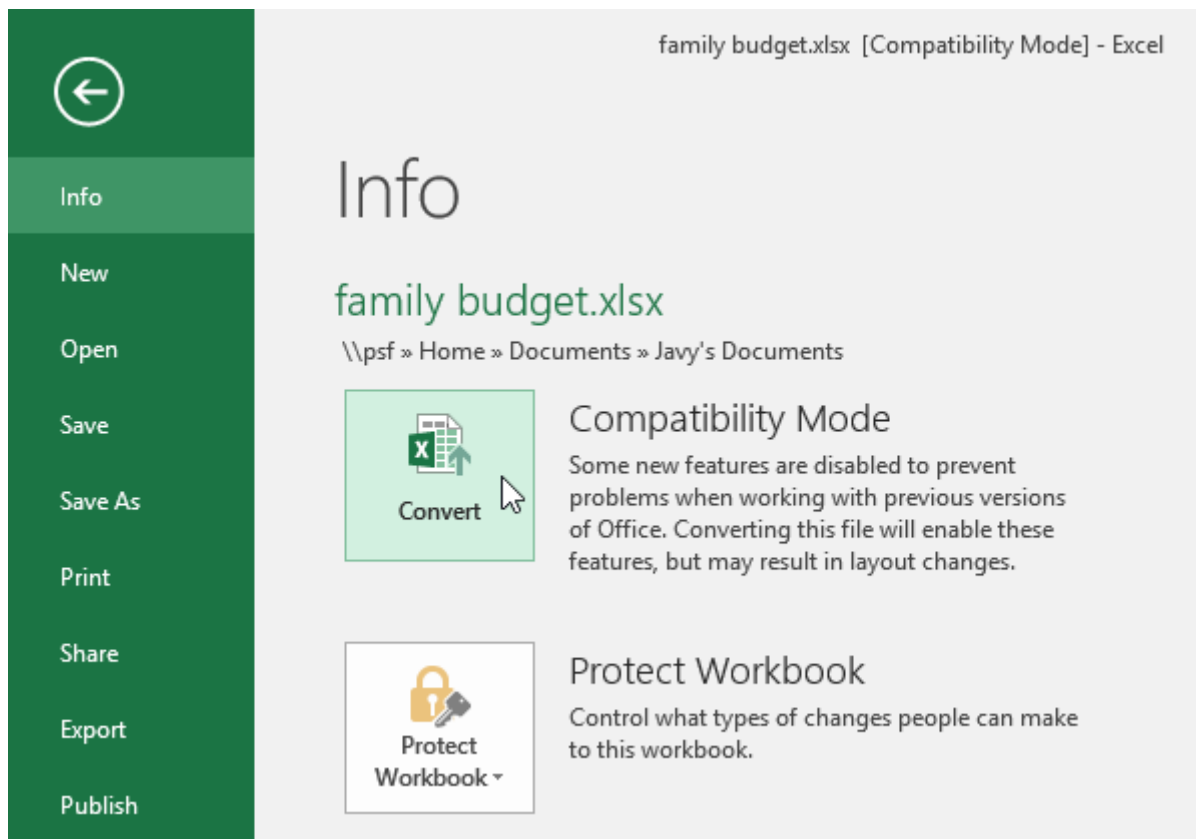
To convert a workbook:

If you want access to all of the Excel 2016 features, you can convert the workbook to the 2016 file format. Note that converting a file may cause some changes to the original layout of the workbook.

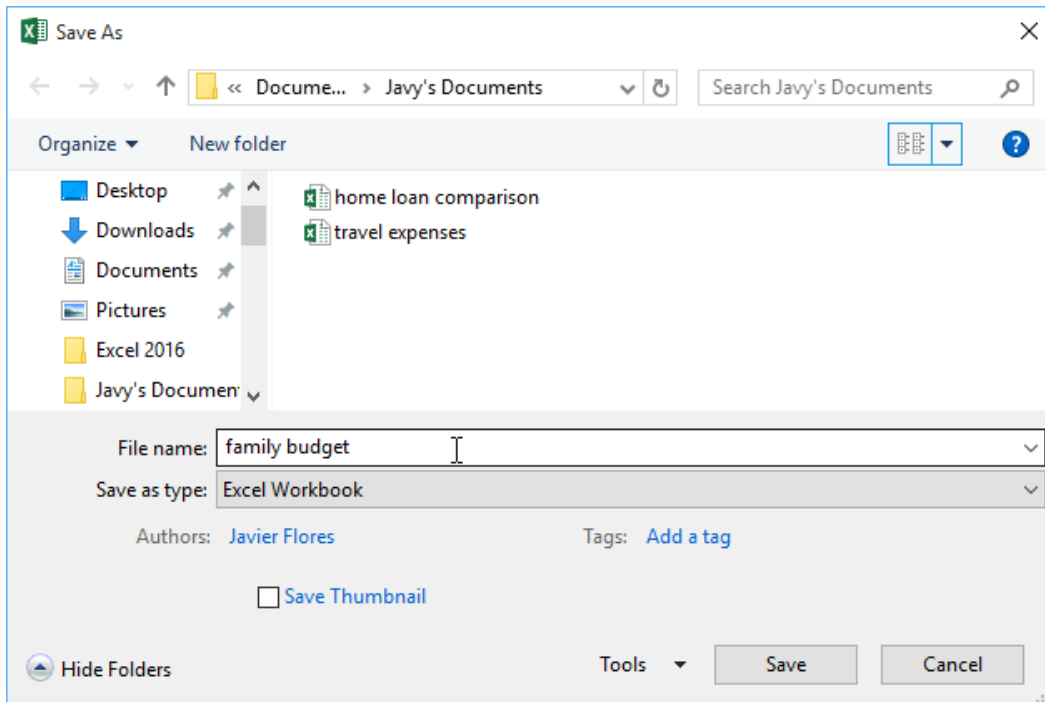
1. Click the File tab to access Backstage view.



2. Locate and select Convert command.



3. The Save As dialog box will appear. Select the location where you want to save the workbook, enter a file name for the workbook, and click Save.



4. The workbook will be converted to the newest file type.

Saving and Sharing Workbooks

Whenever you create a new workbook in Excel, you'll need to know how to save it in order to access and edit it later. As with previous versions of Excel, you can save files locally to your computer. But unlike older versions, Excel 2016 also lets you save a workbook to the cloud using OneDrive. You can also export and share workbooks with others directly from Excel.

Save and Save As

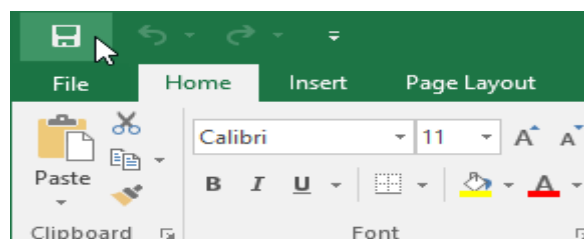
Excel offers two ways to save a file: Save and Save As. These options work in similar ways, with a few important differences:

- **Save:** When you create or edit a workbook, you'll use the Save command to save your changes. You'll use this command most of the time. When you save a file, you'll only need to choose a file name and location the first time. After that, you can just click the Save command to save it with the same name and location.
- **Save As:** You'll use this command to create a copy of a workbook while keeping the original. When you use Save As, you'll need to choose a different name and/or location for the copied version.

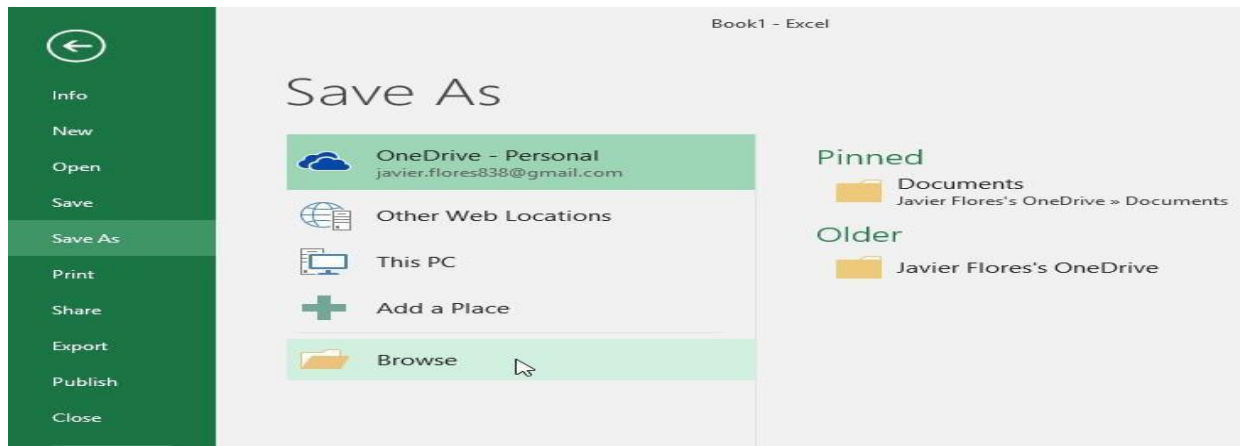
To save a workbook:

It's important to save your workbook whenever you start a new project or make changes to an existing one. Saving early and often can prevent your work from being lost. You'll also need to pay close attention to where you save the workbook so it will be easy to find later.

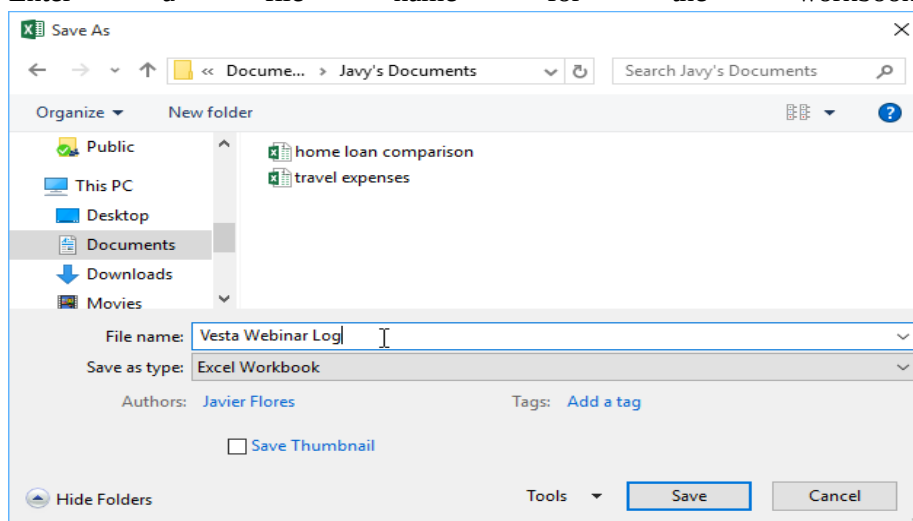
1. Locate and select the Save command on the Quick Access Toolbar.



2. If you're saving the file for the first time, the Save As pane will appear in Backstage view.
3. You'll then need to choose where to save the file and give it a file name. To save the workbook to your computer, select Computer, then click Browse. Alternatively, you can click OneDrive to save the file to your OneDrive.
4. The Save As dialog box will appear. Select the location where you want to save the workbook.



5. Enter a file name for the workbook, then click Save.



6. The workbook will be saved. You can click the Save command again to save your changes as you modify the workbook.

You can also access the Save command by pressing Ctrl+S on your keyboard. Using Save As to make a copy

If you want to save a different version of a workbook while keeping the original, you can create a copy. For example, if you have a file named Sales Data, you could save it as Sales Data 2 so you'll be able to edit the new file and still refer back to the original version.

To do this, you'll click the Save As command in Backstage view. Just like when saving a file for the first time, you'll need to choose where to save the file and give it a new file name.

Restaurant's Weekly Payroll Summary					
NAME	HOURS	WAGE	GROSS PAY	TAXES	NET PAY
Abigail	40.5	14.62			
Brianna	39.5	12.45			
Callie	27.75	11.09			
Charlotte	38	14.88			
Delilah	29	15.95			
Dorothy	27	12.95			
Elizabeth	14	13.45			
Evelyn	20	11.85			
TOTALS:					

EXERCISE:

Q1. Create the following spreadsheet in Microsoft Excel and be sure to use formulas to calculate the following:

1. Each employee's gross pay under the column labelled GROSS PAY. An employee's gross pay can be calculated by multiplying the hours worked by the hourly wage.
2. The amount of tax paid under the column labelled TAXES. Taxes are calculated by taking 35% of the gross pay.
3. The net-pay under the column labelled NET PAY, which is calculated by subtracting the taxes from the gross pay.
4. The totals of the last three columns.

Your spreadsheet should look something like this:



Question no: 1

Restaurant's Weekly Payroll Summary

EMPLOYEE NAME	HOURS WORKED	HOURLY WAGE	GROSS PAY	TAXES	NET PAY
Zara Fatima	40	Rs 180.00	Rs 7,200.00	Rs 2,520.00	Rs 4,680.00
Mariam Khan	35	Rs 150.00	Rs 5,250.00	Rs 1,837.50	Rs 3,412.50
Saad Ali	42	Rs 160.00	Rs 6,720.00	Rs 2,352.00	Rs 4,368.00
Adeel Raza	38	Rs 170.00	Rs 6,460.00	Rs 2,261.00	Rs 4,199.00
Bilal Hussain	45	Rs 175.00	Rs 7,875.00	Rs 2,756.25	Rs 5,118.75
Hamza Tariq	30	Rs 190.00	Rs 5,700.00	Rs 1,995.00	Rs 3,705.00
Sana Ahmed	50	Rs 160.00	Rs 8,000.00	Rs 2,800.00	Rs 5,200.00
Fatima Sohail	37	Rs 185.00	Rs 6,845.00	Rs 2,395.75	Rs 4,449.25
TOTALS:			Rs 54,050.00	Rs 18,917.50	Rs 35,132.50

Q2. Create the following spreadsheet in Microsoft Excel and be sure to use formulas to calculate the following:

1. Each student's overall average by adding the marks on each student's assignments and tests and dividing it by 6. Be sure to use the AVERAGE function.
2. The class average for each assignment and test.
3. The class median for each assignment and test. Use the MEAN function.
4. The highest mark for each assignment and test. Use the MAX function.
5. The lowest mark for each assignment and test. Use the MIN function.
6. The overall class average, the overall class median, the overall highest mark in the class, and the overall lowest mark in the class.

Your spreadsheet should look something like this:

NAME	A1	T1	A2	T2	A3	T3	A4	T3	AVERAGE
Abigail	63	66	88	79	74	85	91	74	
Brianna	56	84	74	82	88	35	44	81	
Callie	75	74	55	38	44	12	65	92	
Charlotte	45	66	44	90	41	49	98	53	
Delilah	95	35	65	77	25	68	87	64	
Dorothy	65	15	39	62	35	12	57	54	
Elizabeth	35	25	74	31	39	35	68	45	
Evelyn	58	55	66	44	45	17	45	75	
CLASS AVERAGE:									
CLASS MEDIAN:									
HIGHEST MARK:									
LOWEST MARK:									

ARTMENT OF
ENGINEERING

Question no: 2

STUDENT NAME	ASSIGNMENT 1	TEST 1	ASSIGNMENT 2	TEST 2	ASSIGNMENT 3	TEST 3	AVERAGE
Ali Raza	78	84	75	80	85	90	82.0
Fatima Sohail	92	88	90	94	96	98	93.0
Bilal Khan	70	65	68	72	74	76	70.8
Hina Ahmed	81	90	84	80	82	85	83.7
Usman Malik	60	67	63	65	70	69	65.7
Sara Tariq	85	90	87	76	88	91	86.2
Hamza Ali	75	72	78	97	79	81	80.3
Ayesha Khan	95	91	92	89	94	99	93.3
CLASS AVERAGE:	79.5	80.9	79.6	81.6	83.5	86.1	81.9
CLASS MEDIAN:	79.5	86	81	80	83.5	87.5	82.3
HIGHEST MARK:	95	91	92	97	96	99	99
LOWEST MARK:	60	65	63	65	70	69	60

LAB SESSION 04

Getting Started with MS Power Point

THEORY

PowerPoint Basics

PowerPoint is a presentation program that allows you to create dynamic slide presentations. These presentations can include animation, narration, images, videos, and much more. In this lesson, you'll learn your way around the PowerPoint environment, including the Ribbon, Quick Access Toolbar, and Backstage view.

Getting to know PowerPoint

PowerPoint 2016 is similar to PowerPoint 2013 and PowerPoint 2010. If you've previously used these versions, PowerPoint 2016 should feel familiar. But if you are new to PowerPoint or have more experience with older versions, you should first take some time to become familiar with the PowerPoint 2016 interface.

The PowerPoint interface

When you open PowerPoint for the first time, the Start Screen will appear. From here, you'll be able to create a new presentation, choose a template, and access your recently edited presentations. From the Start Screen, locate and select Blank Presentation to access the PowerPoint interface.

PROCEDURE:

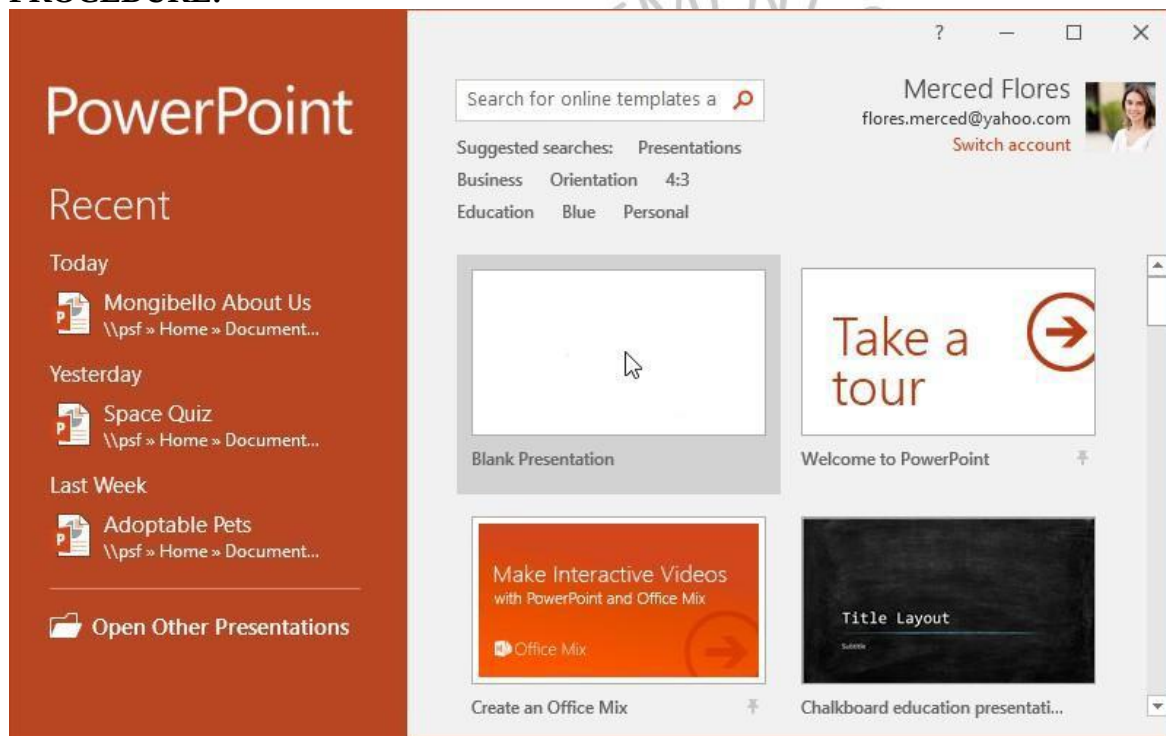
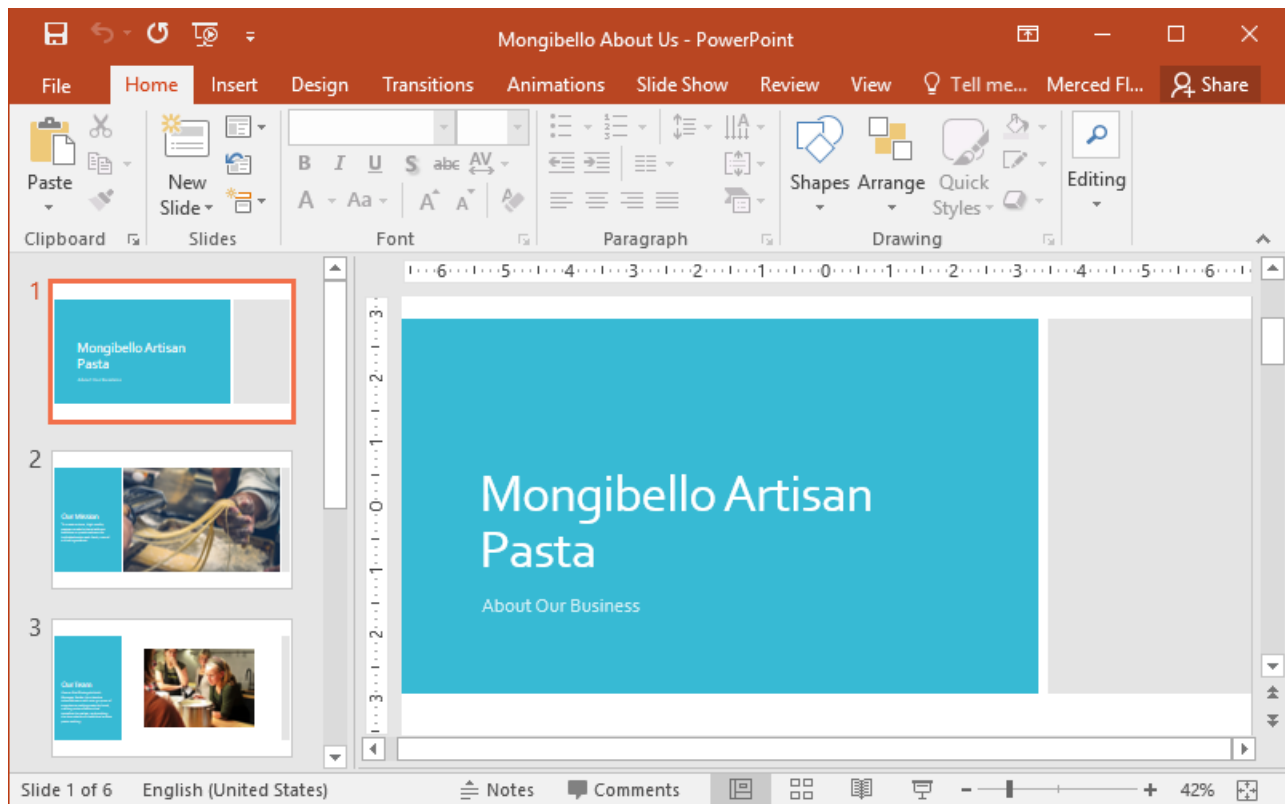


Fig 4.1 Power point Interface

Click the buttons in the interactive below to become familiar with the PowerPoint interface.

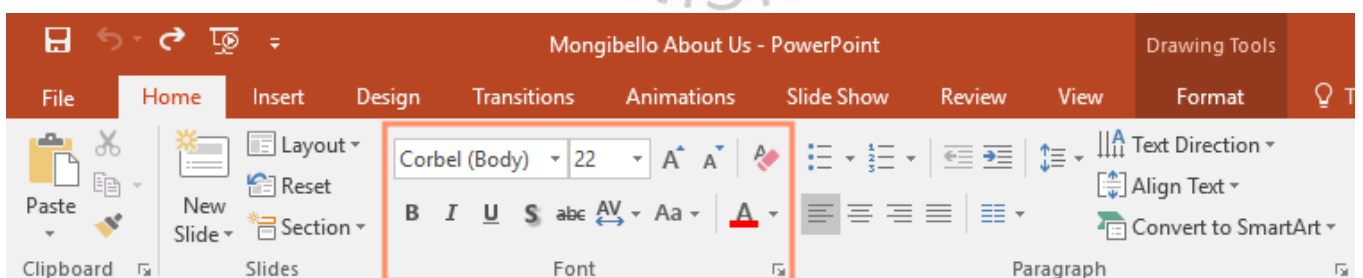


Working with the PowerPoint environment

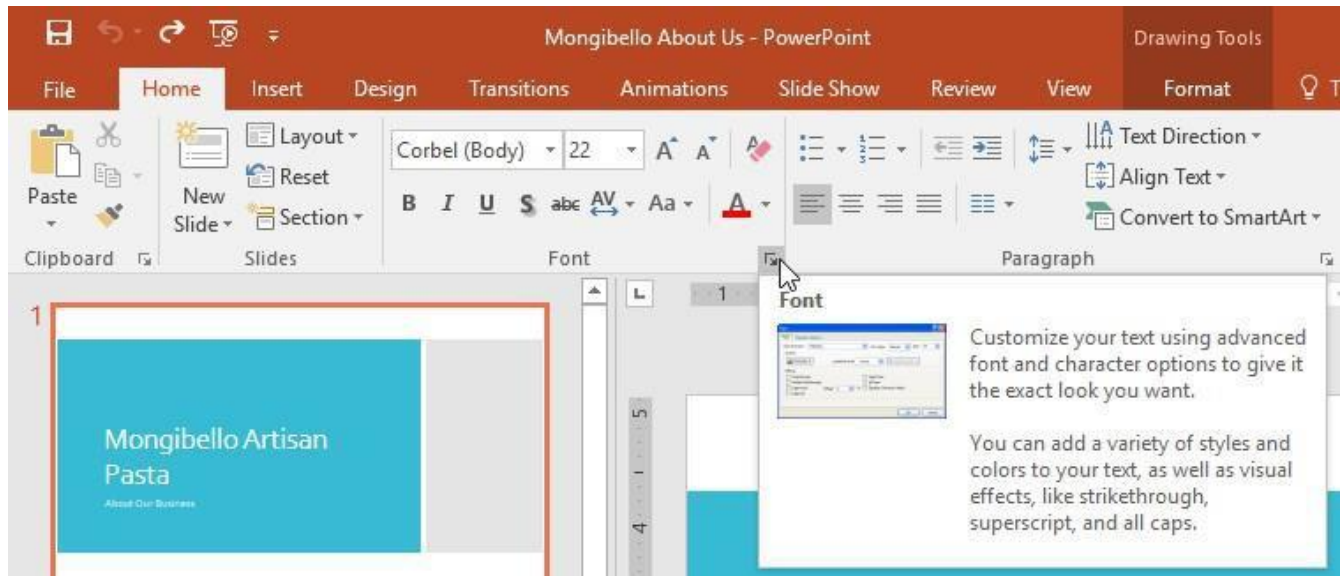
The Ribbon and Quick Access Toolbar are where you will find the commands to perform common tasks in PowerPoint. Backstage view gives you various options for saving, opening a file, printing, and sharing your document.

The Ribbon

PowerPoint uses a **tabbed Ribbon system** instead of traditional menus. **The Ribbon** contains **multiple tabs**, each with several **groups of commands**. For example, the Font group on the Home tab contains commands for formatting text in your document.

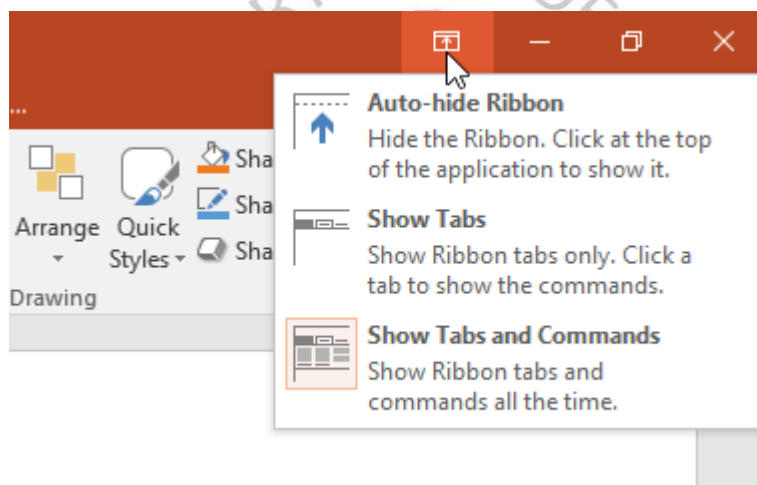


Some groups also have a **small arrow** in the bottom-right corner that you can click for even more options.



Showing and hiding the Ribbon

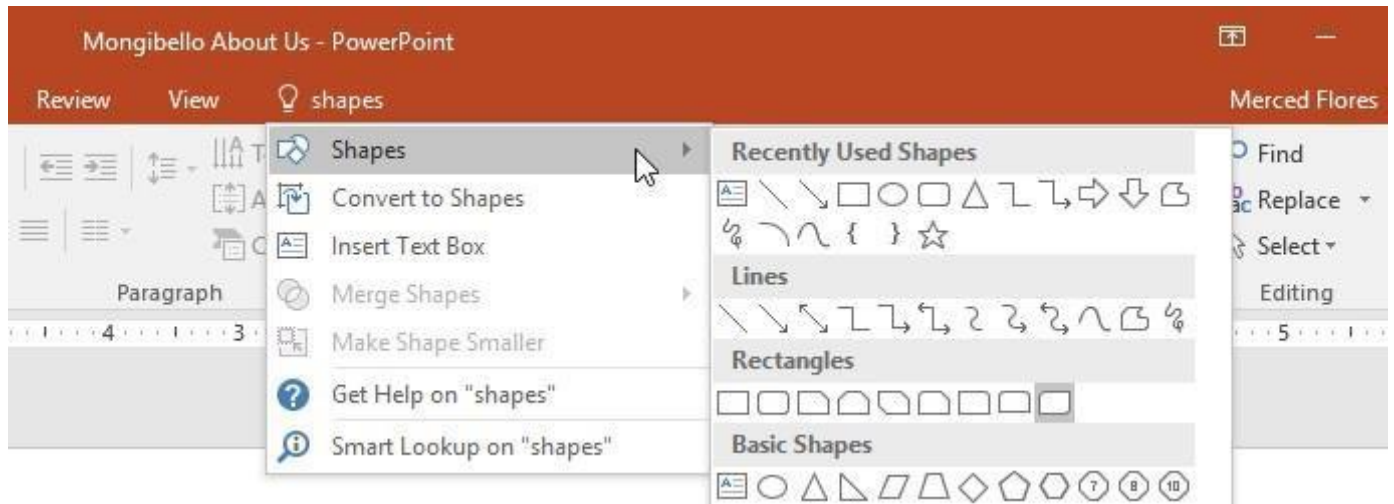
The Ribbon is designed to respond to your current task, but you can choose to **minimize** it if you find that it takes up too much screen space. Click the **Ribbon Display Options** arrow in the upper-right corner of the Ribbon to display the drop-down menu.



- **Auto-hide Ribbon:** Auto-hide displays your workbook in full-screen mode and completely hides the Ribbon. To **show the Ribbon**, click the **Expand Ribbon** command at the top of screen.
- **Show Tabs:** This option hides all command groups when they're not in use, but **tabs** will remain visible. To **show the Ribbon**, simply click a tab.
- **Show Tabs and Commands:** This option maximizes the Ribbon. All of the tabs and commands will be visible. This option is selected by default when you open PowerPoint for the first time.

Using the Tell me feature

If you're having trouble finding command you want, the **Tell Me** feature can help. It works just like a regular search bar: Type what you're looking for, and a list of options will appear. You can then use the command directly from the menu without having to find it on the Ribbon.

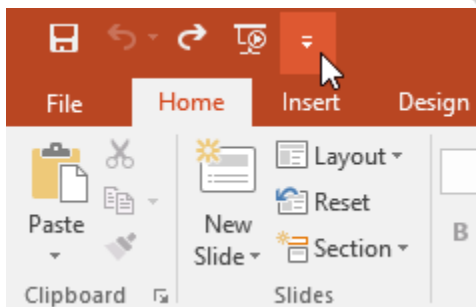


The Quick Access Toolbar

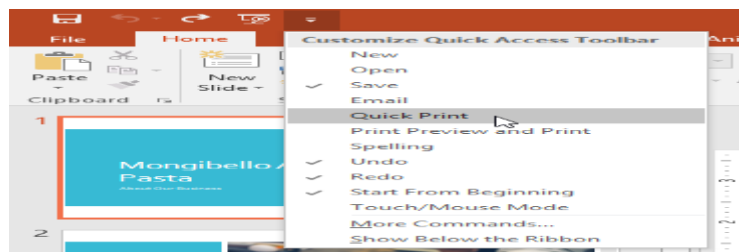
Located just above the Ribbon, the **Quick Access Toolbar** lets you access common commands no matter which tab is selected. By default, it includes the **Save**, **Undo**, **Redo**, and **Start from Beginning** commands. You can add other commands depending on your preference.

To add commands to the Quick Access Toolbar:

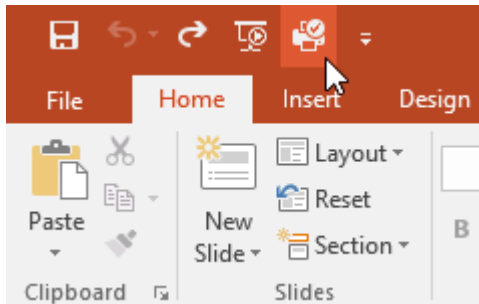
1. Click the **drop-down arrow** to the right of the **Quick Access Toolbar**.



2. Select the **command** you want to add from the drop-down menu. To choose from more commands, select **More Commands**.

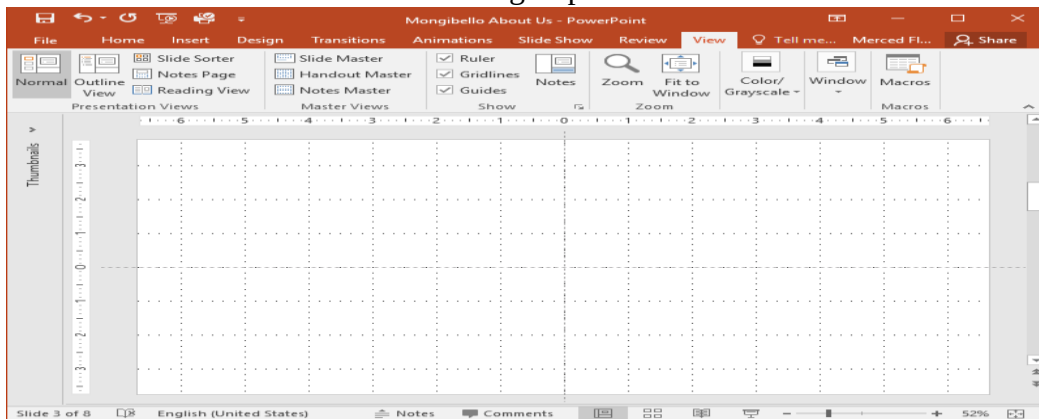


3. The command will be added to the Quick Access Toolbar.



The Ruler, guides, and gridlines

PowerPoint includes several tools to help organize and arrange content on your slides, including the **Ruler**, **guides**, and **gridlines**. These tools make it easier to **align objects** on your slides. Simply click the **check boxes** in the **Show** group on the **View** tab to show and hide these tools.

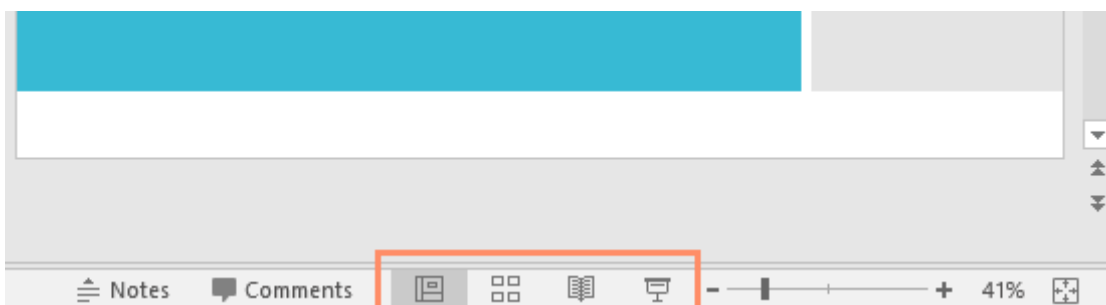


Zoom and other view options

PowerPoint has a variety of viewing options that change how your presentation is displayed. You can choose to view your presentation in **Normal** view, **Slide Sorter** view, **Reading** view, or **Slide Show** view. You can also **zoom in and out** to make your presentation easier to read.

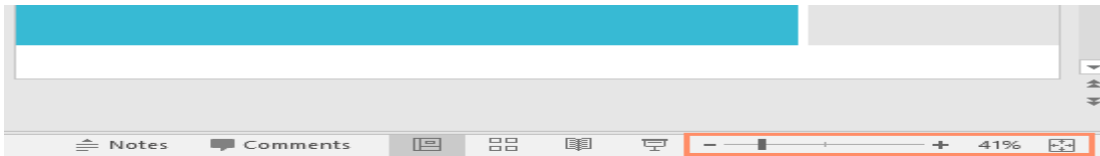
Switching slide views

Switching between different slide views is easy. Just locate and select the desired **slide view command** in the bottom-right corner of the PowerPoint window.



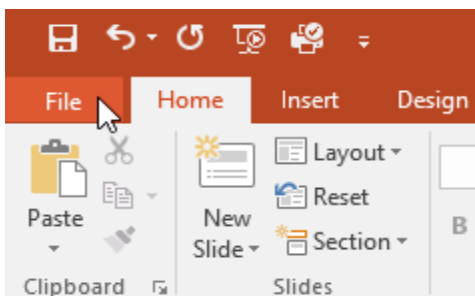
Zooming in and out

To zoom in or out, click and drag the **zoom control slider** in the bottom-right corner of the PowerPoint window. You can also select the + or - **commands** to zoom in or out by smaller increments. The number next to the slider displays the current **zoom percentage**, also called the **zoom level**.

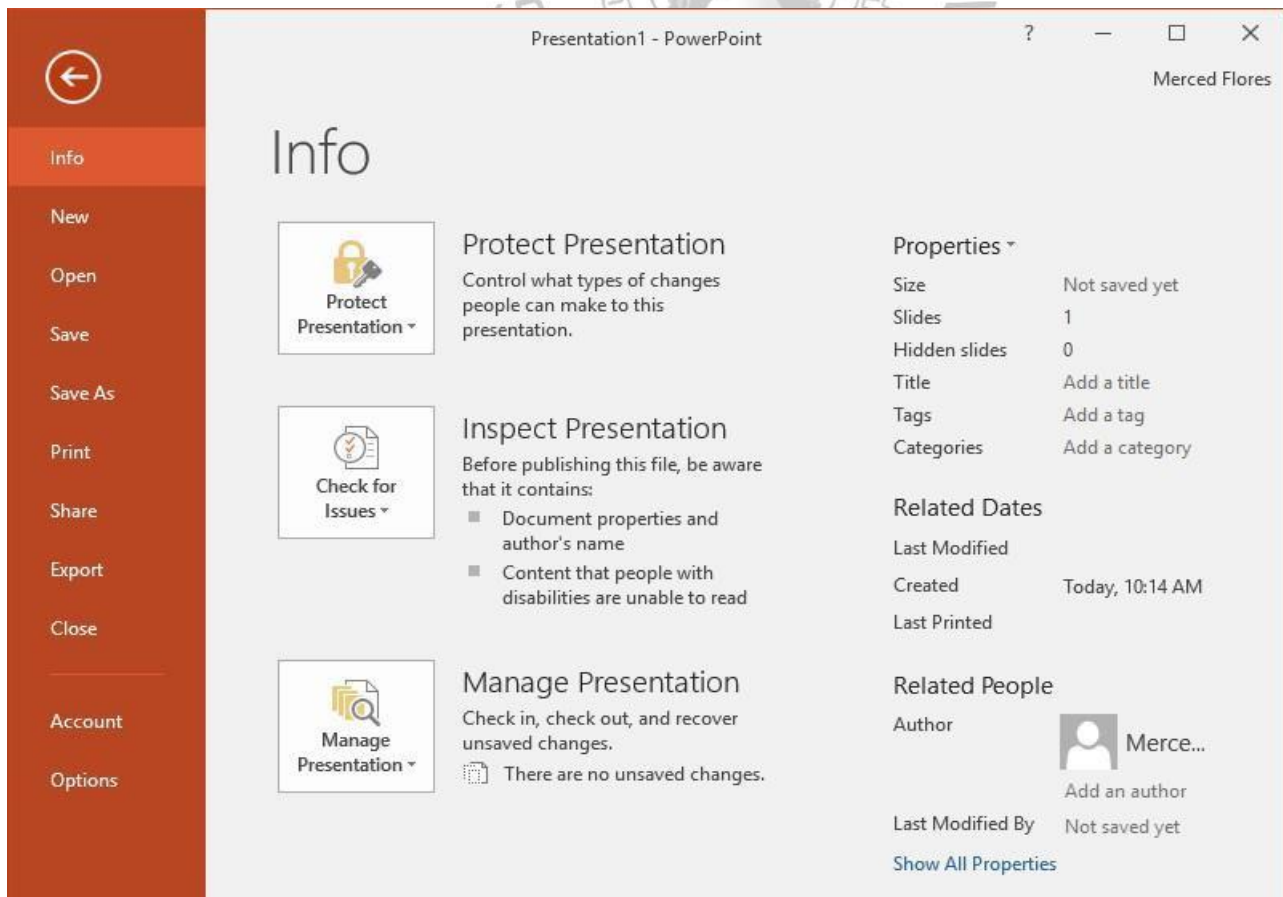


Backstage view

Backstage view gives you various options for saving, opening, printing, and sharing your presentations. To access Backstage view, click the **File** tab on the **Ribbon**.



Click the buttons in the interactive below to learn more about using Backstage view.



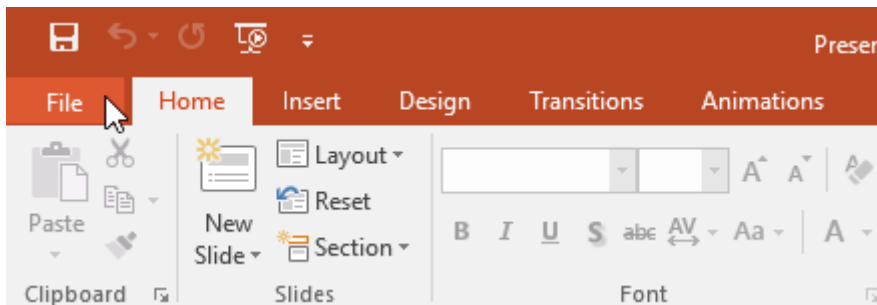
Creating and Opening Presentations

PowerPoint files are called presentations. Whenever you start a new project in PowerPoint, you'll need to create a new presentation, which can either be blank or from a template. You'll also need to know how to open an existing presentation.

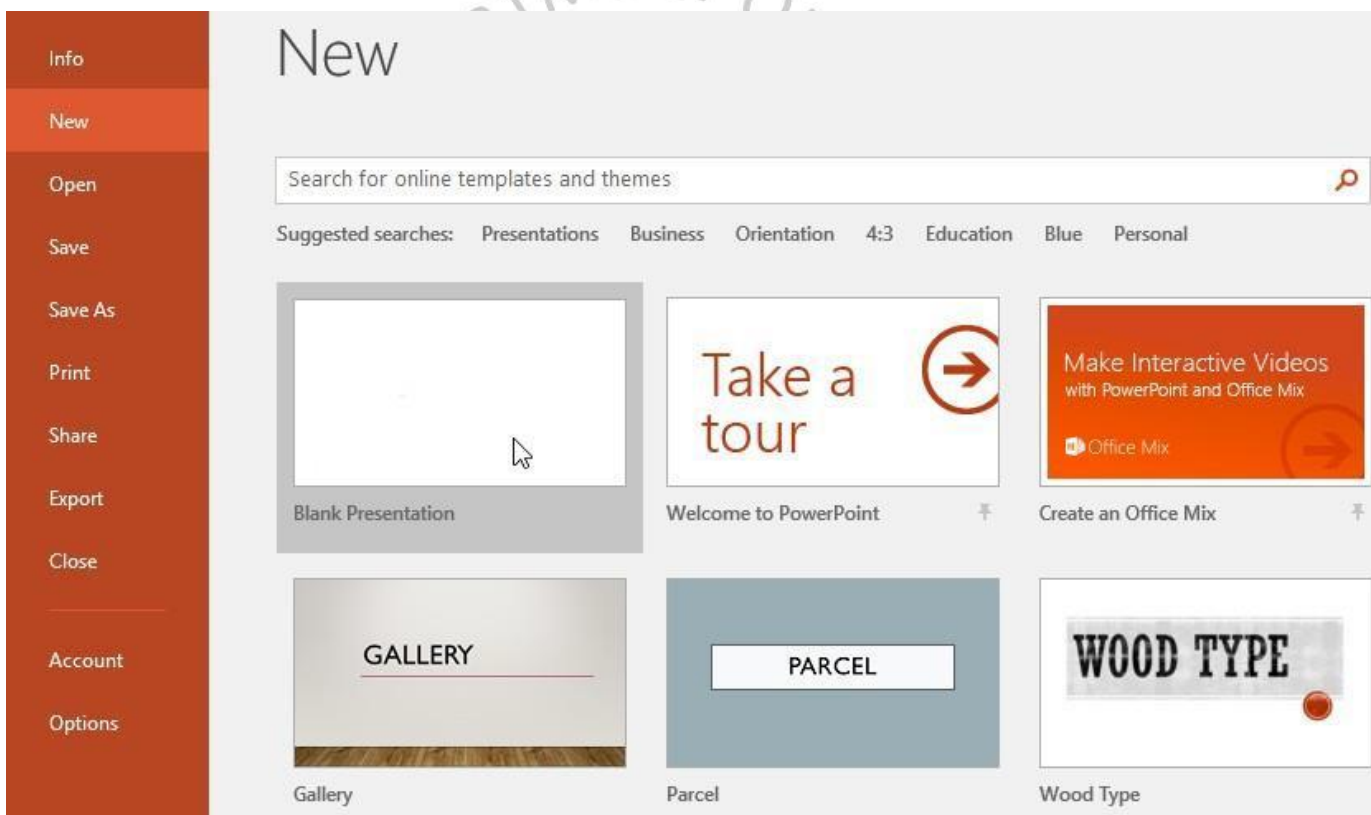
To create a new presentation:

When beginning a new project in PowerPoint, you'll often want to start with a new blank presentation.

1. Select the **File** tab to go to **Backstage view**.



2. Select **New** on the left side of the window, then click **Blank Presentation**.

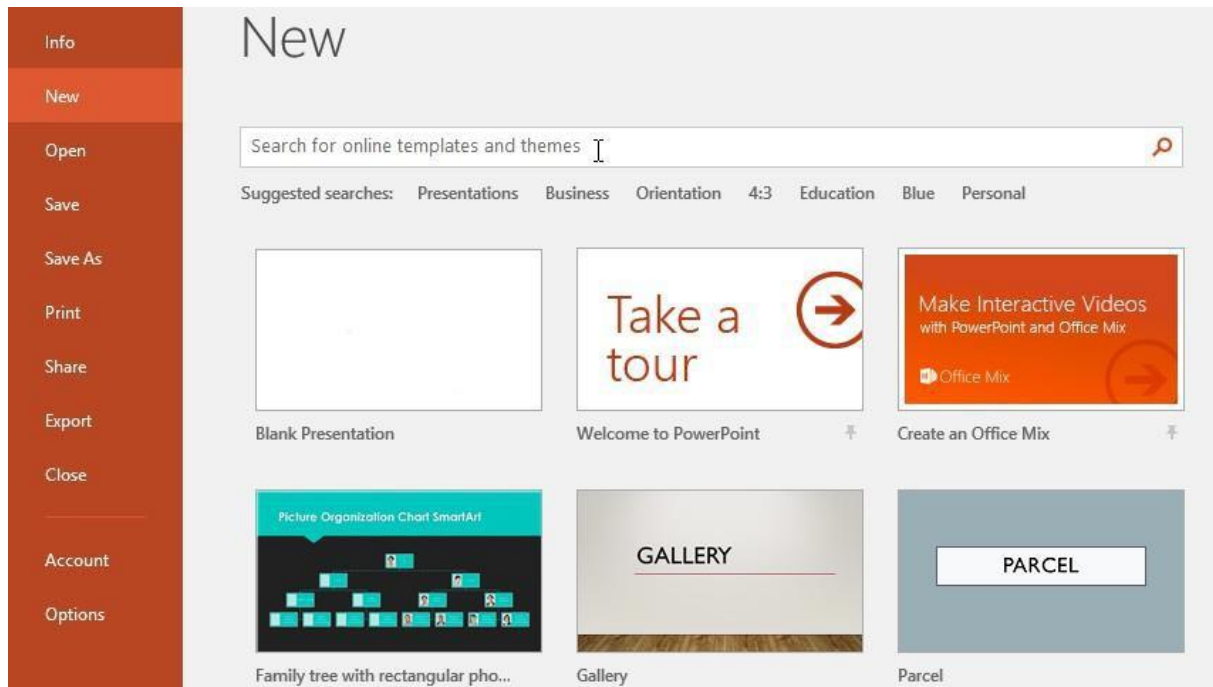


3. A new presentation will appear.

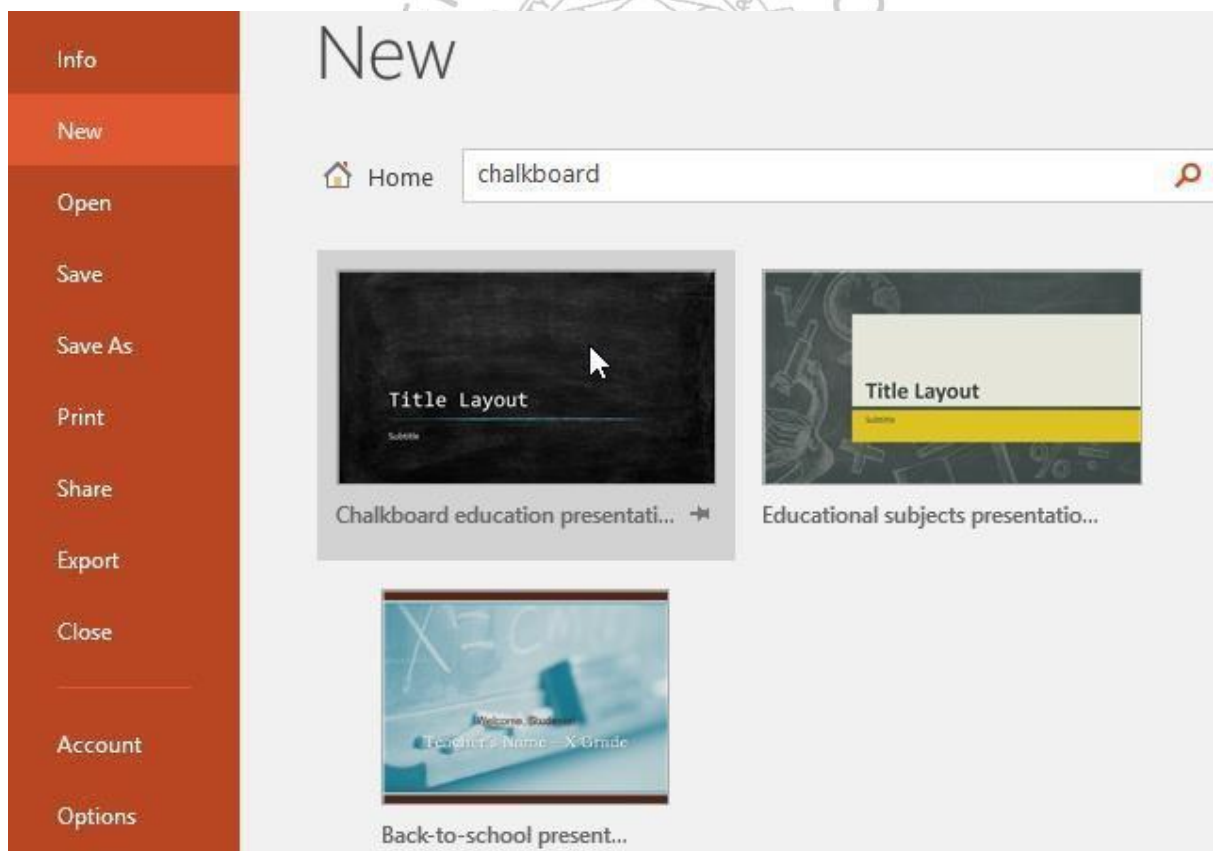
To create a new presentation from a template:

A **template** is a **predesigned presentation** you can use to create a new slide show quickly. Templates often include **custom formatting** and **designs**, so they can save you a lot of time and effort when starting a new project.

1. Click the **File** tab to access **Backstage view**, then select **New**.
2. You can click a suggested search to find templates or use the **search bar** to find something more specific. In our example, we'll search for the keyword **chalkboard**.



3. Select a **template** to review it.



4. A **preview** of the template will appear, along with **additional information** on how the template can be used.
5. Click **Create** to use the selected template.



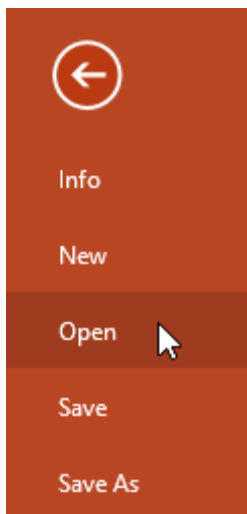
6. A new presentation will appear with the **selected template**.

It's important to note that not all templates are created by Microsoft. Many are created by third-party providers and even individual users, so some templates may work better than others.

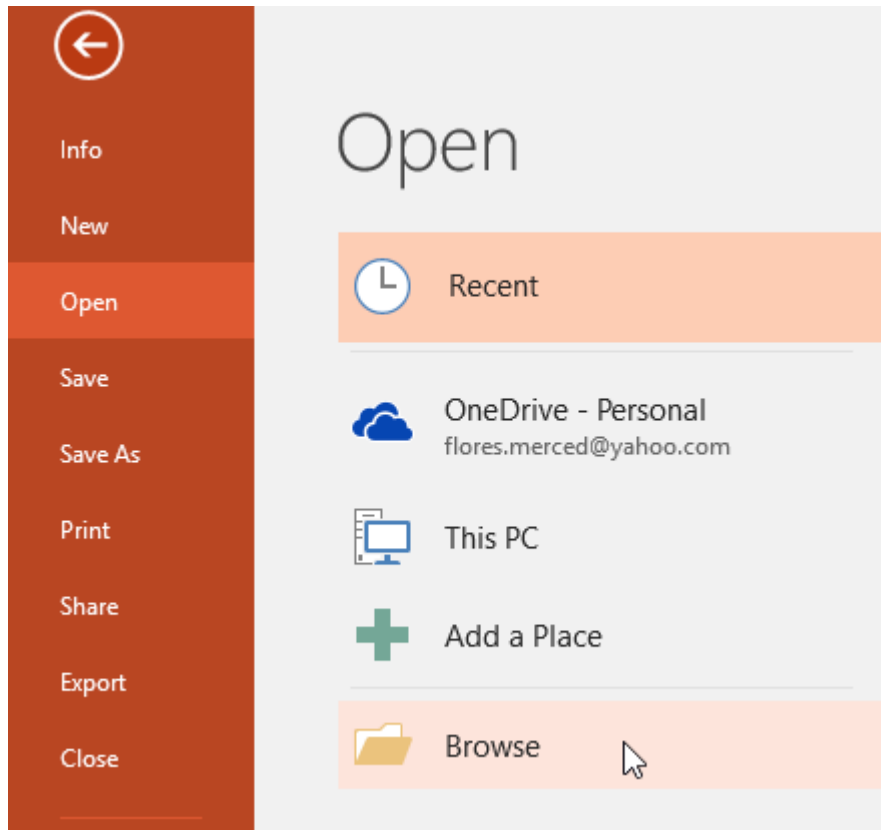
To open an existing presentation:

In addition to creating new presentations, you'll often need to open a presentation that was previously saved.

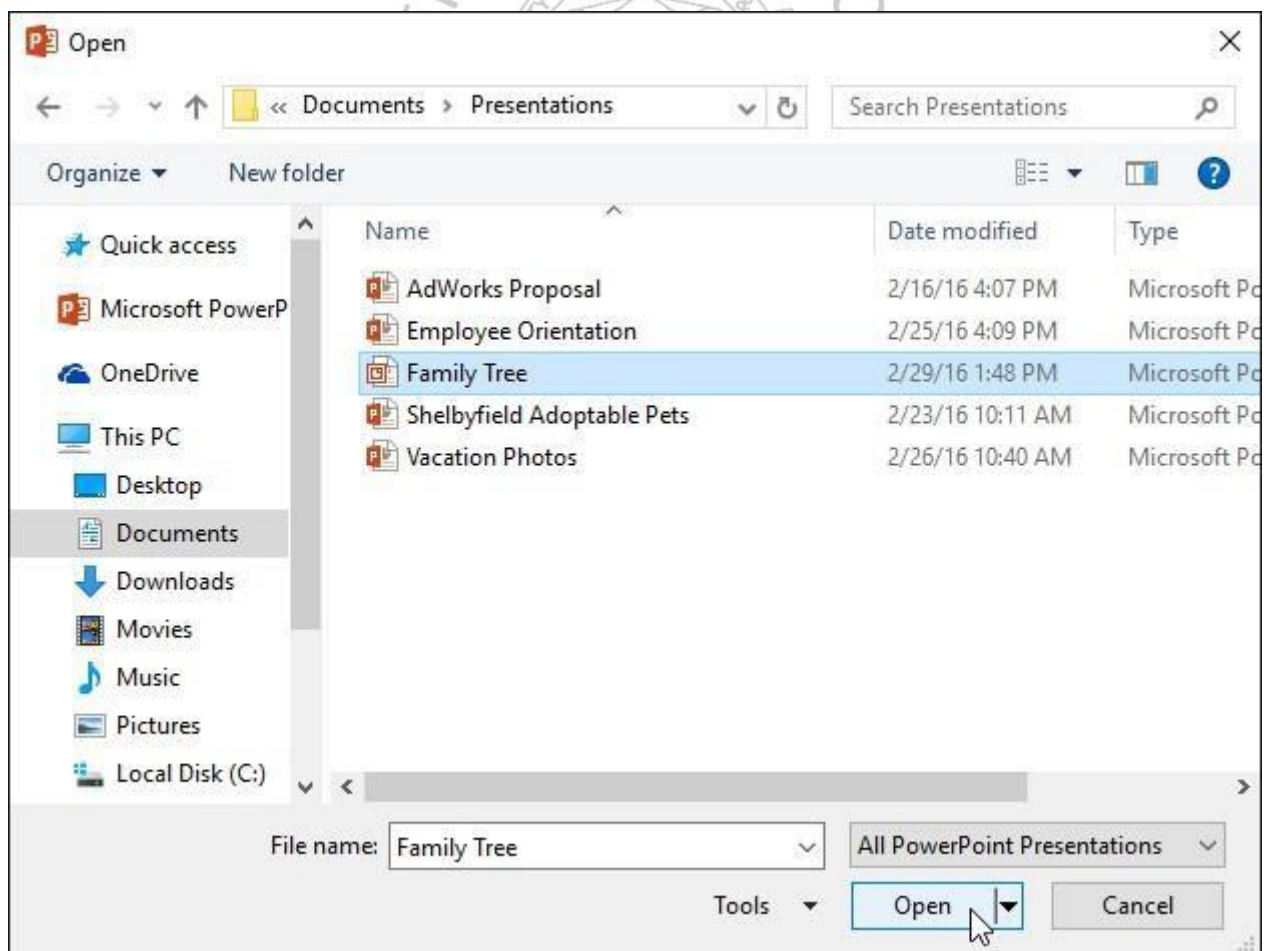
1. Select the **File** tab to go to **Backstage view**, then click **Open**.



2. Click **Browse**. Alternatively, you can choose **OneDrive** to open files stored on your OneDrive.



3. The **Open** dialog box will appear. Locate and select your **presentation**, then click **Open**.

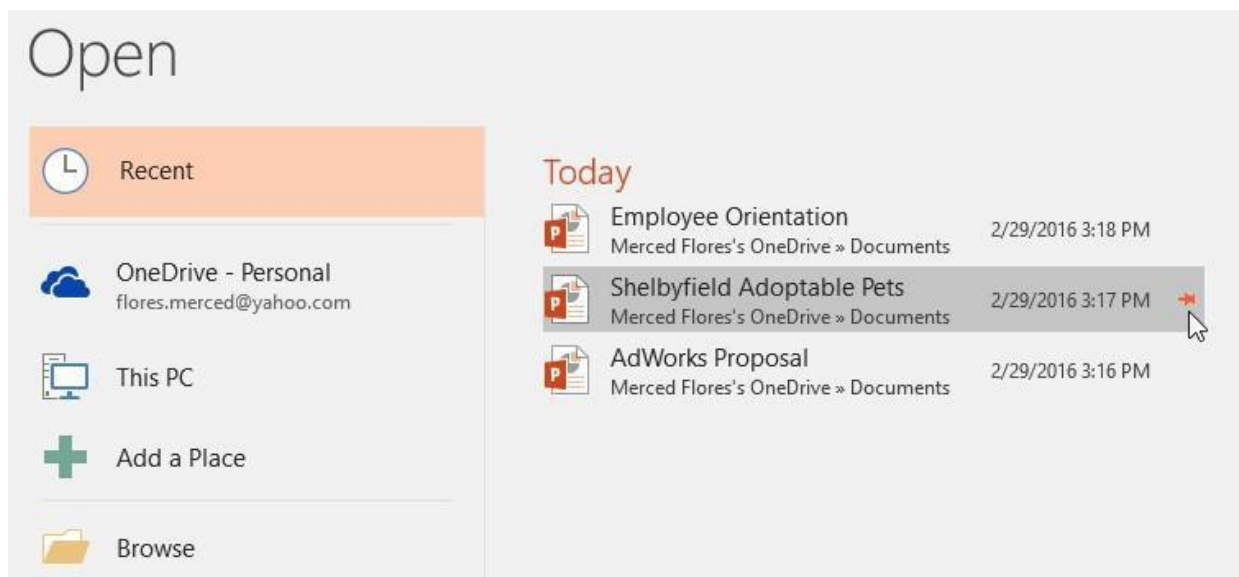


Most features in Microsoft Office, including PowerPoint, are geared toward saving and sharing documents **online**. This is done with **OneDrive**, which is an online storage space for your documents and files.

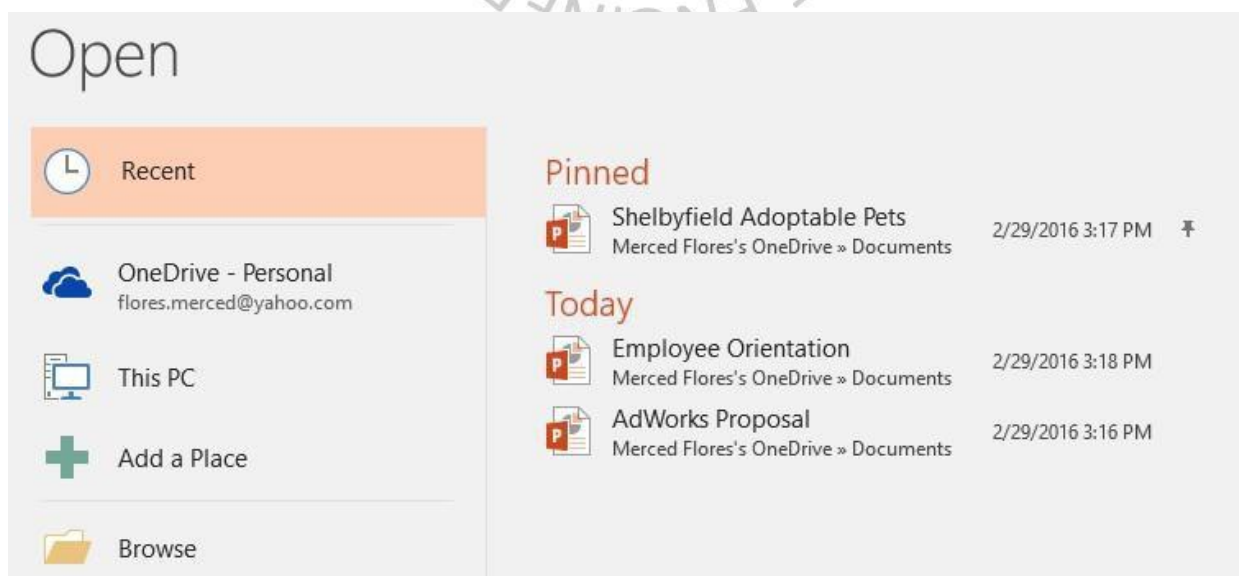
To pin a presentation:

If you frequently work with the **same presentation**, you can **pin it** to Backstage view for easy access.

1. Select the **File** tab to go to **Backstage view**, then click **Open**. Your **Recent Presentations** will appear.
2. Hover the mouse over the **presentation** you want to pin, then click the **pushpin icon**.



3. The presentation will stay in the **Recent presentations** list until it is unpinned. To **unpin** a presentation, click the pushpin icon again.

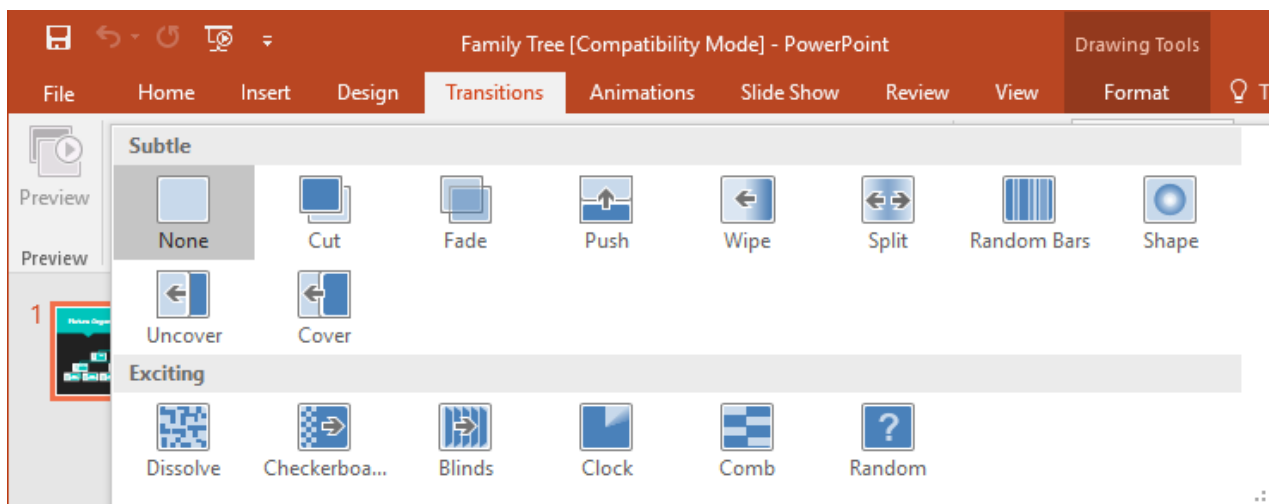


Compatibility Mode

Sometimes you may need to work with presentations that were created in earlier versions of PowerPoint, like PowerPoint 2003 or PowerPoint 2000. When you open these types of presentations, they will appear in **Compatibility Mode**.

Compatibility Mode **disables** certain features, so you'll only be able to access commands found in the program that was used to create the presentation. For example, if you open a presentation created in PowerPoint 2003, you can only use tabs and commands found in PowerPoint 2003.

In the image below, you can see at the top of the window that the presentation is in Compatibility Mode. This will disable some PowerPoint 2016 features, including newer types of slide transitions.



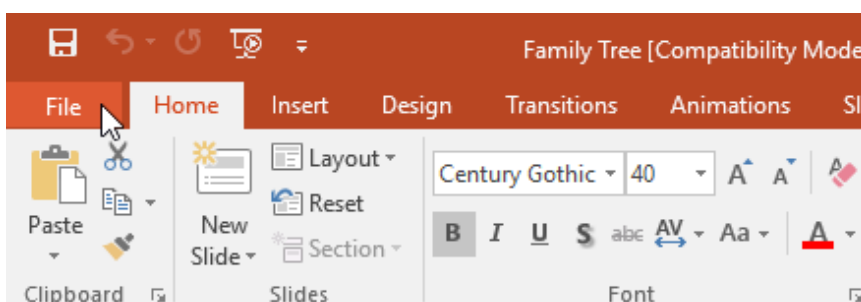
To exit Compatibility Mode, you'll need to **convert** the presentation to the current version type. However, if you're collaborating with others who only have access to an earlier version of PowerPoint, it's best to leave the presentation in Compatibility Mode so the format will not change.

To convert a presentation:

If you want access to all PowerPoint 2016 features, you can **convert** the presentation to the 2016 file format.

Note that converting a file may cause some changes to the **original layout** of the presentation.

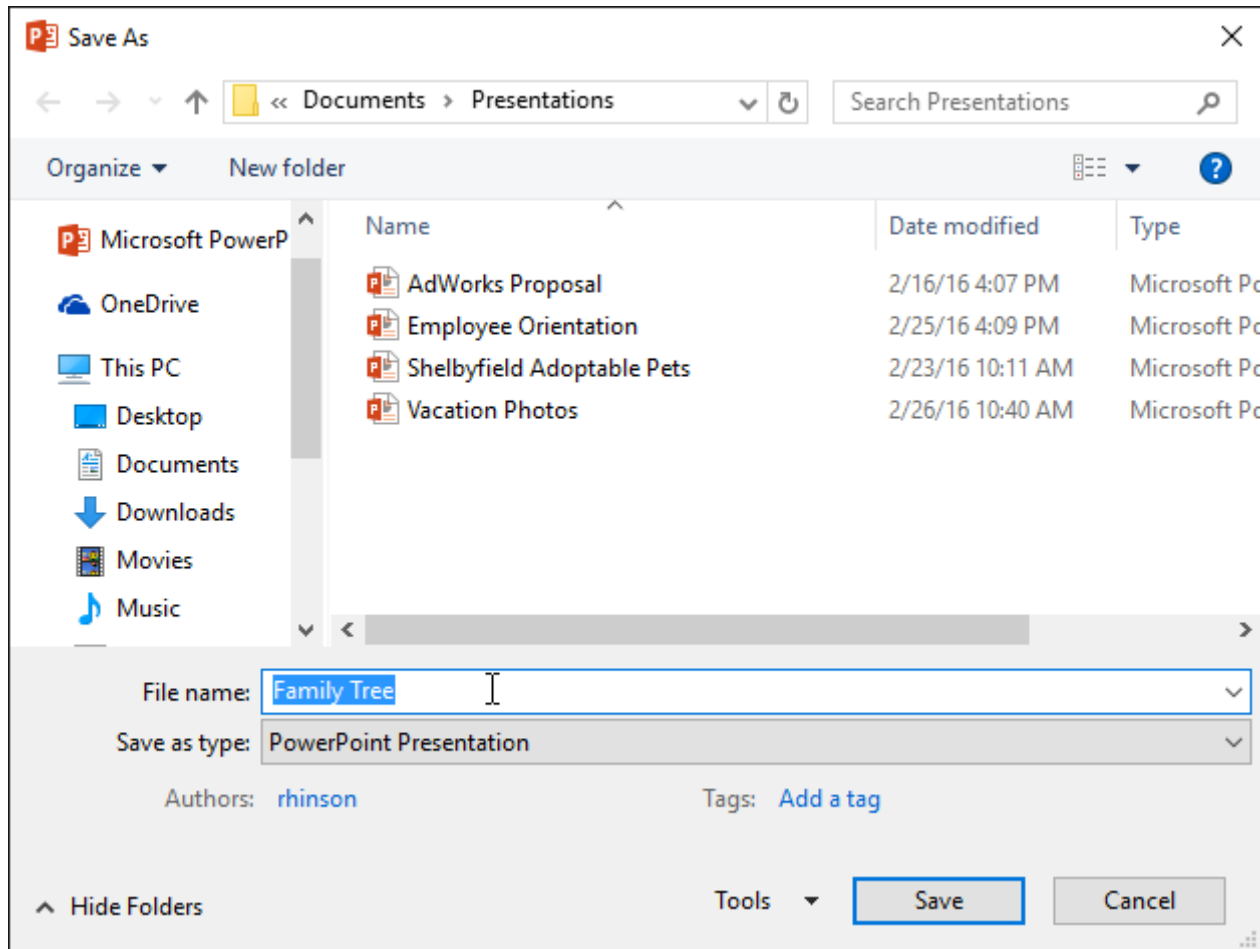
1. Click the **File** tab to access Backstage view.



2. Locate and select the **Convert** command.



3. The **Save As** dialog box will appear. Select the **location** where you want to save the presentation, enter a **file name**, and click **Save**.



4. The presentation will be converted to the newest file type.

Saving Presentations

Whenever you create a new presentation in PowerPoint, you'll need to know how to save in order to access and edit it later. As with previous versions of PowerPoint, you can save files to your computer. If you prefer, you can also save files to the cloud using OneDrive. You can even export and share presentations directly from PowerPoint.

Save and Save As

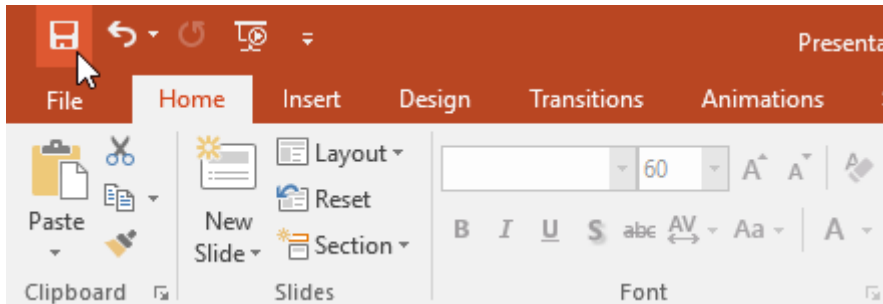
PowerPoint offers two ways to save a file: **Save** and **Save As**. These options work in similar ways, with a few important differences.

- **Save:** When you create or edit a presentation, you'll use the **Save** command to save your changes. You'll use this command most of the time. When you save a file, you'll only need to choose a file name and location the first time. After that, you can just click the Save command to save it with the same name and location.
- **Save As:** You'll use this command to create a **copy** of a presentation while keeping the original. When you use Save As, you'll need to choose a different name and/or location for the copied version.

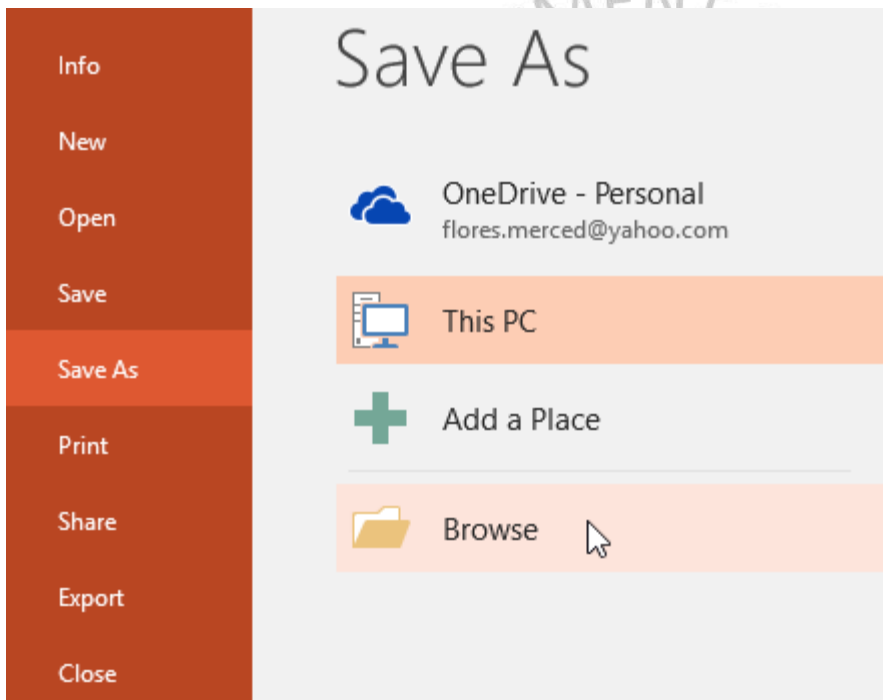
To save a presentation:

It's important to **save your presentation** whenever you start a new project or make changes to an existing one. Saving early and often can prevent your work from being lost. You'll also need to pay close attention to **where you save** the presentation so it will be easy to find later.

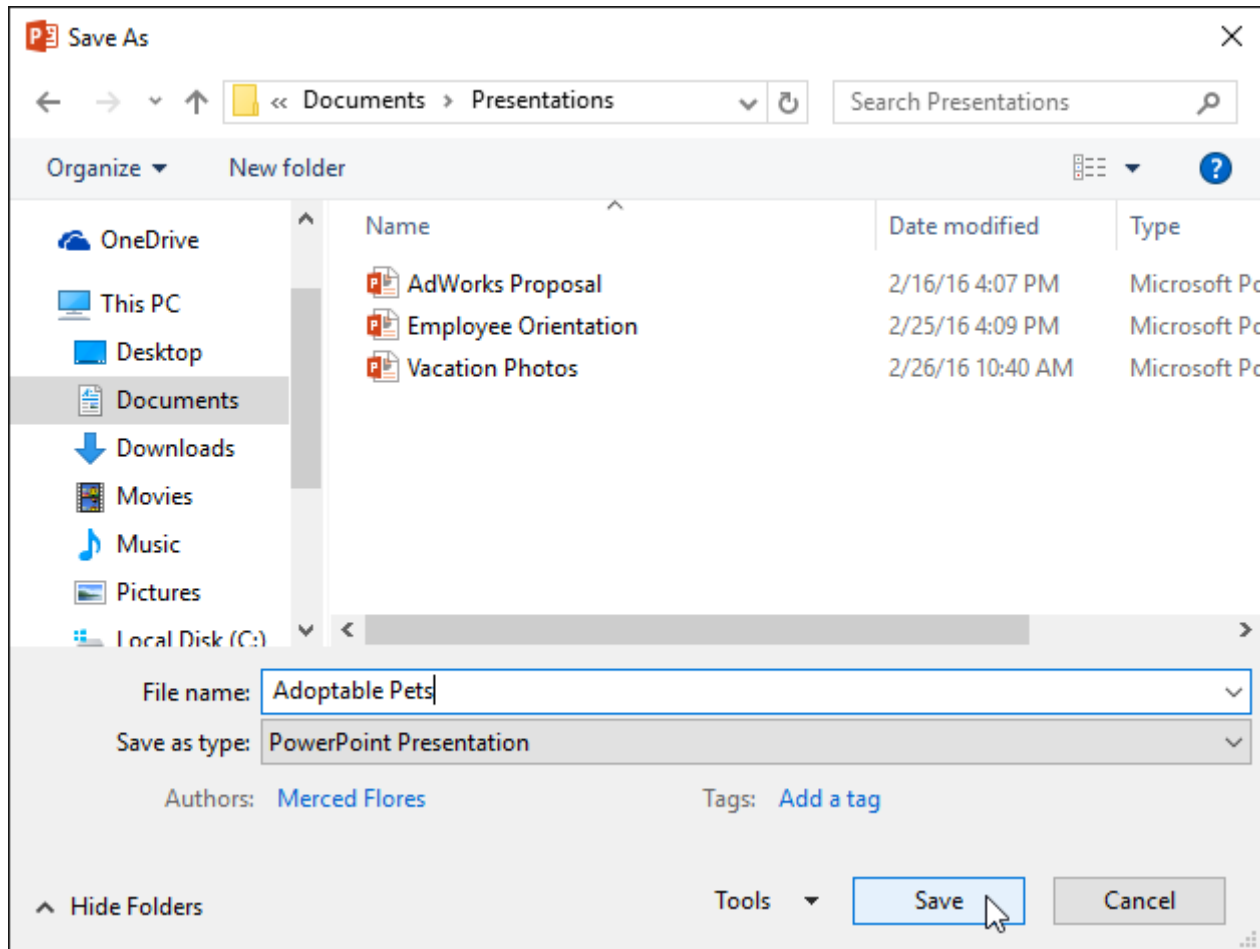
1. Locate and select the **Save** command on the **Quick Access Toolbar**.



2. If you're saving the file for the first time, the **Save As** pane will appear in **Backstage view**.
3. You'll then need to choose **where to save** the file and give it a **file name**. Click **Browse** to select a location on your computer. Alternatively, you can click **OneDrive** to save the file to your OneDrive.



4. The **Save As** dialog box will appear. Select the **location** where you want to save the presentation.
5. Enter a **file name** for the presentation, then click **Save**.



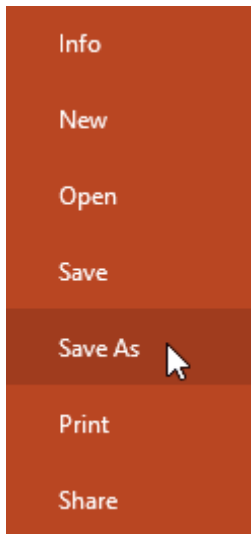
6. The presentation will be **saved**. You can click the **Save** command again to save your changes as you modify the presentation.

You can also access the **Save** command by pressing **Ctrl+S** on your keyboard.

Using Save As to make a copy

If you want to save a **different version** of a presentation while keeping the original, you can create a **copy**. For example, if you have a file named **Client Presentation** you could save it as **Client Presentation 2** so you'll be able to edit the new file and still refer back to the original version.

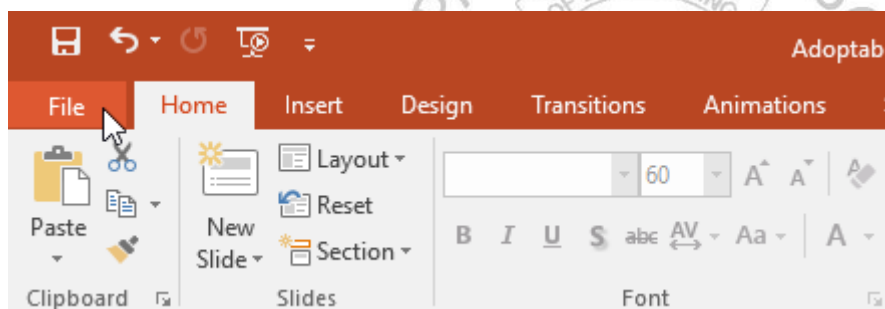
To do this, you'll click the **Save As** command in Backstage view. Just like when saving a file for the first time, you'll need to choose **where to save** the file and give it a new **file name**.



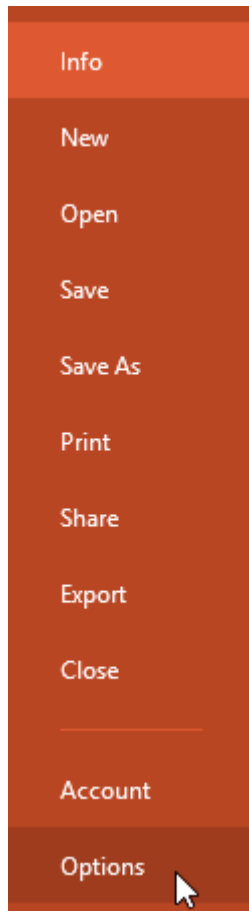
To change the default save location:

If you don't want to use **OneDrive**, you may be frustrated that OneDrive is selected as the default location when saving. If you find this inconvenient, you can change the **default save location** so **This PC** is selected by default.

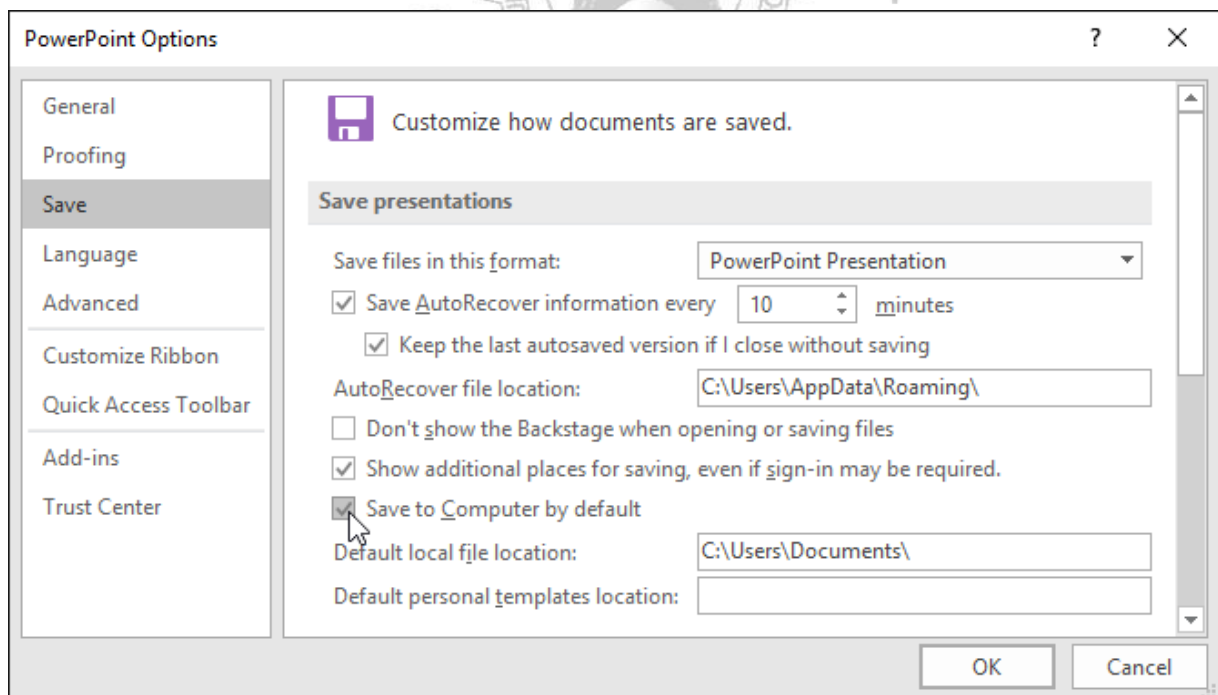
1. Click the **File** tab to access **Backstage view**.



2. Click **Options**.



3. The **PowerPoint Options** dialog box will appear. Select **Save**, check the box next to **Save to Computer by default**, then click **OK**. The default save location will be changed.

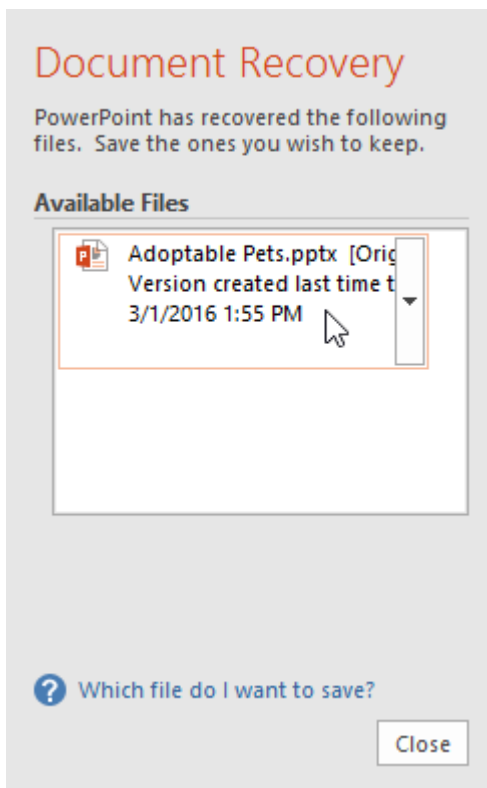


Using AutoRecover

PowerPoint automatically saves your presentations to a temporary folder while you are working on them. If you forget to save your changes or if PowerPoint crashes, you can restore the file using **AutoRecover**.

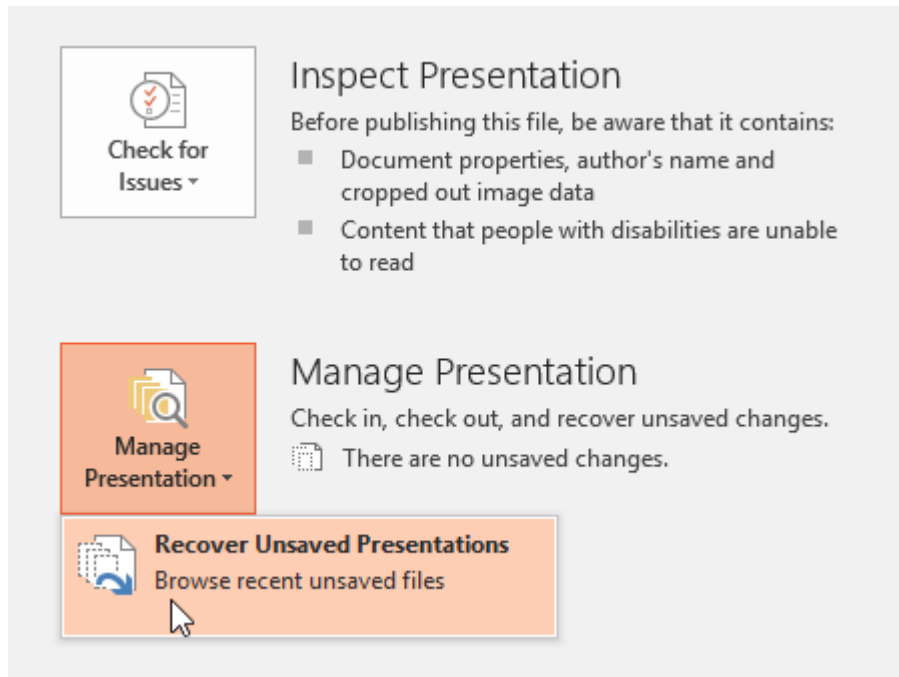
To use AutoRecover:

1. Open PowerPoint. If **autosaved versions** of a file are found, the **Document Recovery** pane will appear.
2. Click to **open** an available file. The presentation will be recovered.



By default, PowerPoint autosaves every 10 minutes. If you are editing a presentation for less than 10 minutes, PowerPoint may not create an autosaved version.

If you don't see the file you need, you can browse all autosaved files from **Backstage view**. Just select the **File** tab, click **Manage Presentation**, then choose **Recover Unsaved Presentations**.



Exporting presentations

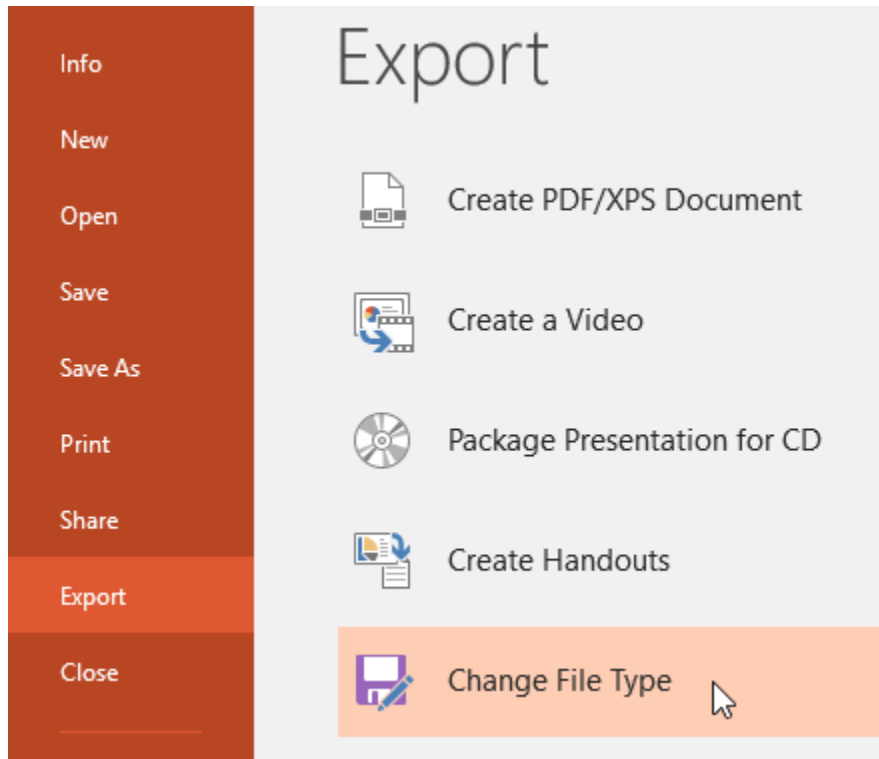
By default, PowerPoint presentations are saved in the **.pptx** file type. However, there may be times when you need to use **another file type**, such as a **PDF** or **PowerPoint 97-2003 presentation**. It's easy to **export** your presentation from PowerPoint in a variety of file types.

- **PDF:** Saves the presentation as a **PDF document** instead of a PowerPoint file
- **Video:** Saves the presentation as a video
- **Package for CD:** Saves the presentation in a folder along with the Microsoft PowerPoint Viewer, a special slide show player anyone can download
- **Handouts:** Prints a handout version of your slides
- **Other file type:** Saves in other file types, including **PNG** and **PowerPoint 97-2003**

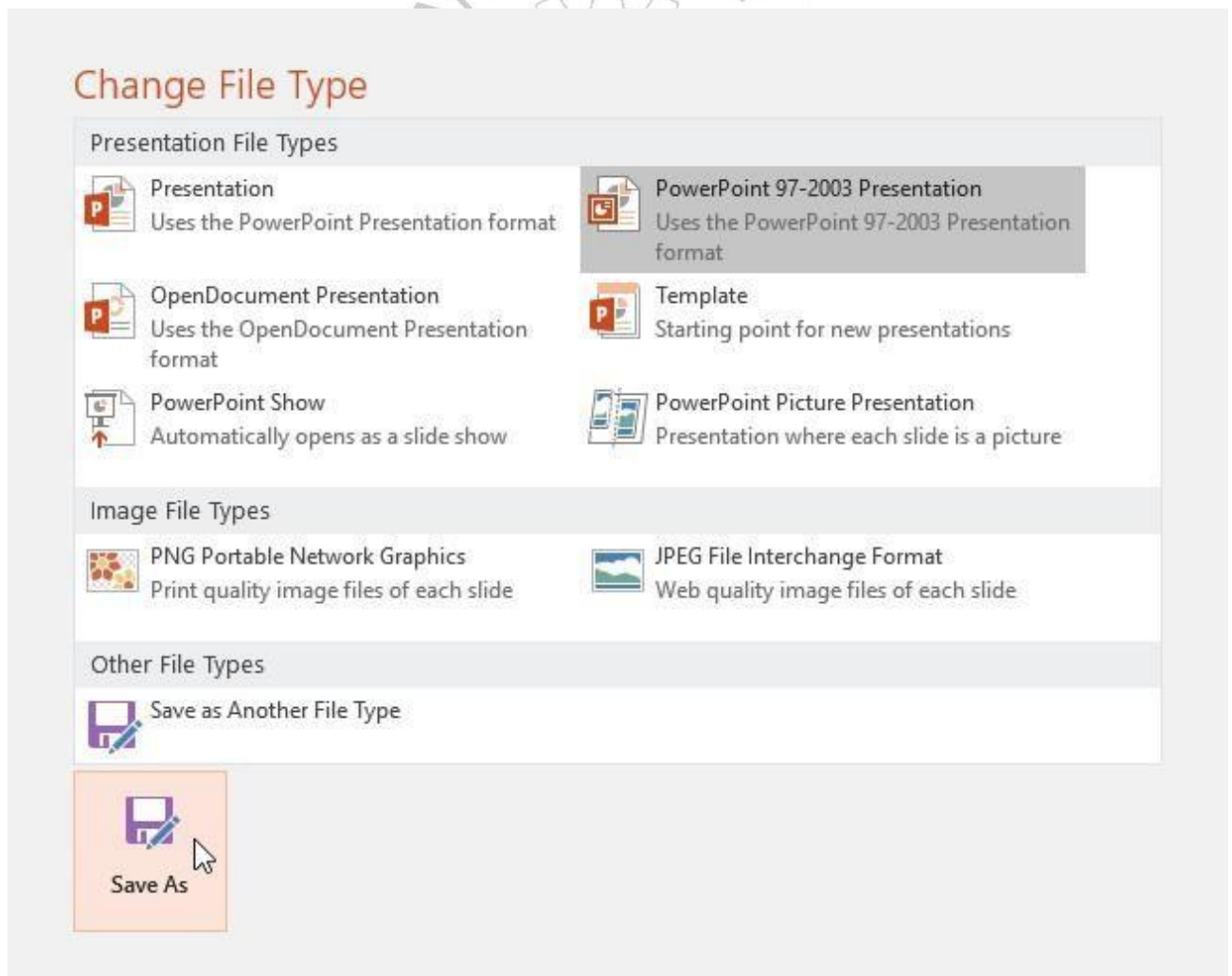
To export a presentation:

In our example, we'll save the presentation as a **PowerPoint 97-2003** file.

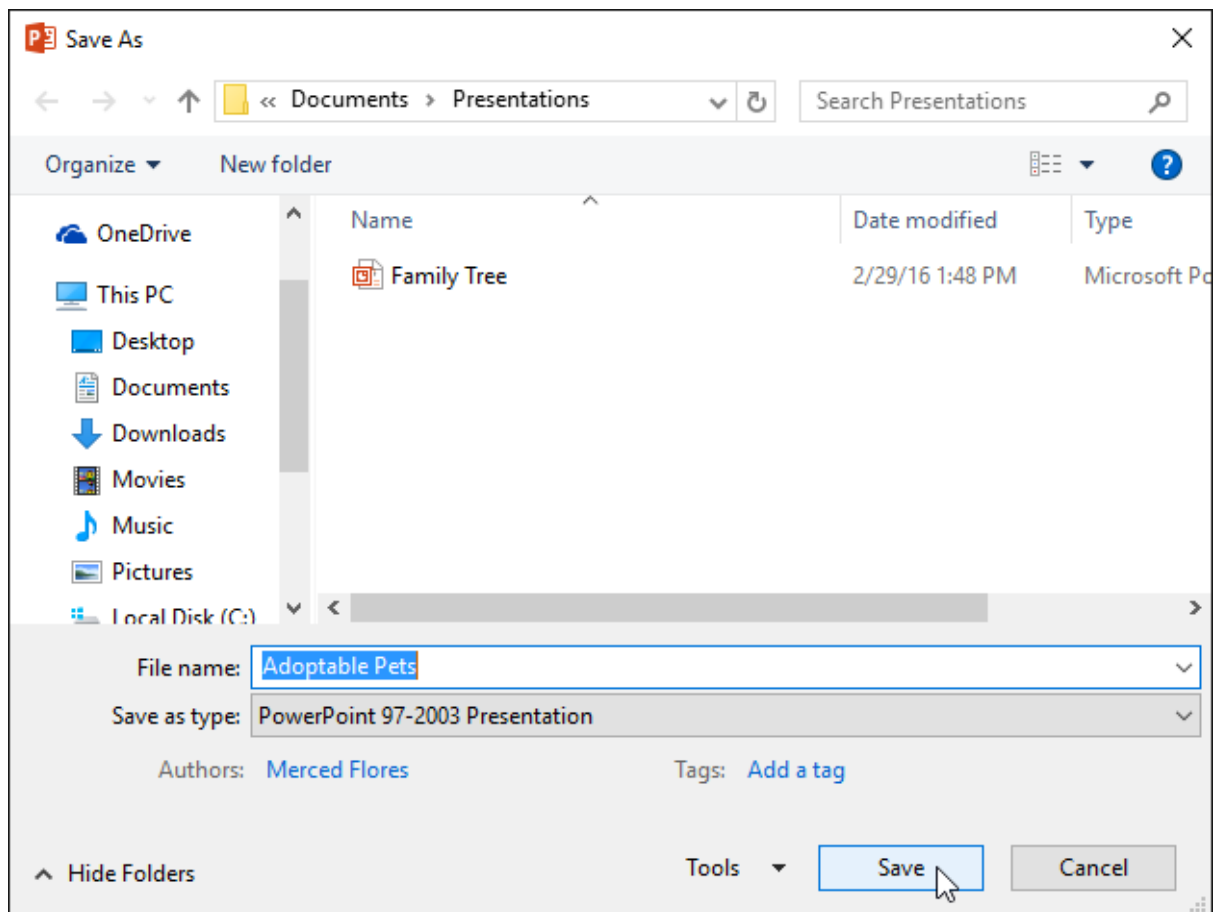
1. Click the **File** tab to access **Backstage view**.
2. Click **Export**, then choose the desired option. In our example, we'll select **Change File Type**.



3. Select a **file type**, then click **Save As**.



4. The **Save As** dialog box will appear. Select the **location** where you want to export the presentation, type a **file name**, then click **Save**.



You can also use the **Save as type** drop-down menu in the **Save As** dialog box to save presentations in a variety of file types. Be careful to choose a file type others will be able to open.

EXERCISE:

Create a PowerPoint presentation with Title “Global Warming”, include the transitions, slide layout and images in the slides.

- Minimum five slides
- First slide must contain “Title” of your presentation topic, your name; roll number, batch and department.
- The next slide must contain Introduction about the topic.
- Last slide must contain “Conclusion”.
- Your slides must have slide number.
- Try as much features as possible.

Name: MUHAMMAD KAYHAN KHAN
Roll No: SE-095
Batch: 2025
Department: Software Engineering

Global Warming

A Journey from Crisis to Hope



1

Understanding Our Changing Planet:

Global warming represents an unprecedented shift in Earth's climate system. Over the past century, our planet's average temperature has risen approximately 1.1°C, driven primarily by human activities. This presentation explores the science, consequences, and solutions to one of humanity's greatest challenges.



2

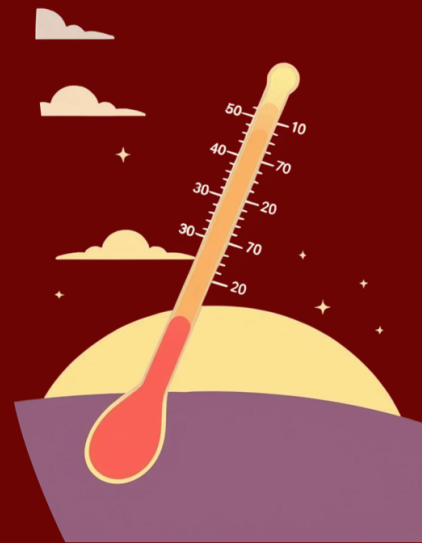
What is Global Warming?

Definition:

Global warming is the long-term increase in Earth's average surface temperature caused by greenhouse gas emissions, primarily from fossil fuel combustion and deforestation.

The Scale:

Since 1880, global temperatures have increased by 1.1°C. Most warming has occurred in the last 50 years, with the decade of 2010-2020 being the warmest on record.



3

Primary Causes of Global Warming:



Industrial Emissions:

Manufacturing and energy production release massive quantities of CO₂ into the atmosphere.



Transportation:

Fossil fuel-powered vehicles emit greenhouse gases, accounting for roughly 27% of total emissions.



Deforestation:

Removing forests eliminates CO₂-absorbing trees and releases stored carbon back into the atmosphere.



Agriculture:

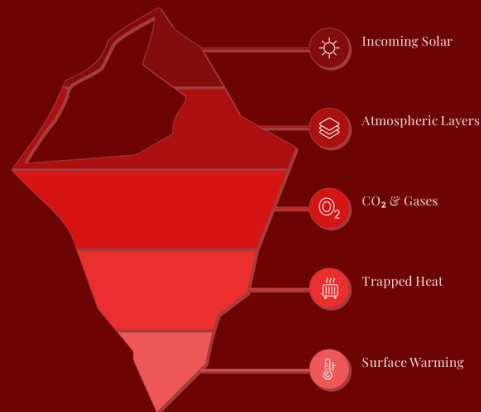
Livestock farming and crop production generate methane and nitrous oxide, potent greenhouse gases.



4

The Greenhouse Effect:

The greenhouse effect is a natural process essential for life, but human activities have intensified it dramatically. Greenhouse gases—primarily carbon dioxide, methane, and nitrous oxide—trap heat in Earth's atmosphere like a blanket, preventing thermal radiation from escaping to space.



5

Consequences for Our Planet



Melting Ice

Arctic sea ice and glaciers are disappearing at accelerating rates, raising global sea levels.



Extreme Weather

Intensified hurricanes, droughts, floods, and wildfires devastate ecosystems and communities worldwide.



Rising Sea Levels

Thermal expansion and melting ice sheets threaten island nations and coastal populations globally.



Ocean Acidification

Increased CO₂ absorption alters ocean chemistry, threatening marine ecosystems and food security.

6

Human and Ecological Impact

On Humanity

- Displacement of millions from climate-vulnerable regions
- Food and water security threats for billions
- Disproportionate impact on developing nations and marginalized communities
- Increased disease vectors and health crises

On Ecosystems

- Species extinction rates accelerating rapidly
- Habitat loss and ecosystem fragmentation
- Disruption of migration patterns and breeding cycles
- Collapse of coral reefs and marine biodiversity



7

Solutions and Preventive Measures:

Renewable Energy

Transitioning to solar, wind, and hydroelectric power eliminates fossil fuel dependency and reduces emissions by up to 80%.

Sustainable Practices

Regenerative agriculture and circular economy models reduce waste and sequester carbon in soil.



Reforestation

Planting billions of trees absorbs CO₂, restores ecosystems, and provides long-term carbon storage.

Sustainable Transport

Electric vehicles and public transit systems reduce transportation emissions by 50-70% over their lifetime.

Energy Efficiency

Smart buildings, efficient manufacturing, and better insulation can cut energy consumption by 30-40%.

8

Future Outlook: Two Paths

Our choices today determine the trajectory of our future. The contrast is stark.

Path of Inaction



Path of Action



9



Conclusion: A Call to Action

"The Earth is what we all have in common." — Wendell Berry

Global warming is not inevitable—it is a choice. Every fraction of a degree matters. Through renewable energy adoption, sustainable practices, policy change, and individual action, we can reverse course and build a thriving future. The time for action is now. Our generation holds the power to write a story of recovery, resilience, and hope for the planet we call home.

10





Criteria	Level of Attainment					
	(5)	(4)	(3)	(2)	(1)	(0)
Problem Identification	Able to correctly identify the problem(s) and all required data (input and output).	Able to correctly identify the problem(s) but not all the required data (input and output).	Able to partially identify both the problem(s) and the required data (input and output).	Able to partially identify the problem(s) but cannot identify required data (input and output).	Not able to correctly identify the problem(s) and all required data (input and output).	Not tried to identify the problem(s) and all required data (input and output).
Problem Solving Skills	Uses an effective strategy to solve the problem(s) and clearly identifies the steps.	Uses an effective strategy to solve the problem(s) but cannot clearly identify the steps.	Uses an adequate strategy to solve the problem(s) but misses some details.	Uses an inadequate strategy to solve the problem(s) and misses most details.	Uses an ineffective strategy to solve the problem(s) and completely misses the steps.	Didn't use any strategy to solve the problem(s) and completely misses the steps.
Efficiency in Tool Usage	Efficient implementation and excellent tool usage skills.	Efficient implementation and good tool usage skills.	Adequate implementation and fair tool usage skills.	Inadequate implementation and tool usage skills.	Incomplete implementation and ineffective tool usage skills.	No implementation and tool usage skills. .
Deliverable(s)	Generates correct and complete output.	Generates correct but partially complete output.	Generates partially correct and partially complete output.	Generates partially correct and incomplete output.	Cannot generate the desired output.	No output generated.
Depth of Knowledge	Thorough grasp of the concepts covered in the lab and can apply it in real-life situations.	Good grasp of the concepts covered in the lab but cannot effectively apply it in real-life situations.	Fair grasp of the concepts covered in the lab and can vaguely identify a real-life application.	Limited grasp of the concepts covered in the lab and cannot identify any real-life application.	Does not grasp the concepts covered in the lab and cannot identify any real-life application.	Didn't try to grasp the concepts covered in the lab and cannot identify any real-life application.
Weighted CLO Score						
Remarks and Signature						

LAB SESSION 05

Getting Started with MS Publisher

THEORY

Microsoft Publisher is a program designed to help you create **publications**, or documents that you can **print** and **distribute**.

In this lesson, you'll learn about the **advantages** and **disadvantages** of using Publisher. You'll also learn about the Publisher **interface**, including the **Ribbon**, **rulers** and **guides**, and **Backstage view**.

Even if you've never used Publisher before, you'll probably find its interface familiar, as it has many of the same features as other Microsoft Office programs. However, it does include some features specifically designed for creating publications.

Advantages and disadvantages of Publisher

Publisher offers templates and other tools to help you create a variety of publications, including brochures, newsletters, business cards, and menus.

PROCEDURE:

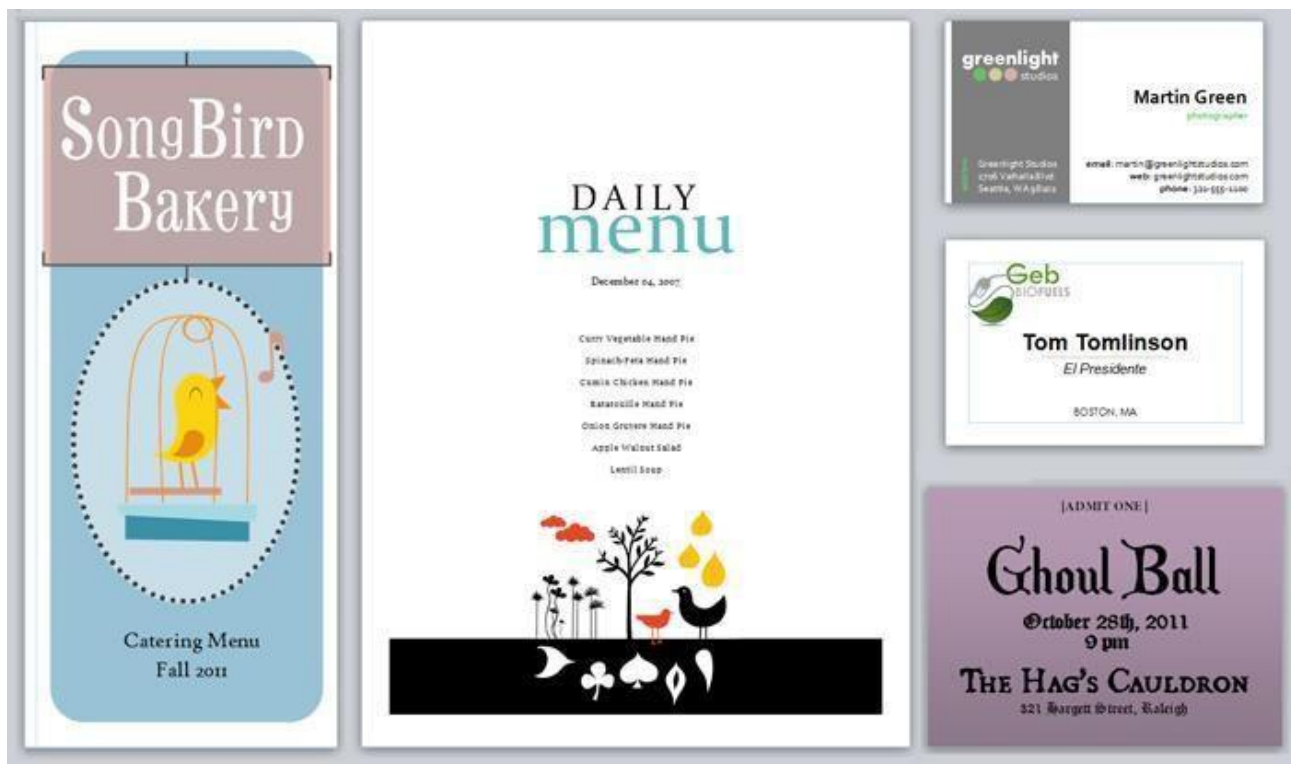


Fig 5.1: Publisher Interface

As you might know, you can do most—if not all—of the tasks mentioned above in Microsoft Word. So why not create publications using that program instead?

One of the advantages of Publisher is that it offers more control over certain aspects of your publication. For instance, Publisher offers a work environment that makes it easy to work with documents in a variety of sizes and shapes. It also has a larger variety of tools to help you arrange and align text, images, and other objects within the page margins.

However, while Publisher does have certain advantages, it also has the disadvantage of being the

least developed program in the Office 2010 suite. It does not include the theme colors, advanced image editing tools, or many of the other new features present in Word and PowerPoint 2010. In fact, in terms of these features Publisher 2010 is more similar to Publisher 2007 and 2003 than it is to other current programs in the Office suite. As you begin to use Publisher 2010, you may find that you prefer to create certain types of publications in Word.

Introduction to the Publisher 2010 environment

If you are familiar with Publisher 2007 or 2003, you'll notice a few changes to the 2010 interface. The main change is the addition of the Ribbon, which appears in all Office 2010 programs. Publisher also added Backstage view, which we'll cover later in this lesson.

In most other ways, the Publisher environment is much the same as in past editions. If you are new to Publisher, you should take some time to familiarize yourself with its interface.

Review the interactive to learn more about the Ribbon and other features in the Publisher 2010 environment.

Working with the Publisher environment

One of the first things you might notice when you open Publisher is that the rulers and guidelines that are used to line up document components are slightly different from those in other Office programs. Other features of the Publisher environment might seem more familiar. For instance, you'll recognize the Ribbon, where you can find the commands you need to complete common tasks in Publisher.

The Ribbon

The Ribbon contains multiple tabs, each with several groups of commands. Some tabs, like Text Box Tools or Picture Tools, may appear only when you're working with certain items like text boxes or images.

Viewing tools

Publisher offers a group of **viewing tools** to help you control the layout of your text, images, and objects on the page. These viewing tools are for your editing purposes only. They can be turned on and off and **will not appear** in your printed publication.

- To access and choose viewing tools, select the **View** tab, then locate the **Show** group. Click the check boxes of the various view options to turn them on and off.

dfsdg

- **Orientation**

Do you want your publication to be in **landscape orientation** (wider than tall) or **portrait orientation** (taller than wide)? Your choice will probably be influenced by standard design conventions. For instance, **business cards** are usually printed in **landscape** orientation.

- **Margins**

Margins are areas of blank space that line the edges of a printed document. While professional printers can print publications where the text and images stretch out all the way to the edge of the page, most home and office printers require your publication to have margins, even if they're narrow ones.

The **width** of your page margins can affect the **look** and **feel** of your publication. For example, extremely **wide margins**, like those often found in menus for fancy restaurants, can look sophisticated. However, wide margins don't leave much room for images and text. In contrast, **narrow margins** give you a large amount of space to work with.

Publishing electronically

Depending on the purpose of your publication, you may decide to publish it **electronically** and distribute it **online**. Publisher offers two ways to do this. You can either:

- Publish as a **PDF** and attach the file to an email or upload it to a website
- Publish as **HTML**, which Publisher then **embeds** directly into an email

Although Publisher promotes the HTML option as useful, **it is almost always better to convert your publication into a PDF**. This is because the PDF option is better at presenting your publication the way you designed it. Publications saved as HTML can lose some of their formatting and may not include custom fonts and images.

However, PDFs aren't perfect either. Depending on the type of publication you're working with, the PDF version may not be easy to browse and read. For example, while you technically can convert a brochure to a PDF, the panels will not be lined up as they would be.

EXERCISE:

1. DESIGN A BUSINESS CARD

Design and publish a business card with complete information using Microsoft Publisher.



2. DESIGN A Brochure

Design and create a brochure using Microsoft Publisher. Include relevant content in the form of text, images and other options.

WE BUILD EVERY HOME AS IF IT'S OUR HOME

**SIGMA BUILDERS**

BUILDING WITH TRUST



Who we are

Sigma Builders is a construction services company in Karachi, that provides assistance with constructing buildings like houses. We have been constructing buildings for 5 years now and constructed a total of 1000 houses in BTK. The company aims to expand through other cities.

What we do

Construct buildings or infra-structures of what the company and their clients agreed on their clients' requests.

Contact us!

Dominion Business Center-2, Bahria Town Karachi, Karachi
sigmabuilders@gmail.com
www.sigmabuilders.com

For more info
Please call 0300-0000000

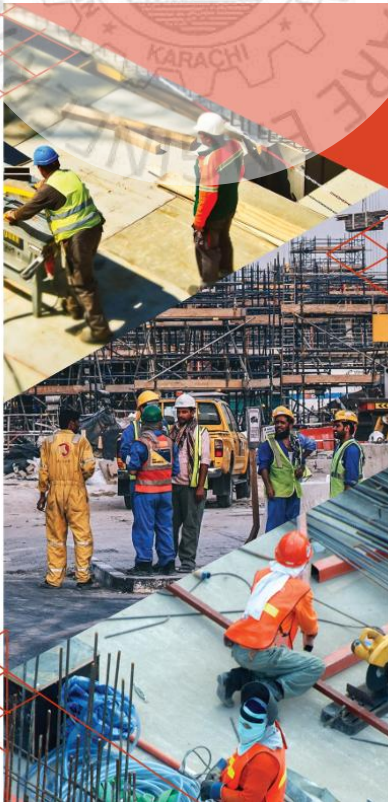
Our goals

The company aims to construct buildings with strong and firm structures that will last for more than 5 centuries.

Our services

We offer high-quality services to our clients because we believe the best marketing strategy is a satisfied customer. Here are the services you can avail from us.

IF YOU DREAM IT, WE CAN BUILD IT



Our advantages

The buildings constructed by the company has survived for more than 5 decades. We offer promos occasionally for our clients to be able for them to afford.

Our properties

We don't rely on not trustworthy tools, because we want to provide quality products to our clients. With all these, we can achieve the home you always dream of. Work with us and experience one of a kind services.

LAB SESSION 06

Getting Familiar with Microsoft Visio

THEORY:

Microsoft Office Visio is drawing and diagramming software that helps transform concepts into a visual representation. Here we are going to focus on how to create a logical diagram and flow chart when drawn in Visio.

1. BASIC FLOWCHARTING SHAPES AND SYMBOLS

Flowcharts use special shapes to represent different types of actions or steps in a process. Lines and arrows show the sequence of the steps, and the relationships among them.



Start/End

The terminator symbol marks the starting or ending point of the system. It usually contains the word "Start" or "End."



Action or Process

A box can represent a single step ("add two cups of flour"), or an entire sub-process ("make bread") within a larger process.



Document

A printed document or report.



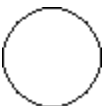
Decision

A decision or branching point. Lines representing different decisions emerge from different points of the diamond.



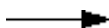
Input/Output

Represents material or information entering or leaving the system, such as customer order (input) or a product (output).



Connector

Indicates that the flow continues where a matching symbol (containing the same letter) has been placed.



Flow Line

Lines indicate the sequence of steps and the direction of flow.

PROCEDURE:

Flow chart

Step 1: From the categories areas select Basic Flow Chart, and select Flow Chart.

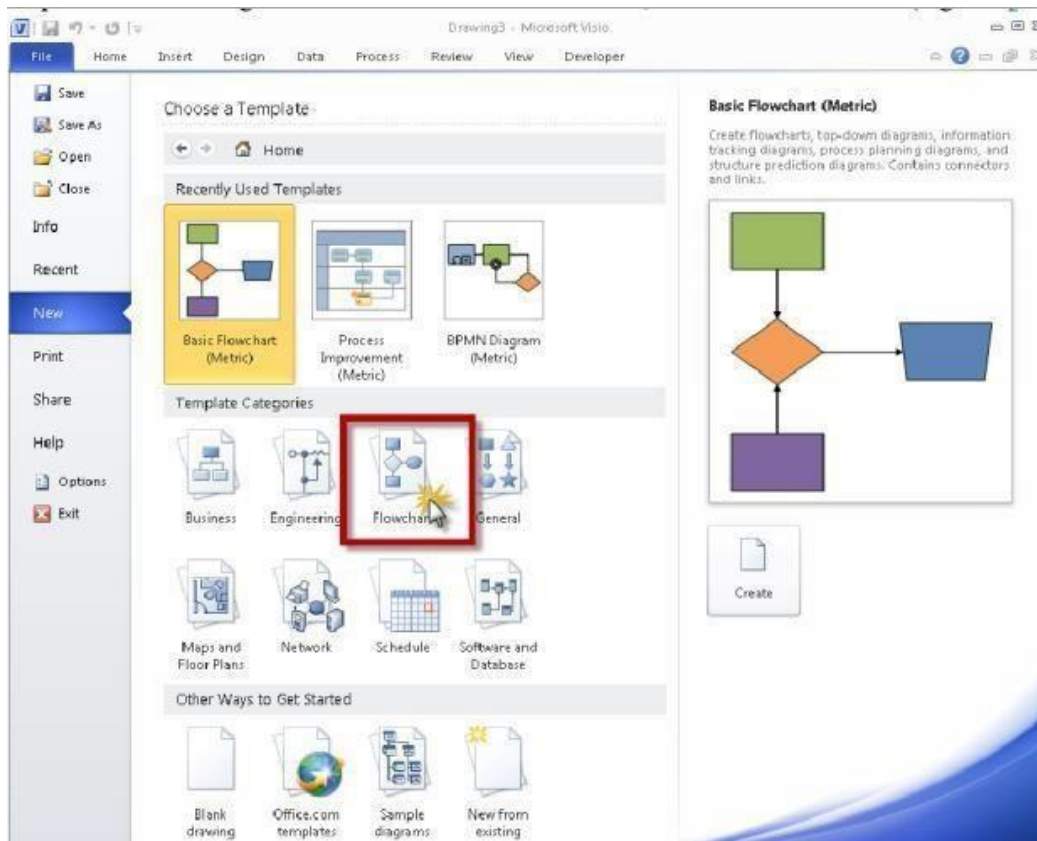
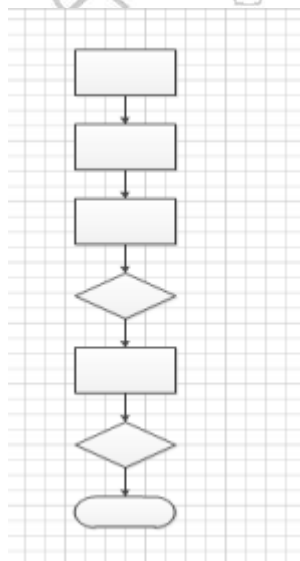


Fig 6.1 Microsoft Visio Environment

Step 2: Double click on Flow Chart, you will see Visio Screen Layout

Working with Shapes from a Basic Flow Chart Template:

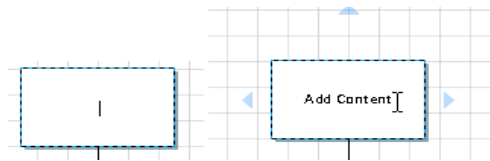


Layout your screen as follows

Adding Text to the shape

Adding the Connector

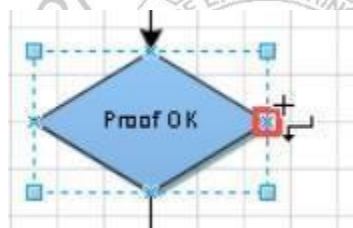
a. Select Connector



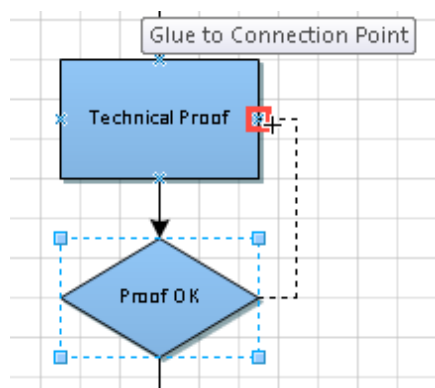
b. Select the connector tool button from the home tab



c. Move the mouse to the position shown below note the cursor has changed to reveal the connector tool. As you position to a point you can connect to the shape a small red square will appear.



d. Depress the left mouse key and drag the mouse pointer to the center left edge connection point of the shape above.

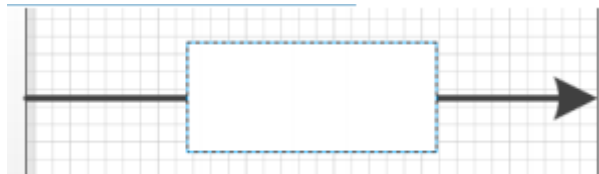


e. Release the mouse Button and you will see a connector from the diamond to the rectangle above and shown.

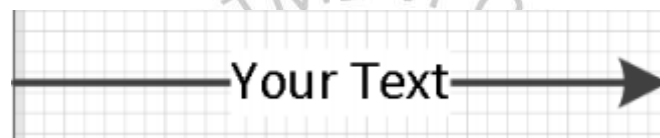


Add text to connectors

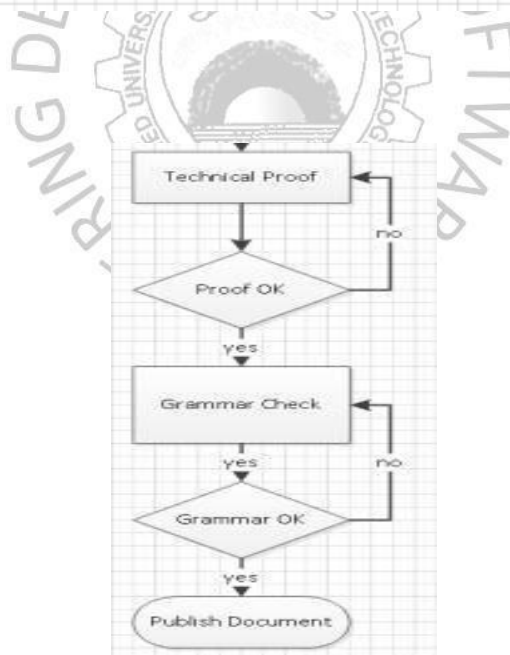
- a. Double click on the connector line to type in text.



- b. Enter your required text then click away from the Line



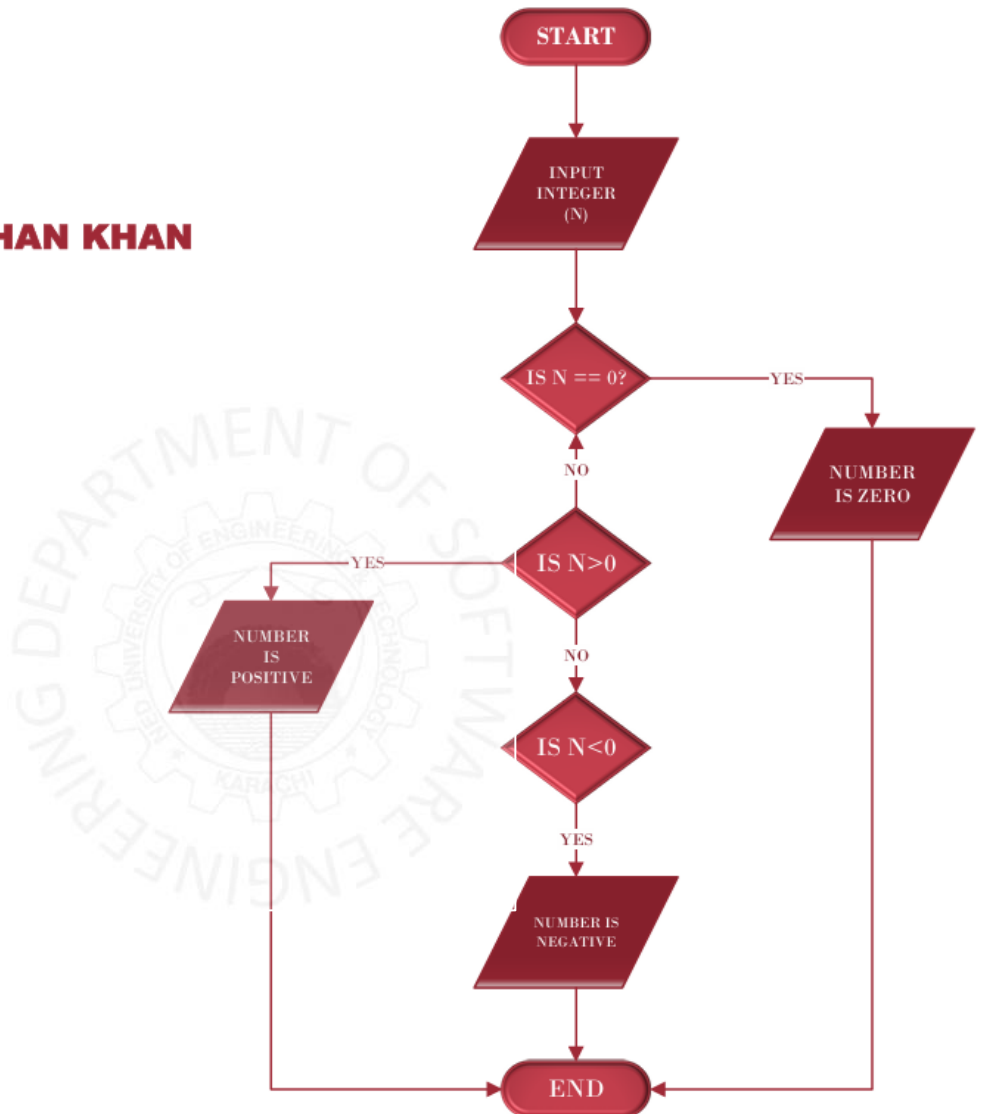
- c. Layout Document as below.



EXERCISE:

1. Draw a flowchart for a program that prompts the user to input an integer value and tests if the input value is positive or negative. The program displays an output message according to the following cases:
 - a. Input value is 0
 - b. Input value is positive
 - c. Input value is negative

NAME: MUHAMMAD KAYHAN KHAN
ROLL NO: SE-095
SECTION: "B"





NED University of Engineering & Technology
Department of Software Engineering
EF-101 IT Fundamentals And Applications

Criteria	Level of Attainment					
	(5)	(4)	(3)	(2)	(1)	(0)
Problem Identification	Able to correctly identify the problem(s) and all required data (input and output).	Able to correctly identify the problem(s) but not all the required data (input and output).	Able to partially identify both the problem(s) and the required data (input and output).	Able to partially identify the problem(s) but cannot identify required data (input and output).	Not able to correctly identify the problem(s) and all required data (input and output).	Not tried to identify the problem(s) and all required data (input and output).
Problem Solving Skills	Uses an effective strategy to solve the problem(s) and clearly identifies the steps.	Uses an effective strategy to solve the problem(s) but cannot clearly identify the steps.	Uses an adequate strategy to solve the problem(s) but misses some details.	Uses an inadequate strategy to solve the problem(s) and misses most details.	Uses an ineffective strategy to solve the problem(s) and completely misses the steps.	Didn't use any strategy to solve the problem(s) and completely misses the steps.
Efficiency in Tool Usage	Efficient implementation and excellent tool usage skills.	Efficient implementation and good tool usage skills.	Adequate implementation and fair tool usage skills.	Inadequate implementation and tool usage skills.	Incomplete implementation and ineffective tool usage skills.	No implementation and tool usage skills. .
Deliverable(s)	Generates correct and complete output.	Generates correct but partially complete output.	Generates partially correct and partially complete output.	Generates partially correct and incomplete output.	Cannot generate the desired output.	No output generated.
Depth of Knowledge	Thorough grasp of the concepts covered in the lab and can apply it in real-life situations.	Good grasp of the concepts covered in the lab but cannot effectively apply it in real-life situations.	Fair grasp of the concepts covered in the lab and can vaguely identify a real-life application.	Limited grasp of the concepts covered in the lab and cannot identify any real-life application.	Does not grasp the concepts covered in the lab and cannot identify any real-life application.	Didn't try to grasp the concepts covered in the lab and cannot identify any real-life application.
Weighted CLO Score						
Remarks and Signature						

LAB SESSION 07

Getting Familiar with Microsoft Project

THEORY:

Microsoft Project is a project management software program developed and sold by Microsoft which is designed to assist project managers in developing plans, assigning resources to tasks, tracking progress, managing budgets and analyzing workloads.

Project creates budgets based on assignment work and resource rates. As resources are assigned to tasks and assignment work estimated, the program calculates the cost, equal to the work times the rate, which rolls up to the task level and then to any summary tasks and finally to the project level.

Example

This is the view you will get when opening **Microsoft Project** for the first time, it is called the **Gantt view**. It contains the columns for listening tasks and durations.

The second view contains a calendar for viewing the duration of specific tasks. You have different views accessible from the toolbar. An important one is table which has several options for different views of the current category (resources, tasks, etc.).

PROCEDURE:

Create and code a WBS in MS Project

Inserting tasks

Now we are going to fill in the information. The project will be about a Mobile Reporting System. We will also fill in the deliverables. We will have to fill in the “Performance Requirements” and in order to do that we need to “Interview Users” and “Define Requirements”.

Check the picture for the entire list of tasks we will be using. If you forgot a task in the list, click where you want to insert the task and go to “Insert > New” task or press “Insert”.

	i	Task Name	Duration	Start	Finish
1		Mobile Reporting Sysem	1 day?	Mon 9/11/09	Mon 9/11/09
2		Performance Requirement:	1 day?	Mon 9/11/09	Mon 9/11/09
3		Interview Users	1 day?	Mon 9/11/09	Mon 9/11/09
4		Define the Reqs	1 day?	Mon 9/11/09	Mon 9/11/09
5		Software	1 day?	Mon 9/11/09	Mon 9/11/09
6		Design Logic	1 day?	Mon 9/11/09	Mon 9/11/09
7		Design Database	1 day?	Mon 9/11/09	Mon 9/11/09
8		Code A	1 day?	Mon 9/11/09	Mon 9/11/09
9		Code B	1 day?	Mon 9/11/09	Mon 9/11/09
10		Hardware	1 day?	Mon 9/11/09	Mon 9/11/09
11		Purchase Hardware	1 day?	Mon 9/11/09	Mon 9/11/09
12		Assemble Hardware	1 day?	Mon 9/11/09	Mon 9/11/09
13		Prototype	1 day?	Mon 9/11/09	Mon 9/11/09
14		Integration	1 day?	Mon 9/11/09	Mon 9/11/09
15		Integrate SW/HW	1 day?	Mon 9/11/09	Mon 9/11/09
16		Test	1 day?	Mon 9/11/09	Mon 9/11/09

Outlining structure

Next we will be creating an outline of our project. The first level is the project name itself. By



selecting all the deliverables underneath and clicking the indent button they will all line up. Next we will outline for each of the major deliverables the tasks below. You will notice the deliverables turning bold as we created a 3 level breakdown.

	i	Task Name	Duration	Start	Finish
1		Mobile Reporting Sysem	1 day?	Mon 9/11/09	Mon 9/11/09
2		Performance Requirements	1 day?	Mon 9/11/09	Mon 9/11/09
3		Interview Users	1 day?	Mon 9/11/09	Mon 9/11/09
4		Define the Reqs	1 day?	Mon 9/11/09	Mon 9/11/09
5		Software	1 day?	Mon 9/11/09	Mon 9/11/09
6		Design Logic	1 day?	Mon 9/11/09	Mon 9/11/09
7		Design Database	1 day?	Mon 9/11/09	Mon 9/11/09
8		Code A	1 day?	Mon 9/11/09	Mon 9/11/09
9		Code B	1 day?	Mon 9/11/09	Mon 9/11/09
10		Hardware	1 day?	Mon 9/11/09	Mon 9/11/09
11		Purchase Hardware	1 day?	Mon 9/11/09	Mon 9/11/09
12		Assemble Hardware	1 day?	Mon 9/11/09	Mon 9/11/09
13		Prototype	1 day?	Mon 9/11/09	Mon 9/11/09
14		Integration	1 day?	Mon 9/11/09	Mon 9/11/09
15		Integrate SW/HW	1 day?	Mon 9/11/09	Mon 9/11/09
16		Test	1 day?	Mon 9/11/09	Mon 9/11/09

Assign numerical code to tasks

Now we have created a work breakdown structure we will assign a numerical code to it. We will go to “tools” in the task bar. Under “options” we will simply check underneath “outline options” “show outline numbers”

☒ **Show outline number** and click ok. And you will notice the task view has changed and have a numerical code to our work breakdown structure.

Each deliverable can be identified by the code. As an example with 1.3.1. Purchase Hardware the 1.3 means hardware. 1.2. means hardware and so on.

	i	Task Name	Duration	Start	Finish
1		1 Mobile Reporting Sysem	1 day?	Mon 9/11/09	Mon 9/11/09
2		1.1 Performance Requiremer	1 day?	Mon 9/11/09	Mon 9/11/09
3		1.1.1 Interview Users	1 day?	Mon 9/11/09	Mon 9/11/09
4		1.1.2 Define the Reqs	1 day?	Mon 9/11/09	Mon 9/11/09
5		1.2 Software	1 day?	Mon 9/11/09	Mon 9/11/09
6		1.2.1 Design Logic	1 day?	Mon 9/11/09	Mon 9/11/09
7		1.2.2 Design Database	1 day?	Mon 9/11/09	Mon 9/11/09
8		1.2.3 Code A	1 day?	Mon 9/11/09	Mon 9/11/09
9		1.2.4 Code B	1 day?	Mon 9/11/09	Mon 9/11/09
10		1.3 Hardware	1 day?	Mon 9/11/09	Mon 9/11/09
11		1.3.1 Purchase Hardware	1 day?	Mon 9/11/09	Mon 9/11/09
12		1.3.2 Assemble Hardware	1 day?	Mon 9/11/09	Mon 9/11/09
13		1.4 Prototype	1 day?	Mon 9/11/09	Mon 9/11/09
14		1.4.1 Integration	1 day?	Mon 9/11/09	Mon 9/11/09
15		1.4.2 Integrate SW/HW	1 day?	Mon 9/11/09	Mon 9/11/09
16		1.4.3 Test	1 day?	Mon 9/11/09	Mon 9/11/09

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Create a Microsoft project schedule

In this **tutorial** you will be learning how to create a project schedule by entering estimated duration of tasks and enter predecessor information.

Entering estimated duration

Estimates entered in the project are based upon experience, talking to experts and the people performing the tasks. For this example we will assume we have done that and are ready to enter the information. We will go to the duration column and to the lowest specific task such as “Interview

Users” which will take 7 days or “Define Requirements” which will only take one day and so on. We only enter estimated duration for the tasks and not the deliverables. MS Project will fill these in for you giving the times of the tasks below that deliverable. The duration will not be counted up at the deliverable only the longest task is reflected, this will cause all the tasks start on the same date. Also notice the duration default is days. You can change this with shortcut keys to:

1. m: minutes
2. h: hours
3. d: days
4. w: weeks
5. mo: months
6. y: years

		Task Name	Duration	Start	Finish
1		1 Mobile Reporting Sysem	7 days	Mon 9/11/09	Tue 17/11/09
2		1.1 Performance Requiremer	7 days	Mon 9/11/09	Tue 17/11/09
3		1.1.1 Interview Users	7 days	Mon 9/11/09	Tue 17/11/09
4		1.1.2 Define the Reqs	1 day	Mon 9/11/09	Mon 9/11/09
5		1.2 Software	7 days	Mon 9/11/09	Tue 17/11/09
6		1.2.1 Design Logic	3 days	Mon 9/11/09	Wed 11/11/09
7		1.2.2 Design Database	2 days	Mon 9/11/09	Tue 10/11/09
8		1.2.3 Code A	7 days	Mon 9/11/09	Tue 17/11/09
9		1.2.4 Code B	5 days	Mon 9/11/09	Fri 13/11/09
10		1.3 Hardware	7 days	Mon 9/11/09	Tue 17/11/09
11		1.3.1 Purchase Hardware	7 days	Mon 9/11/09	Tue 17/11/09
12		1.3.2 Assemble Hardware	2 days	Mon 9/11/09	Tue 10/11/09
13		1.4 Prototype	3 days	Mon 9/11/09	Wed 11/11/09
14		1.4.1 Integration	0 days	Mon 9/11/09	Mon 9/11/09
15		1.4.2 Integrate SW/HW	3 days	Mon 9/11/09	Wed 11/11/09
16		1.4.3 Test	1 day	Mon 9/11/09	Mon 9/11/09

Entering predecessor information

Next we will enter the predecessor information for the tasks. You might need to expand the task view and so increase the calendar view to see the predecessor column. Our first task is “Interview Users” which doesn’t have a predecessor. For the next task we will enter the number 3 as predecessor. This number comes from the row association with “Interview Users”.

We will fill this out for all the tasks. Task “Code A” and task “Code B” can have the same predecessor, this means they will start at the same time. “Integration” and “Integrate SW/HW” have multiple predecessors, this means all of the predecessors have to be finished before this task can start. On the calendar view you will notice the bars have extend out as they represent the schedule and too the start and finish dates have changed as they reflect the predecessors. Only fill in predecessor information under tasks at the lowest level and not under summary tasks.

	i	Task Name	Duration	Start	Finish
1		1 Mobile Reporting System	7 days	Mon 9/11/09	Tue 17/11/09
2		1.1 Performance Requirement	7 days	Mon 9/11/09	Tue 17/11/09
3		1.1.1 Interview Users	7 days	Mon 9/11/09	Tue 17/11/09
4		1.1.2 Define the Reqs	1 day	Mon 9/11/09	Mon 9/11/09
5		1.2 Software	7 days	Mon 9/11/09	Tue 17/11/09
6		1.2.1 Design Logic	3 days	Mon 9/11/09	Wed 11/11/09
7		1.2.2 Design Database	2 days	Mon 9/11/09	Tue 10/11/09
8		1.2.3 Code A	7 days	Mon 9/11/09	Tue 17/11/09
9		1.2.4 Code B	5 days	Mon 9/11/09	Fri 13/11/09
10		1.3 Hardware	7 days	Mon 9/11/09	Tue 17/11/09
11		1.3.1 Purchase Hardware	7 days	Mon 9/11/09	Tue 17/11/09
12		1.3.2 Assemble Hardware	2 days	Mon 9/11/09	Tue 10/11/09
13		1.4 Prototype	3 days	Mon 9/11/09	Wed 11/11/09
14		1.4.1 Integration	0 days	Mon 9/11/09	Mon 9/11/09
15		1.4.2 Integrate SW/HW	3 days	Mon 9/11/09	Wed 11/11/09
16		1.4.3 Test	1 day	Mon 9/11/09	Mon 9/11/09

Inserting lag

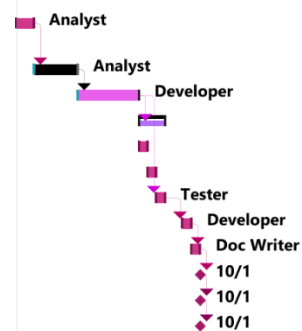
When we look closer at our projects we notice that the 7 days for purchasing hardware isn't accurate. It only takes 1 day to order it and 7 days to deliver it. There is a lag relationship between purchasing and assembling hardware. We need to double click the assembling hardware task for its property window to open. We will fill in the lag under the predecessor box. We will be changing the lag from 0 to 6 for the shipping to take place. You also notice a predecessor has a type for when to start.

1. Finished-to-Start (FS)—Task (B) cannot start until another task (A) finishes.
2. Start-to-start (SS)—Task (B) cannot start until another task (A) starts.
3. Finish-to-finish (FF)—Task (B) cannot finish until another task (A) finishes.
4. Start-to-finish (SF)—Task (B) cannot finish until another task (A) starts.

EXERCISE:

1. Create work breakdown structure of a Project that should consists of tasks and time duration carried by a project take example of your own project you have done in this semester.
2. Include milestones and summary tasks in the project plan.
3. Add work resources that will be used in the project

ID	i	Task Mode	Task Name	Duration	Start	Finish	Predecessors	per 2025	October 2025
1			1 PROJECTILE MOTION - PHYSICS	29 days	Wed 9/10/25	Mon 10/20/25		7 10 13 16 19 22 25 28	1 4 7 10 13 16 19 22
2			1.1 Requirement Gathering	2 days	Wed 9/10/25	Thu 9/11/25			
3			1.2 Feasibility study	3 days	Fri 9/12/25	Tue 9/16/25	2		
4			1.3 Development	5 days	Wed 9/17/25	Tue 9/23/25	3		
5			1.4 coding	3 days	Wed 9/24/25	Fri 9/26/25	4		
6			1.4.1 logic	1 day	Wed 9/24/25	Wed 9/24/25			
7			1.4.2 impliment	1 day	Thu 9/25/25	Thu 9/25/25			
8			1.4.3 Testing	1 day	Fri 9/26/25	Fri 9/26/25	4		
9			1.5 Deployment	1 day	Mon 9/29/25	Mon 9/29/25	8		
10			1.6 documentation	1 day	Tue 9/30/25	Tue 9/30/25	9		
11			1.7 final presentati	0 days	Wed 10/1/25	Wed 10/1/25	10		
12			1.8 project approv	0 days	Wed 10/1/25	Wed 10/1/25	11		
13			1.9 project closed	0 days	Wed 10/1/25	Wed 10/1/25	12		



LAB SESSION 08

Open Ended Lab

OBJECTIVE

The objective of this lab is to assess students on the knowledge and skills acquired during the previous seven lab sessions. Perceive the given task and perform within stipulated time.

EVALUATION METHOD

Students will be evaluated through open-ended problem-solving tasks using the OEL rubrics (5 marks) with criteria levels 0 and 1. Evaluation will be based on Task completion, Process taken along and Rubrics based assessment.

TASKS

Tasks will be based on all the prior 7 labs performance and will be technically observed according to the assessment of Rubrics.

NED University of Engineering & Technology
Department of Software Engineering
OEL RUBRICS
ITFA (EF-101)



Criteria	Achieved (1)	Not Achieved (0)	Marks
Problem Identification	Able to correctly identify the problem(s) and all required data (input and output)	Not able to correctly identify the problem(s) and all required data (input and output).	
Problem Solving Skills	Uses an effective strategy to solve the problem(s) and clearly identifies the steps	Uses an ineffective strategy to solve the problem(s) and completely misses the steps	
Implementation & Debugging	Efficient implementation and excellent debugging skills. Solution works without any error	Incomplete implementation and ineffective debugging skills. Major errors in the implementation.	
Deliverable(s)	Generates correct and complete output.	Cannot generate the desired output.	
Depth of Knowledge	Thorough grasp of the concepts covered in the lab and can apply it in real-life situations	Does not grasp the concepts covered in the lab and cannot identify any real-life application	
Weighted CLO Score			
Remarks and Signature			

LAB SESSION 09

Getting Familiar with HTML

THEORY:

HTML for Absolute Beginners

While many guides on the internet attempt to teach HTML using a lot of mind-boggling theory, this tutorial will instead focus on giving you the practical skills to build your first site.

The aim is to show you ‘how’ to create your first web page without spending the entire tutorial focusing too much on the “why.”

By the end of this tutorial, you will have the know-how to create a basic website and we hope that this will inspire you to delve further into the world of HTML using our follow-on guides.

What is HTML?

Okay, so this is the only bit of mandatory theory. In order to begin to write HTML, it helps if you know what you are writing.

HTML is the language in which most websites are written. HTML is used to create pages and make them functional.

The code used to make them visually appealing is known as CSS and we shall focus on this in a later tutorial. For now, we will focus on teaching you how to build rather than design.

HTML is derived from a language SGML (Standard Graphics Markup Language). HTML is a computer language that is used to create documents on the World Wide Web. HTML is very simple, and logical. It reads from left to right, top to bottom and uses plain text. HTML is NOT a programming language, but a mark-up language that uses <Tags> like this. The websites you view on the internet are actually text files that consist of HTML Tags.

The History of HTML

HTML was first created by Tim Berners-Lee, Robert Cailliau, and others starting in 1989. It stands for Hyper Text Markup Language.

Hypertext means that the document contains links that allow the reader to jump to other places in the document or to another document altogether. The latest version is known as HTML5.

A Markup Language is a way that computers speak to each other to control how text is processed and presented. To do this HTML uses two things: tags and attributes.

What are Tags and Attributes?

Tags and attributes are the basis of HTML.

They work together but perform different functions – it is worth investing 2 minutes in differentiating the two.

What Are HTML Tags?

Tags are used to mark up the start of an HTML element and they are usually enclosed in angle brackets. An example of a tag is: <h1>.

Most tags must be opened <h1> and closed </h1> in order to function.

What are HTML Attributes?

Attributes contain additional pieces of information. Attributes take the form of an opening tag and additional info is placed inside.

An example of an attribute is:

```

```

In this instance, the image source (src) and the alt text (alt) are attributes of the tag.

Golden Rules To Remember

1. The vast majority of tags must be opened (<tag>) and closed (</tag>) with the element information such as a title or text resting between the tags.
2. When using multiple tags, the tags must be closed in the order in which they were opened. For example:

This is really important!

Creating Your First HTML Webpage

First off, you need to open your HTML editor, where you will find a clean white page on which to write your code. From there you need to layout your page with the following tags.

Basic Construction of an HTML Page

These tags should be placed underneath each other at the top of every HTML page that you create.

- `<!DOCTYPE html>` — This tag specifies the language you will write on the page. In this case, the language is HTML 5.
- `<html>` — This tag signals that from here on we are going to write in HTML code.
- `<head>` — This is where all the metadata for the page goes — stuff mostly meant for search engines and other computer programs.
- `<body>` — This is where the content of the page goes.



This is how your average HTML page is structured visually.

Further Tags

Inside the `<head>` tag, there is one tag that is always included: `<title>`, but there are others that are just as important:

`<title>`

This is where we insert the page name as it will appear at the top of the browser window or tab.

`<meta>`

This is where information about the document is stored: character encoding, name (page context), description.

Let's try out a basic <head> section:

```
<head>

<title>My First Webpage</title>

<meta charset="UTF-8">

<meta name="description" content="This field contains information about your page. It is usually around two sentences long.">
```

Adding Content

Next, we will make <body> tag.

The HTML <body> is where we add the content which is designed for viewing by human eyes.

This includes text, images, tables, forms and everything else that we see on the internet each day.

How to Add HTML Headings To Your Web Page

In HTML, headings are written in the following elements:

- <h1>
- <h2>
- <h3>
- <h4>
- <h5>
- <h6>

As you might have guessed <h1> and <h2> should be used for the most important titles, while the remaining tags should be used for sub-headings and less important text.

Search engine bots use this order when deciphering which information is most important on a page.

Creating Your Heading

Let's try it out. On a new line in the HTML editor, type:

```
<h1>Welcome to My Page</h1>
```

And hit save. We will save this file as "index.html" in a new folder called "my webpage."

How To Add Text In HTML

Adding text to our HTML page is simple using an element opened with the tag <p> which creates a new paragraph. We place all of our regular text inside the element <p>.

When we write text in HTML, we also have a number of other elements we can use to control the text or make it appear in a certain way.

Other Key Elements

They are as follows:

Element	Meaning	Purpose
---------	---------	---------

	Bold	Highlight important information
	Strong	Similarly to bold, to highlight key text
<i>	Italic	To denote text
	Emphasised Text	Usually used as image captions
<mark>	Marked Text	Highlight the background of the text
<small>	Small Text	To shrink the text

Element	Meaning	Purpose
<strike>	Striked Out Text	To place a horizontal line across the text
<u>	Underlined Text	Used for links or text highlights
<ins>	Inserted Text	Displayed with an underline to show an inserted text
<sub>	Subscript Text	Typographical stylistic choice
<sup>	Superscript Text	Another typographical presentation style

These tags must be opened and closed around the text in question.

Let's try it out. On a new line in the HTML editor, type the following HTML code:

```
<p>Welcome to <em>my</em> brand new website. This site will be my <strong>new</strong> home on the web. </p>
```

Don't forget to hit save and then refresh the page in your browser to see the results.

How To Add Links In HTML

As you may have noticed, the internet is made up of lots of links.

Almost everything you click on while surfing the web is a link takes you to another page within the website you are

visiting or to an external site.

Links are included in an attribute opened by the <a> tag. This element is the first that we've met which uses an attribute and so it looks different to previously mentioned tags.

The Anchor Tag

The <a> (or anchor) opening tag is written in the format:

```
<a href="https://blogging.com/how-to-start-a-blog/">Your Link Text Here </a>
```

 The first

part of the attribute points to the page that will open once the link is clicked.

Meanwhile, the second part of the attribute contains the text which will be displayed to a visitor in order to entice them to click on that link.

If you are building your own website then you will most likely host all of your pages on professional web hosting. In this case, internal links on your website will Link Here.

Let's Create An Anchor Tag

Let's try it out. Make a duplicate of the code from your current index.html page. Copy / paste it into a new window in your HTML editor.

Save this new page as "page2.html" and ensure that it is saved in the same folder as your index.html page. On

page2.html add the following code:

```
<a href="http://www.google.com">Google</a>
```

This will create a link to Google on page 2. Hit save and return to your index.html page. On a new

line on index.html add the following code:

```
<a href="*/folder(s)*/page2.html">Page2</a>
```

Ensure the folder path to the file (page2.html) is correct. Hit save and preview index.html in your browser.

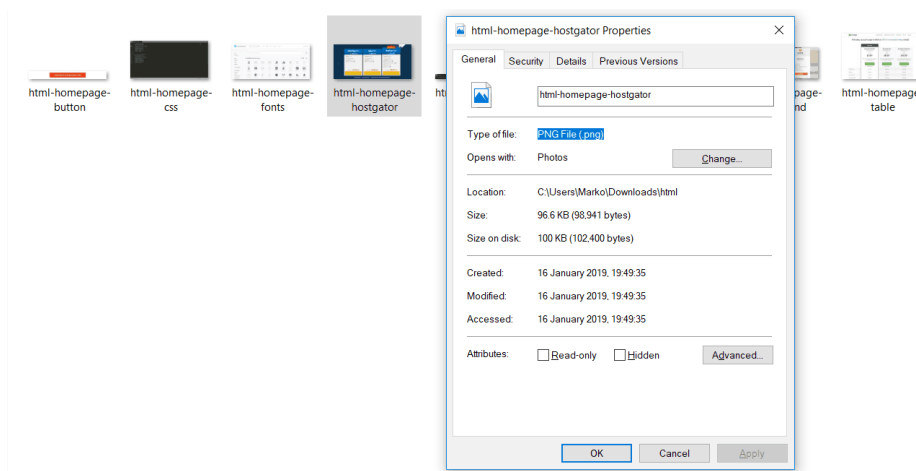
If everything is correct then you will see a link which will take you to your second page. On the second page, there will be a link that will take you to google.com.

How To Add Images In HTML To Your Website

In today's modern digital world, images are everything. The tag has everything you need to display images on your site. Much like the <a> anchor element, also contains an attribute.

The attribute features information for your computer regarding the source, height, width and alt text of the image.

Styling and Formats



You can check the file type of an image by right-clicking the item and selecting 'Properties'.

You can also define borders and other styles around the image using the class attribute. However, we shall cover this in a later tutorial.

The file types generally used for image files online are: .jpg, .png, and (less and less) .gif.

Alt text is important to ensure that your site is ranked correctly on search sites and also for visually impaired visitors to your site.

The tag normally is written as follows:

```

```

Let's try it out.

How To Make an HTML List

In web design, there are 3 different types of lists which you may wish to add to your site.

Ordered List

The first is an : This is an ordered list of contents. For example:

1. An item
2. Another item
3. Another goes here.

Inside the tag we list each item on the list inside tags. For

example:

```
<ol>

<li>An item </li>

<li>Another item </li>

<li>Another goes here </li>
```

Unordered List

The second type of list that you may wish to include is an unordered list. This is better known as a bullet point list and contains no numbers.

An example of this is:

```
<ul>

<li>This is </li>

<li>An Unordered </li>

<li>List </li>
```

Definition List

Finally, you may wish to include a definition list <dl> on your page. An example of a <dl> list is as follows:

HTML

Hypertext markup language is a programming language used to create web pages and is rendered by a web browser.

The code used for the above is as follows:


```

<dl>

<dt>Item</dt>

<dd>The definition goes here</dd>

```

Let's try it out. Open index.html and on a new line, enter the following HTML:

```

<p>This website will have the following benefits for my business:</p>

<ul>

<li>Increased traffic </li>

<li>Global Reach</li>

<li>Promotional Opportunities</li>

```

Now hit save and check out the results in your browser. If everything worked out, then it will display a bullet- pointed table displaying the information above.

How To Add Tables In HTML

The screenshot shows a code editor with two panels: HTML and CSS (SCSS). The HTML panel contains the following code:

```

1 <table class="flat-table">
2   <tbody>
3     <tr>
4       <th>Head 1</th>
5       <th>Head 2</th>
6       <th>Head 3</th>
7       <th>Head 4</th>
8       <th>Head 5</th>
9     </tr>
10    <tr>
11      <td>Example 1</td>
12      <td>Example 2</td>
13      <td>Example 3</td>
14      <td>Example 4</td>
15      <td>Example 5</td>
16    </tr>
17    <tr>
18      <td>Example 6</td>

```

The CSS panel contains the following code:

```

1 .flat-table {
2   display: block;
3   font-family: sans-serif;
4   -webkit-font-smoothing: antialiased;
5   font-size: 115%;
6   overflow: auto;
7   width: auto;
8
9   th {
10    background-color: rgb(112, 196, 105);
11    color: white;
12    font-weight: normal;
13    padding: 20px 30px;
14    text-align: center;
15  }
16  td {
17    background-color: rgb(238, 238, 238);
18    color: reb(111, 111, 111);

```

Below the code editor, a live preview of the rendered table is shown:

Head 1	Head 2	Head 3	Head 4	Head 5
Example 1	Example 2	Example 3	Example 4	Example 5
Example 6	Example 7	Example 8	Example 9	Example 10

Tables can be styled in various ways – Codepen.io offers live previews to keep track of changes. Another way to keep your website looking neat and orderly is through the use of a table.

What Does a Table Consist Of?

When drawing a table we must open an element with the `<table>` opening tag. Inside this tag, we structure the table using the table rows, `<tr>`, and cells, `<td>`.

An example of an HTML table is as follows:

```
<table>

<tr>

<td>Row 1 - Column 1</td>

<td>Row 1 - Column 2 </td>

<td>Row 1 - Column 3 </td>

</tr>

<tr>

<td>Row 2 - Column 1</td>

<td>Row 2 - Column 2</td>
```

This will produce a 2-row table with 3 cells in each row.

Tables can get quite complicated, so be sure to check out our special HTML tables tutorial.

Table Tags

However, watch out for these tags so that you can recognize them and use them as your skills develop. Here are the tables tags presented in a table – pun totally intended.

Table Tag	Meaning	Location
<code><thead></code>	Table Head	Top of the table
<code><tbody></code>	Table Body	Content of the table
<code><tfoot></code>	Table Foot	Bottom of the table
<code><colgroup></code>	Column Group	Within the table
<code><th></code>	Table Header	Data cell for the table header

Let's Make a Table

Go to a new line on the index.html page within your text editor. Enter the following HTML code:

```

<table>

<tr>

<td>Row 1 - Column 1</td>

<td>Row 1 - Column 2 </td>

</tr>

<tr>

<td>Row 2 - Column 1</td>

```

Hit save and preview it in your browser. Congratulations: You did it!

How to Close an HTML Document

You've reached the end of our absolute beginners HTML tutorial.

The final step we need to complete is to close the `<body>` and `<html>` tags at the end of each page using the following HTML code:

```

</body>

</html>

```

EXERCISE:

1. Create a webpage that prints your name on the screen.

Source Code:

```

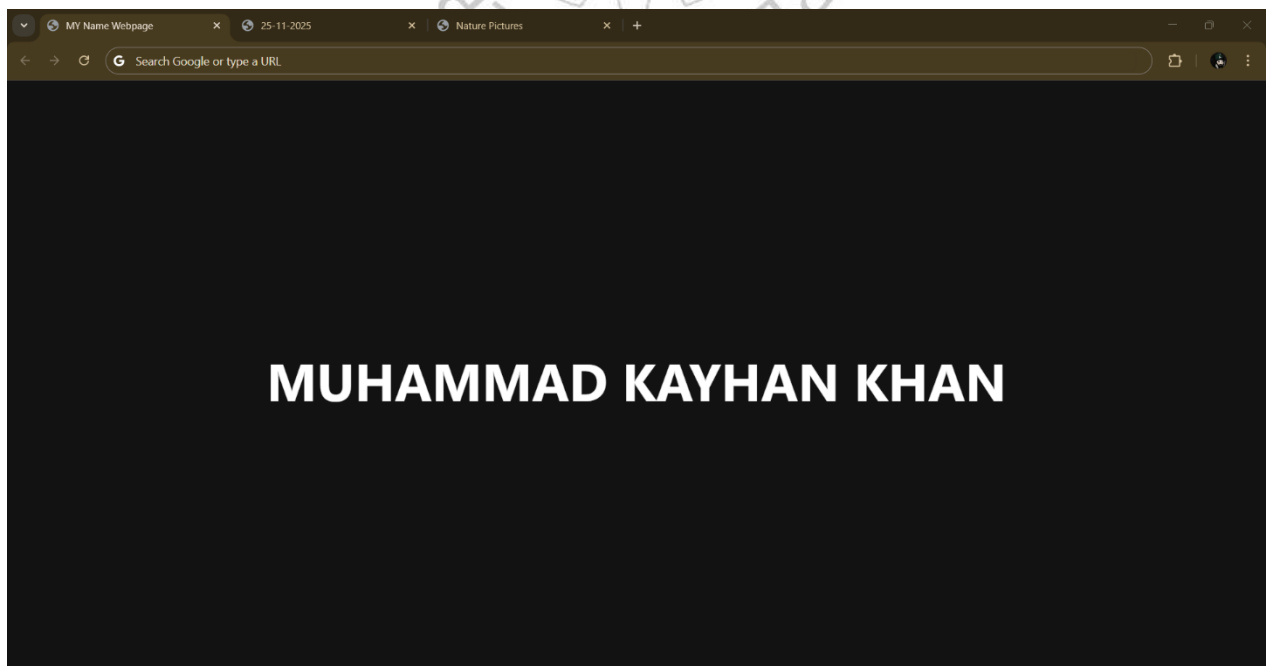
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="utf-8" />
  <title>MY Name Webpage</title>
  <meta name="viewport" content="width=device-width, initial-scale=1" />
  <style>
    :root {
      color-scheme: light dark;
    }

```

```
body {  
  margin: 0;  
  min-height: 100vh;  
  display: grid;  
  place-items: center;  
  font-family: system-ui, -apple-system, Segoe UI, Roboto,  
Helvetica, Arial, sans-serif;  
}  
  
h1 {  
  font-size: clamp(2rem, 6vw, 4rem);  
  margin: 0;  
}  
</style>  
</head>  
  
<body>  
  <h1>MUHAMMAD KAYHAN KHAN</h1>  
</body>  
  
</html>
```

Output:



2. Create a webpage that prints the message "When was this webpage created? Set the title of the page to the current date.

Source Code:

```
<!DOCTYPE html>
<html lang="en">

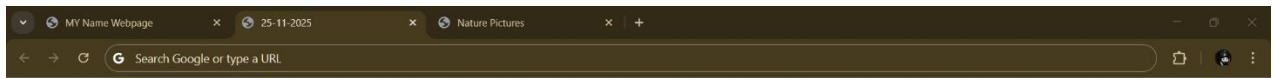
<head>
  <meta charset="utf-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1" />
  <title></title>
  <script>
    (function () {
      var now = new Date();
      var dd = String(now.getDate()).padStart(2, '0');
      var mm = String(now.getMonth() + 1).padStart(2, '0');
      var yyyy = now.getFullYear();
      document.title = dd + '-' + mm + '-' + yyyy;
    })();
  </script>
  <style>
    body {
      margin: 0;
      min-height: 100vh;
      display: grid;
      place-items: center;
      font-family: system-ui, -apple-system, Segoe UI, Roboto, Helvetica,
Arial, sans-serif;
    }

    h1 {
      font-size: clamp(1.5rem, 6vw, 3rem);
      margin: 0;
    }
  </style>
</head>

<body>
  <h1>"When was this webpage created?"</h1>
</body>

</html>
```

Output:



"When was this webpage created?"

3. Display five different images. Skip two lines between each image. Each image should have a title.

Source Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Nature Pictures</title>
</head>
<body>
  <h1>Five Nature Images</h1>

  <h3>Mountains</h3>
  

  <br><br>

  <h3>Ocean</h3>
  
  <br><br>

  <h3>Forest</h3>
  
```

```
<br><br>
```

```
<h3>Waterfall</h3>
```

```

```

```
<br><br>
```

```
<h3>Desert</h3>
```

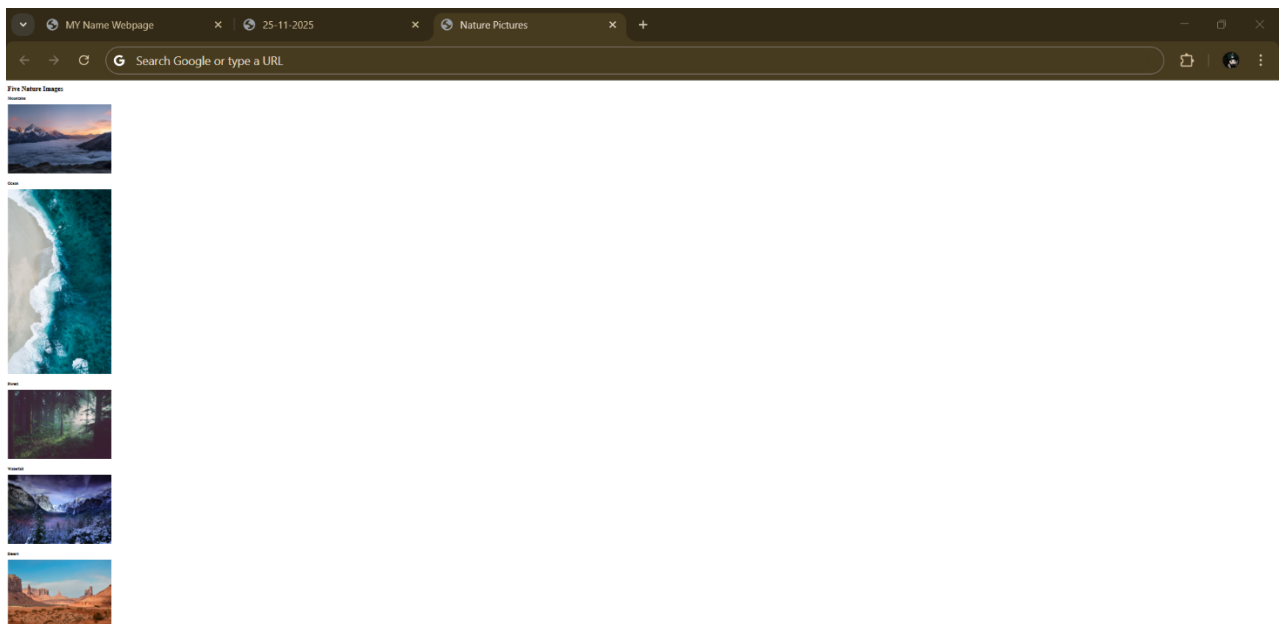
```

```

```
</body>
```

```
</html>
```

Output:



LAB SESSION 10

Creating HTML Forms

THEORY:

HTML Forms

An HTML form is used to collect user input. The user input is most often sent to a server for processing.

What is HTML Form

HTML Forms are required to collect different kinds of user inputs, such as contact details like name, email address, phone numbers, or details like credit card information, etc.

PROCEDURE:

Forms contain special elements called controls like inputbox, checkboxes, radio-buttons, submit buttons, etc. Users generally complete a form by modifying its controls e.g. entering text, selecting items, etc. and submitting this form to a web server for further processing.

The `<form>` tag is used to create an HTML form. Here's a simple example of a login form:

Example

```
<form>
  <label>Username: <input type="text"></label>
  <label>Password: <input type="password"></label>
  <input type="submit" value="Submit">
</form>
```

The following section describes different types of controls that you can use in your form.

Input Element

This is the most commonly used element within HTML forms.

It allows you to specify various types of user input fields, depending on the `type` attribute. An input element can be of type text field, password field, checkbox, radio button, submit button, reset button, file select box, as well as several new input types introduced in HTML5.

The most frequently used input types are described below.

Text Fields

Text fields are one line areas that allow the user to input text.

Single-line text input controls are created using an `<input>` element, whose `type` attribute has a value

of `text`. Here's an example of a single-line text input used to take username:

Example

```
<form>
  <label for="username">Username:</label>
  <input type="text" name="username" id="username">
</form>
```

— The output of the above example will look something like this:

Username:

Note: The `<label>` tag is used to define the labels for `<input>` elements. If you want your user to enter several lines you should use a `<textarea>` instead.

Password Field

Password fields are similar to text fields. The only difference is; characters in a password field are masked, i.e. they are shown as asterisks or dots. This is to prevent someone else from reading the password on the screen. This is also a single-line text input controls created using an `<input>` element whose `type` attribute has a value of `password`.

Here's an example of a single-line password input used to take user password:

Example

```
<form>
  <label for="user-pwd">Password:</label>
  <input type="password" name="user-password" id="user-pwd">
</form>
```

— The output of the above example will look something like this:

Password:

Radio Buttons

Radio buttons are used to let the user select exactly one option from a pre-defined set of options. It is created using an `<input>` element whose `type` attribute has a value of `radio`.

Here's an example of radio buttons that can be used to collect user's gender information:

Example

```
<form>
  <input type="radio" name="gender" id="male">
  <label for="male">Male</label>
  <input type="radio" name="gender" id="female">
  <label for="female">Female</label>
</form>
```

— The output of the above example will look something like this:

☐ Male ☐ Female

Checkboxes

Checkboxes allows the user to select one or more option from a pre-defined set of options. It is created using an `<input>` element whose `type` attribute has a value of `checkbox`.

Here's an example of checkboxes that can be used to collect information about user's hobbies:

Example

```
<form>
  <input type="checkbox" name="sports" id="soccer">
  <label for="soccer">Soccer</label>
  <input type="checkbox" name="sports" id="cricket">
  <label for="cricket">Cricket</label>
  <input type="checkbox" name="sports" id="baseball">
  <label for="baseball">Baseball</label>
</form>
```

— The output of the above example will look something like this:

☐ Soccer ☐ Cricket ☐ Baseball

Note: If you want to make a radio button or checkbox selected by default, you can add the `checked` attribute to the input element, like `<input type="checkbox" checked>`.

File Select box

The file fields allow a user to browse for a local file and send it as an attachment with the form data. Web browsers such as Google Chrome and Firefox render a file select input field with a Browse button that enables the user to navigate the local hard drive and select a file.

This is also created using an `<input>` element, whose `type` attribute value is set to `file`.

Example

```
<form>
  <label for="file-select">Upload:</label>
  <input type="file" name="upload" id="file-select">
</form>
```

— The output of the above example will look something like this:

Upload: No file chosen

Tip: There are several other input types. Please check out the chapter on HTML5 new input types to learn more about the newly introduced input types.

Textarea

Textarea is a multiple-line text input control that allows a user to enter more than one line of text.

Multi-line text input controls are created using an `<textarea>` element.

Example

```
<form>
  <label for="address">Address:</label>
  <textarea rows="3" cols="30" name="address" id="address"></textarea>
</form>
```

— The output of the above example will look something like this:

Address:

Select Boxes

A select box is a dropdown list of options that allows user to select one or more option from a pull- down list of options. Select box is created using the `<select>` element and `<option>` element.

The `<option>` elements within the `<select>` element define each list item.

Example

```
<form>
  <label for="city">City:</label>
  <select name="city" id="city">
    <option value="sydney">Sydney</option>
    <option value="melbourne">Melbourne</option>
    <option value="cromwell">Cromwell</option>
  </select>
</form>
```

— The output of the above example will look something like this:

City:

Submit and Reset Buttons

A submit button is used to send the form data to a web server. When submit button is clicked the form data is sent to the file specified in the form's `action` attribute to process the submitted data.

A reset button resets all the forms control to default values. Try out the following example by typing your name in the text field, and click on submit button to see it in action.

Example

```
<form action="action.php" method="post">
  <label for="first-name">First Name:</label>
  <input type="text" name="first-name" id="first-name">
  <input type="submit" value="Submit">
  <input type="reset" value="Reset">
</form>
```

First Name:

Type your name in the text field above, and click on submit button to see it in action.

Note: You can also create buttons using the `<button>` element. Buttons created with the `<button>` element function just like buttons created with the input element, but they offer richer rendering possibilities by allowing the embedding of other HTML elements.

Grouping Form Controls

You also group logically related controls and labels within a web form using the `<legend>` element. Grouping form controls into categories makes it easier for users to locate a control which makes the form more user-friendly. Let's try out the following example to see how it works:

Example

```
<form>
  <fieldset>
    <legend>Contact Details</legend>
    <label>Email Address: <input type="email" name="email"></label>
    <label>Phone Number: <input type="text" name="phone"></label>
  </fieldset>
</form>
```

Frequently Used Form Attributes

The following table lists the most frequently used form element's attributes:

Attribute	Description
name	Specifies the name of the form.
action	Specifies the URL of the program or script on the web server that will be used for processing the information submitted via form.
method	Specifies the HTTP method used for sending the data to the web server by the browser. The value can be either get (the default) and post.
target	Specifies where to display the response that is received after submitting the form. Possible values are <code>_blank</code> <code>_self</code> <code>_parent</code> and <code>_top</code> .
enctype	Specifies how the form data should be encoded when submitting the form to the server. Applicable only when the value of the method attribute is post.

There are several other attributes, to know about them please see the `<form>` tag reference.

Note: The name attribute represents the form's name within the forms collection. Its value must be unique among the forms in a document, and must not be an empty string.

Tip: All the data sent via `get` method is visible in the browser's address bar. But, the data sent via `post` is not visible to the user. Please check out the tutorial on [GET vs. POST](#) to learn about the difference between these two HTTP methods in detail.

HTML FORM EXAMPLE

```
<html>
<head>
```

```

<title>POPCORN SALES FORM</title>
</head>
<body>
<h2>Welcome to millennium Gymnastic Booster Club Popcorn sales </h2>
<form action=" ">
<p>
<!-- text boxes for name and address-->
<label> Buyer's Name : <input type="text" name = "name" size="30"/></label>
<br>
<label> Street Address: <input type="text" name = "Street" size="30"/></label>
<br>
<label> City State Zip: <input type="text" name = "city" size="30"/></label>
</p>
<!-- table for item orders -->
<table>
<!--first, the column headings--->
<tr>
<th> Product Name </th>
<th>Price </th>
<th> Quality </th>
</tr>
<!--table data entities-->
<tr>
<td> Unpopped Popcorns (1 lb.)</td>
<td> $3.00 </td>
<td> <input type="text" name="unpop" size ="2"/> </td>
</tr>
<tr>
<td> caramel Popcorns (1 lb.)</td>
<td> $3.50 </td>
<td> <input type="text" name="caramel" size ="2"/> </td>
</tr>
<tr>
<td> Caramel nut Popcorns (1 lb.)</td>
<td> $4.50 </td>
<td> <input type="text" name="caramelnut" size ="2"/> </td>
</tr>
<tr>
<td> Toffee nut Popcorns (1 lb.)</td>
<td> $5.00 </td>
<td> <input type="text" name="Toffeenut" size ="2"/> </td>
</tr>
</table>
<!-- the radio buttons for the payment method-->
<h3> payment method</h3>
<p>
<label><input type="radio" name="payment" value="visa" checked ="checked"/>visa</label>
<br>
<label><input type="radio" name="payment" value="mc" checked ="checked"/> master card</label>

```

```

<br>
<label><input type="radio" name="payment" value="Discover" checked ="checked"/> Discover</label>
<br>
<label><input type="radio" name="payment" value="cheque" checked ="checked"/> cheque
</label>
<br>
</p>
<!-- buttons -->
<p>
<input type="submit" value="submit order">
<input type="reset" value = "clear order form">
</p>
</form>
</body>
</html>

```

Exercise 1

- i. Create a login form using appropriate tags.

Create an account!'."/>



☒ Remember me [Forgot password?](#)

Not registred? [Create an account!](#)

Source Code:

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Login Page</title>
</head>

<body>
  <h2>Login Form</h2>

  <form action="/login" method="POST">

    <label for="username">Username/Email:</label><br>
    <input type="text" id="username" name="username" required><br><br>

    <label for="password">Password:</label><br>
    <input type="password" id="password" name="password"
required><br><br>

    <input type="checkbox" id="remember" name="remember">
    <label for="remember">Remember me</label><br><br>

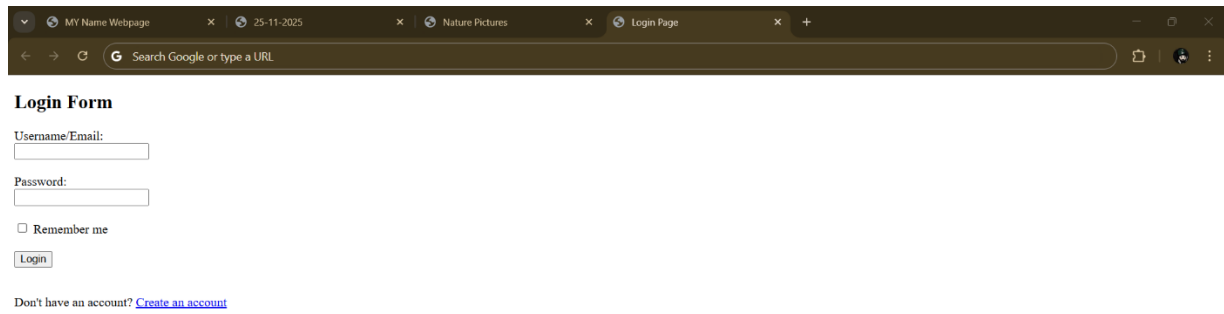
    <button type="submit">Login</button>
  </form>

  <br>

  <p>Don't have an account? <a href="signup.html">Create an
account</a></p>
</body>

</html>
```

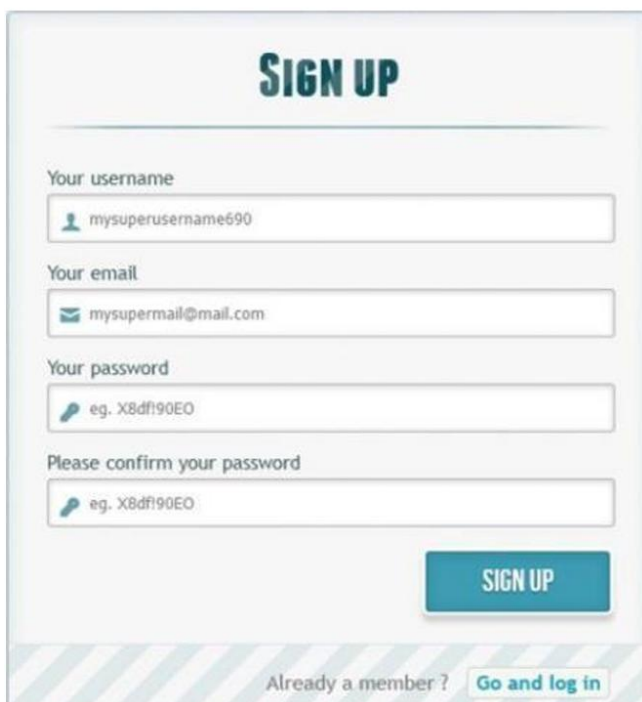
Output:



The screenshot shows a web browser with multiple tabs. The active tab is titled "Login Page". The address bar shows "Search Google or type a URL". The page content includes a "Login Form" with the following elements:

- Username/Email:** A text input field.
- Password:** A text input field.
- ☐ Remember me
- Login** button
- Don't have an account? [Create an account](#)

- ii. "Create an account" Text must be linked to another page i.e. SignUp page.



The screenshot shows a "SIGN UP" form with the following elements:

- SIGN UP** (Header)
- Your username**: Text input field containing "mysuperusername690".
- Your email**: Text input field containing "mysupermail@mail.com".
- Your password**: Text input field containing "eg. X8dfI90EO".
- Please confirm your password**: Text input field containing "eg. X8dfI90EO".
- SIGN UP** button
- Already a member? [Go and log in](#)

Source Code:

```
<!DOCTYPE html>
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Sign Up Page</title>
</head>

<body>
  <h2>Sign Up Form</h2>

  <form action="/signup" method="POST">

    <label for="fullname">Full Name:</label><br>
    <input type="text" id="fullname" name="fullname" required><br><br>

    <label for="email">Email:</label><br>
    <input type="email" id="email" name="email" required><br><br>

    <label for="username">Username:</label><br>
    <input type="text" id="username" name="username" required><br><br>

    <label for="password">Password:</label><br>
    <input type="password" id="password" name="password"
required><br><br>

    <label for="confirmpassword">Confirm Password:</label><br>
    <input type="password" id="confirmpassword" name="confirmpassword"
required><br><br>

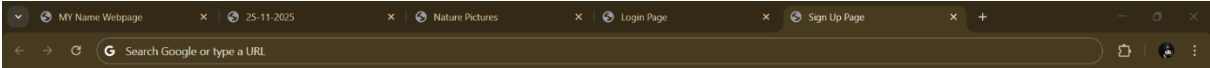
    <button type="submit">Sign Up</button>
  </form>

  <br>

  <p>Already have an account? <a href="login.html">Login here</a></p>
</body>

</html>
```

Output:



The screenshot shows a web browser with multiple tabs open: 'MY Name Webpage', '25-11-2025', 'Nature Pictures', 'Login Page', and 'Sign Up Page'. The address bar contains the text 'Search Google or type a URL'. The main content area displays a 'Sign Up Form' with the following fields and elements:

Sign Up Form

Full Name:

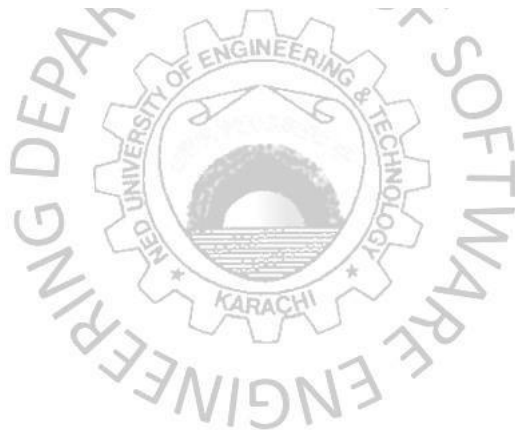
Email:

Username:

Password:

Confirm Password:

Already have an account? [Login here](#)



LAB SESSION 11

Creating Database and executing Queries in MS Access

THEORY:

The real power of a relational database lies in its ability to quickly **retrieve** and **analyze** your data by running a query. **Queries** allow you to **pull information** from one or more tables based on a set of search conditions you define. In this lesson, you will learn how to create a simple **one-table query**.

What are queries?

Queries are a way of **searching** for and **compiling** data from one or more tables. Running a query is like asking a **detailed question** of your database. When you build a query in Access, you are **defining specific search conditions** to find exactly the data you want.

How are queries used?

Queries are far more powerful than the simple searches or filters you might use to find data within a table. This is because queries can draw their information from **multiple** tables. For example, while you could use a **search** in the customers table to find the name of one customer at your business or a **filter** on the orders table to view only orders placed within the past week, neither would let you view both customers and orders at once. However, you could easily run a **query** to find the name and phone number of every customer who's made a purchase within the past week. A well-designed query can give information you might not be able to find out just by examining the data in your tables. When you run a query, the results are presented to you in a table, but when you design one you use a different view. This is called **Query Design view**, and it lets you see how your query is put together.

PROCEDURE:

One-table queries

Let's familiarize ourselves with the query-building process by building the **simplest** query possible: a one- table query.

We will run a query on the **Customers** table of our bakery database. Let's say our bakery is having a special event, and we want to invite our customers who live nearby because they are the most likely to come. This means we need to see a list of all customers who live close by, and **only** those customers.

We want to find our customers who live in the city of **Raleigh**, so we'll search for "**Raleigh**" in the City field. Some customers who live in the suburbs live fairly close by, and we'd like to invite them as well. We'll add their zip code, **27513**, as another criterion.

If you think these sounds a little like applying a filter, you're right. A one-table query is actually just an **advanced filter** applied to a table.

To create a simple one-table query:

1. Select the **Create** tab on the Ribbon, and locate the **Queries** group.
2. Click the **Query Design** command.
3. Access will switch to **Query Design view**. In the **Show Table** dialog box that appears, select the table you want to run a query on. We are running a query on our customers, so we'll select the **Customers** table.
4. Click **Add**, then click **Close**.
5. The selected table will appear as a small window in the **Object Relationship pane**. In the table window, double-click the **field names** you want to include in your query. They will be added to the **design grid** in the bottom part of the screen. In our example, we want to mail invitations to customers who live in a certain area, so we'll include the **First Name**, **Last Name**, **Street Address**, **City**, and **Zip Code** fields.
6. Set the **search criteria** by clicking the cell in the **Criteria:** row of each field you want to filter. Typing

criteria into more than one field in the Criteria: row will set your query to include only results that meet all criteria. If you want to set multiple criteria but don't need the records shown in your results to meet all of them, type the first criteria in the Criteria: row and additional criteria in the **or:** row and the rows beneath it. Because we want to find customers who either live in Raleigh **or** in the 27513 zip code, we'll type "Raleigh" in the **City** field and "27513" into the **or:** row of the **Zip Code** field. The **quotation marks** will search these fields for an **exact match**.

7. After you have set your criteria, **run** the query by clicking the **Run** command on the **Design** tab.
8. The query results will be displayed in the query's **Datasheet view**, which looks like a table. If you want, **save** your query by clicking the **Save** command in the Quick Access Toolbar. When prompted to name it, type the desired name, then click **OK**.

Now you know how to create the simplest type of query with only one table. In the next lesson, you'll learn how to create a query that uses multiple tables.

Designing a Multi-table Query

In the previous lesson, you learned how to create a simple query with one table. Most queries you design in Access will likely use multiple tables, allowing you to answer more complex questions. In this lesson, you'll learn how to design and create a multi-table query.

Queries can be difficult to understand and build if you don't have a good idea of what you're trying to find and how to find it. A one-table query can be simple enough to make up as you go along, but to build anything more powerful you'll need to plan the query in advance.

Planning a query

When planning a query that uses more than one table, you should go through these four steps:

1. **Pinpoint** exactly what you want to know. If you could ask your database any question, what would it be? Building a query is more complicated than just asking a question, but knowing precisely what question you want to answer is essential to building a useful query.
2. **Identify** every type of information you want included in your query results. Which fields contain this information?
3. **Locate** the fields you want to include in your query. Which tables are they contained in?
4. **Determine** the criteria the information in each field needs to meet. Think about the question you asked in the first step. Which fields do you need to search for specific information? What information are you looking for? How will you search for it?

This process might seem abstract at first, but as we go through the process of planning our own multi-table query you should start to understand how planning your queries can make building them a lot easier.

Planning our query

Let's go through this planning process with a query we'll run on our bakery database. As you read through the planning process step by step, think about how each part of the planning process could apply to other queries you might run.

Step 1: Pinpointing the question we want to ask

Our bakery database contains many customers, some of whom have never placed an order but who are in our database because they signed up for our mailing list. Most of them live within the city limits, but others live out of town or even out of state. We want to get our out-of-town customers who've placed orders in the past to come back and give us another try, so we're going to mail them some coupons. We don't actually want our list to include customers who live too far away; sending a coupon to someone who doesn't live in our area probably won't make that person come in. So we just want to find people who don't live in our city but who still live in our area.

In short, the question we want our query to answer is this: Which customers live in our area, are outside the city limits, and have placed an order at our bakery?

Step 2: Identifying the information we need

What information might we want to see in a list about these customers? Obviously, we'll need the customers' names and their contact information—their addresses, phone numbers, and email addresses. But how are we going to know if they've placed orders? Each record of an order identifies the customer who placed that order. If we include the order ID numbers, we should be able to narrow our list down to only customers who have previously placed orders.

Step 3: Locating the tables containing the information we need

In order to write a query, you need to be familiar with the different tables in your database. From working extensively with our own database, we know that the customer information we need is located in fields in the Customers table. Our Order ID numbers are in a field in the Orders table. We only need to include these two tables to find all of the information we need.

Step 4: Determining the criteria our query should search for

When you set criteria for a field in a query, you are basically applying a filter to it that tells the query to retrieve only information that matches your criteria. Review the list of fields we are including in this query. How and where can we set criteria that will best help us answer our question? We don't want customers who live in our town, Raleigh, so we want a criterion that will return all records except for those with Raleigh in the city field. We don't want customers who live too far away, either. All of the phone numbers in the area start with the 919

area code, so we'll also include criteria that will only return records whose entries from the phone number field begin with 919. This should guarantee that we'll only send coupons to customers who live close enough to actually come back and use them.

We won't set criteria for the order ID field or any other fields because we want to see all of the orders made by people who meet the two criteria we just set.

To write queries, you'll need to be able to set criteria in a language that Access **understands**. As you can see in the image above, our criteria requiring phone numbers to begin with 919 must be typed like this: **Like ("919*")**.

Joining tables in queries

The final thing you need to consider when designing a query is the way you link—or **join**—the tables you're working with. When you add two tables to an Access query, this is what you'll see in the **Object Relationship pane**:

The line connecting the two tables is called the **join line**. See how the join line is actually an arrow? This is because it indicates the order in which the query looks at data from the two tables. In the image above, the arrow is pointing from **left** to **right**, which means the query will look at data in the **left** table first, then look at only the data in the **right** table that **relates** to the records it's already seen in the left table.

Your tables won't always be joined this way. Sometimes Access will join them **right to left**. In either case, you might need to **change the direction** of the join to make sure your query includes the correct information. The join direction can affect **which information** your query **retrieves**.

To understand what this means, consider the query we're designing. For our query, we need to see customers who have placed orders, so we've included the **Customers** table and the **Orders** table. Let's take a look at some of the data contained in these tables.

What do you notice when you look at these lists? First of all, every single order in the **Orders** table is linked to someone in the **Customers** table—the customer who placed that order. However, when you look at the Customers table, you'll see that the customers who've placed multiple orders are linked to more than one order, and those who've never placed an order are linked to no orders. As you can see, even when two tables are linked it's possible to have records in one table that have no relationship to any record in the other table.

So what happens when Access tries to run our query with the current join, **left to right**? It pulls every record from the table to the left: our Customers table.

It then retrieves every record from the **right** table that has a relationship with a record Access has already taken from the left table. Because our join began with the **Customers** table, our query will include records for **all** of our customers, including those who've never placed orders. This is more information than we need. We **only** want to see records for **customers who have placed orders**. Fortunately, we can fix this problem by changing the direction of the join line. If we join the tables from **right to left** instead, Access will first retrieve the orders from the **right** table, our **Orders** table: Then Access will look at the left table and retrieve **only** the records of customers who are linked to an order on the right. We now have exactly the information we want: **all** of the customers who have placed an order, and **only** those customers. As you can see, we had to join our tables in the **correct direction** to obtain the information we wanted. Now that we understand which join direction we need to use, we're ready to build our query!

In our query, we needed to use the **right-to-left** join, but the correct join direction for the tables in your queries will depend on **what** information you want to see and **where** that information is stored. When you add tables to a query, Access will automatically join the tables for you, but it often doesn't join them in the correct direction. This is why it's

important to **always review the joins** between your tables before you build a query.

Creating a multi-table query

Now that we've planned our query, we're ready to design and run it. If you have created written plans for your query, be sure to reference them often throughout the query design process.

To create a multi-table query:

1. Select the Query Design command from the Create tab on the Ribbon.
2. In the dialog box that appears, select each table you want to include in your query and click **Add**. You can press and hold the **Ctrl** key on your keyboard to select more than one table. When we planned our query, we decided we needed information from the **Customers** and **Orders** tables, so we'll add these.
3. After you have added all of the tables you want, click **Close**.
4. The tables will appear in the **Object Relationship pane**, linked by a **join line**. Double-click the thin section of the join line between two tables to edit its **join direction**.
5. The **Join Properties** dialog box will appear. Select an option to choose the direction of your join. In our example, we'll choose option 3 because we want a right-to-left join.
6. In the table windows, double-click the **field names** you want to include in your query. They will be added to the **design grid** in the bottom part of the screen. In our example, we'll include most of the fields from the **Customers** table: **First Name**, **Last Name**, **Street Address**, **City**, **State**, **Zip Code**, and **Phone Number**. We'll also include the **ID number** from the **Orders** table.
7. Set field **criteria** by entering the desired criteria in the criteria row of each field. We want to set two criteria: **Not in ("Raleigh")** in the **City** field, and **Like ("919*")** in the **Phone Number** field. This will find customers who do not live in Raleigh but who do live in the 919 area code.
8. After you have set your criteria, **run** the query by clicking the **Run** command on the **Design** tab.
9. The query results will be displayed in the query's **Datasheet view**, which looks like a table. If you want, **save** your query by clicking the **Save** command in the Quick Access Toolbar. When prompted to name it, type the desired name, then click **OK**.

Now you know how to create a **multi-table** query. In the next lesson, we'll talk about more query design options that can make your query even more powerful.

More Query Design Options

Access offers several options that let you design and run queries that return exactly the

information you're looking for. For instance, what if you need to find out how many of something exists within your database? Or what if you would like your query results to automatically be sorted a certain way? If you know how to use query options in Access, you can design almost any query you want.

In this lesson, you'll learn how to modify and sort your queries within Query Design view, as well as how to use the Totals function to create a query that can perform calculations with your data. You'll also learn about additional query-building options offered in Access.

Modifying queries

Access offers several options for making your queries work better for you. In addition to modifying your query criteria and joins after you build your queries, you can choose to sort and hide fields in your query results.

To modify your query:

When you open an existing query in Access, it is displayed in Datasheet view, meaning you will see your query results in a table. To modify your query, you must enter Design view, the view you used when creating it. There are two ways to switch to Design view:

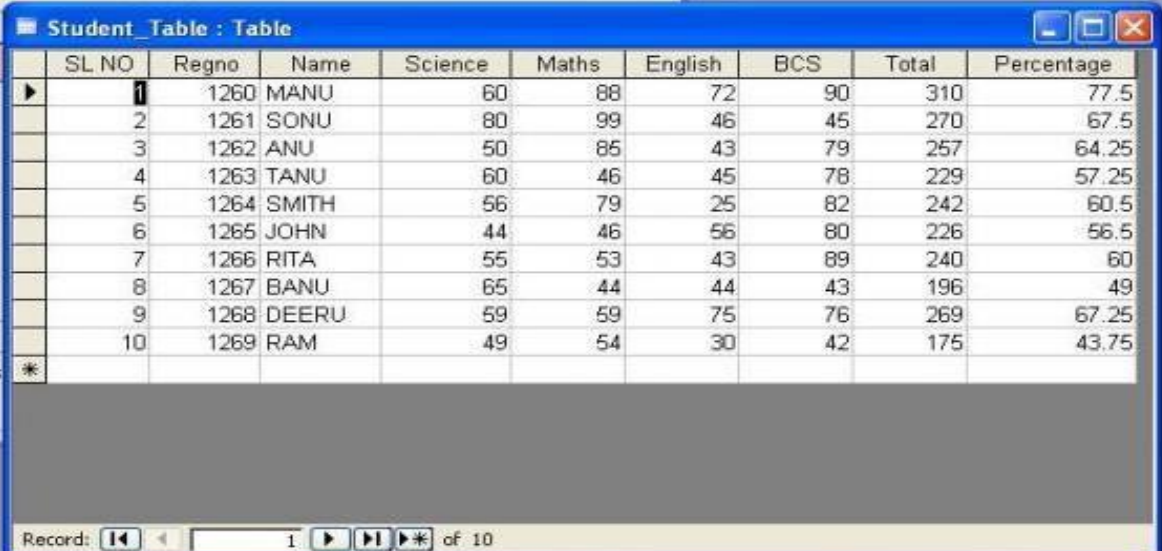
- On the Home tab of the Ribbon, click the View command. Select Design View from the drop-down menu that appears.
- In the bottom-right corner of your Access window, locate the small **view icons**. Click the **Design View** icon, which is the icon farthest to the right.

Once in **Design view**, make the desired changes, then select the **Run** command to view your updated results.

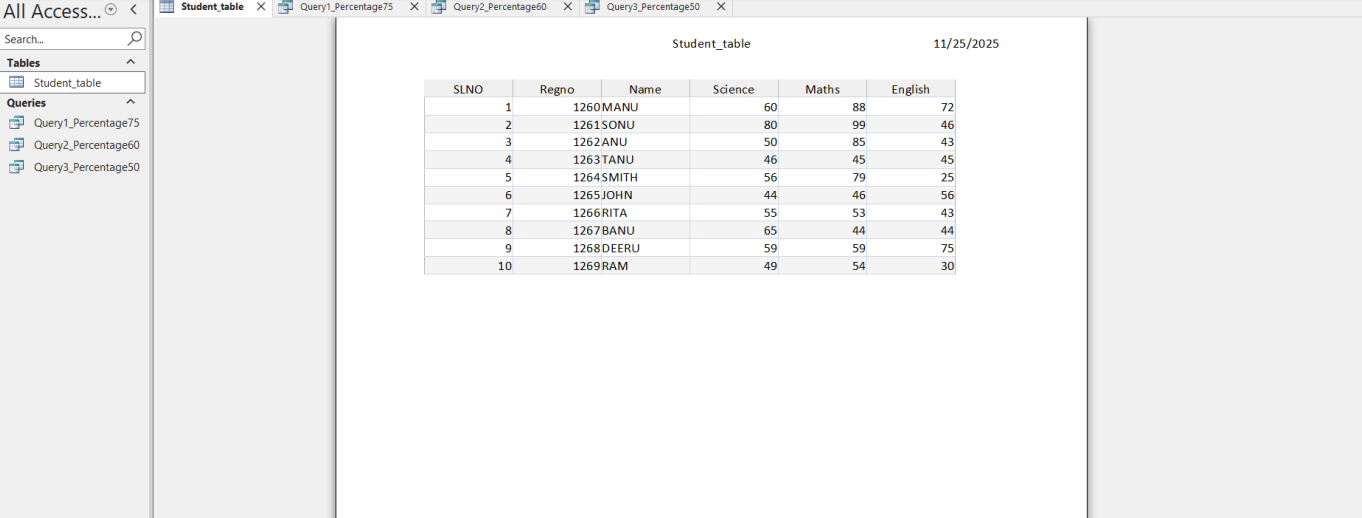
You may notice that Access also offers **SQL view**. You can ignore this. SQL view allows you to create advanced functions that you will not need to use for this tutorial or for most Access functions.

EXERCISE:

Create a Table mention below and apply following Queries:



SL NO	Regno	Name	Science	Maths	English	BCS	Total	Percentage
1	1260	MANU	60	88	72	90	310	77.5
2	1261	SONU	80	99	46	45	270	67.5
3	1262	ANU	50	85	43	79	257	64.25
4	1263	TANU	60	46	45	78	229	57.25
5	1264	SMITH	56	79	25	82	242	60.5
6	1265	JOHN	44	46	56	80	226	56.5
7	1266	RITA	55	53	43	89	240	60
8	1267	BANU	65	44	44	43	196	49
9	1268	DEERU	59	59	75	76	269	67.25
10	1269	RAM	49	54	30	42	175	43.75



SLNO	Regno	Name	Science	Maths	English
1	1260	MANU	60	88	72
2	1261	SONU	80	99	46
3	1262	ANU	50	85	43
4	1263	TANU	46	45	45
5	1264	SMITH	56	79	25
6	1265	JOHN	44	46	56
7	1266	RITA	55	53	43
8	1267	BANU	65	44	44
9	1268	DEERU	59	59	75
10	1269	RAM	49	54	30

- Select * from Student_table where Percentage>=75.

Query1_Percentage75

11/25/2025

SLNO	Regno	Name	Science	Maths	English
1	1260	MANU	60	88	72

- Select * from Student_table where Percentage=60.

Query2_Percentage60

11/25/2025

SLNO	Regno	Name	Science	Maths	English
7	1266	RITA	55	53	43

- Select Regno, Name from Student_table where Percentage<=50.

Query3_Percentage50

11/25/2025

Regno	Name
1267	BANU
1269	RAM

LAB SESSION 12

Creating Reports in MS Access

THEORY:

Creating Reports

1. If you need to share information from your database with someone but don't want that person actually working with your database, consider creating a report. Reports allow you to organize and present your data in a reader-A dialog box will appear. Select the **location** where you want to save the report.
2. Enter a **file name** for the report, then click **Publish**.
3. A dialog box will appear to notify you that your file has been successfully saved. Click **Close** to return to your report.

Advanced Report Options

Access offers several **advanced options** for creating and modifying reports. The **Report Wizard** is a tool that guides you through the process of creating complex reports. Once you've created a report—whether through the Report Wizard or the Report command—you can then **format** it to make it look exactly how you want.

In this lesson, you'll learn how to use the **Report Wizard** to create complex reports. You'll also learn how to use **formatting options** to **format text**, **change report colors** and **fonts**, and **add a logo**.

The Report Wizard

While using the **Report** command is a quick way to create reports from the current object, it's not as helpful if you want to create a report with data from multiple objects. The **Report Wizard** makes it easy to create reports using fields from multiple tables and queries. It even lets you choose how your data will be organized.

To create a report with the Report Wizard:

1. Select the **Create** tab and locate the **Reports** group. Click the **Report Wizard** command.
1. Select either a **portrait** (tall) or **landscape** (wide) orientation for your report.
2. Once you are satisfied with your report layout, click **Next**.
3. Select the text box, and type the **title** you want for your report.
4. Select whether you want to **preview** the report or **modify** its design, then click **Finish**.
5. Your report will be created and saved.

You may have to adjust your field and row **size** and **location** to make sure your data looks the way you want it to. To do this, you'll need to switch to **Design view**. When you're done, switch back to **Report view** to see your changes.

Formatting reports

One of the strengths of reports is that you can modify their appearance to make them look how you want. You can add headers and footers, apply new colors, and even add a logo. All of these things can help you create visually appealing reports.

Modifying report text

The bulk of the information in your report comes straight from the query or table you built it from, which means you can't edit it within the report. However, you can change, add, or delete label text, headers, and footers to make your report clearer and easier to read. For example, in our report we decided we didn't need the field headings to understand our data, so we simply **deleted** them.

Just like other Office programs, Access allows you to modify the text color and font, add shapes, and more.

Depending on your report's design, sometimes you may find that there is no white space in the page header and footer, as in the

image above. If this is the case, you must **resize** the header and footer before you can add anything to them. Simply click and drag the bottom border of the header or footer to make it larger.

Modifying the page header and footer

To view and modify the header and footer that appear on each page of your report, select the View command on the Ribbon and switch to Design view. The header and footer are located in the white space beneath the Page Header and Page Footer bars.

Depending on your report's design, sometimes you may find that there is no white space in the page header and footer, as in the image above. If this is the case, you must resize the header and footer before you can add anything to them. Simply click and drag the bottom border of the header or footer to make it larger.

To add text to a header or footer:

1. Select the Design tab, locate the Controls group, and click the Label command.
2. Click and drag the mouse inside the white area to create your label. Release the mouse when it is the desired size.
3. Click the text box, and type the desired text.

By default, the date and time appear in the **header**. If you would like to move them to the **footer** instead, simply click the date and time boxes and drag them to the desired location.

To add page numbers to a header or footer:

1. Select the **Design** tab, then locate the **Header/Footer** group.
2. Click the **Page Numbers** command.
3. The **Page Numbers** dialog box will appear. Under **Format**, choose **Page N** to display the number of only the current page, or **Page N of M** to display the number of the current page and the number of total pages.
4. Under **Position**, choose **Top of Page** or **Bottom of Page** to control where the page numbers appear.
5. Click the drop-down arrow to select the **alignment** of the page numbers.
6. When you are satisfied with the settings, click **OK**.

Modifying your report's appearance

To add a logo:

1. From the **Design** tab, click the **View** command, then select **Layout View** from the drop-down list.
2. A small version of the image will appear in the header. Click and drag the image border to resize it.
3. If necessary, move your logo to the desired location by clicking and dragging it.

Themes and fonts

A theme is a set of colors and fonts that applies to the entire database to give it a consistent, professional look. By default, databases use the Office theme. When you change the theme, all of the theme fonts and colors in your database change to match the new theme. Designing and modifying reports using theme elements can help you keep the appearance of your reports consistent.

To change the theme:

1. Select the Design tab, locate the Themes group, and click the Themes command.
2. A drop-down menu will appear. Select the desired theme.

To change the theme fonts:

1. Select the **Design** tab, locate the **Themes** group, and click the **Fonts** command.
2. A drop-down menu will appear. Select a set of **theme fonts**.

3. The fonts will be applied to your entire database.

Database Design Tips

After working with your database, you might find that you need to make some changes to the tables that store your data. Access makes it easy to modify your tables to suit your database's needs.

In this lesson, you'll learn how to **create** and **rearrange** table fields. You'll also learn how to ensure your table data is correctly and consistently formatted by setting **validation rules**, **character limits**, and **data types** in your fields. Finally, we will direct you to additional options for performing simple math functions within your tables.

Modifying tables

In addition to making basic modifications to your tables, like **adding** and **moving** fields, you can make more advanced modifications that let you set **rules** for your data. All of these changes can help make your tables even more useful.

Adding and rearranging fields

Access makes it easy to rearrange existing fields and add new ones. When you add a new

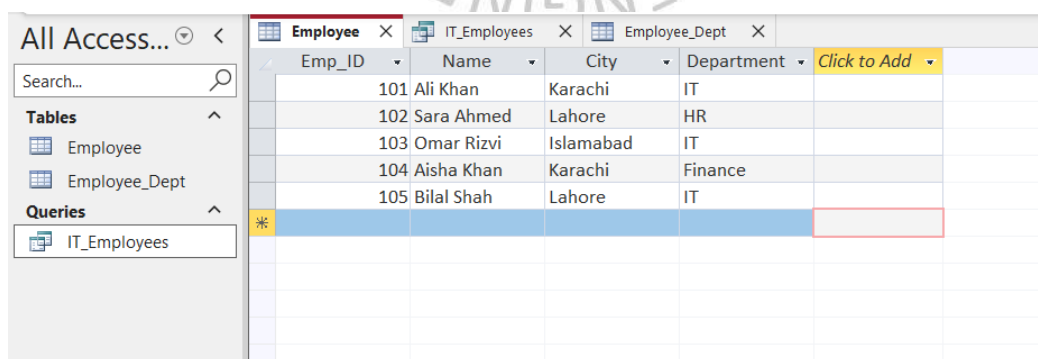
field, you can even set the **data type**, which dictates which **type** of data can be entered into that field.

There are several types of fields you can add to a table:

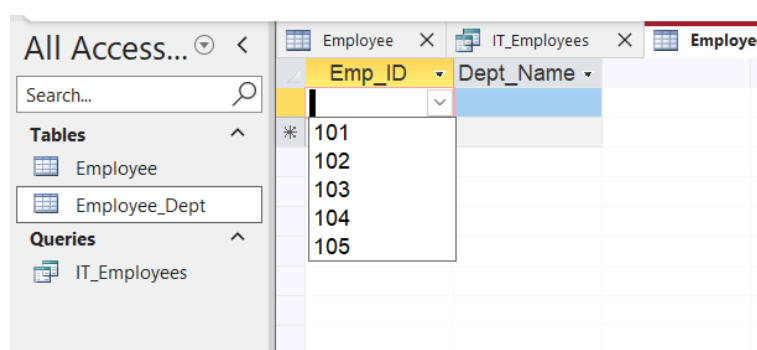
- **Short Text:** This is the default option and is best for most text in Access. You should also choose it for numbers you don't plan to do math with, like postal codes and phone numbers.

EXERCISE:

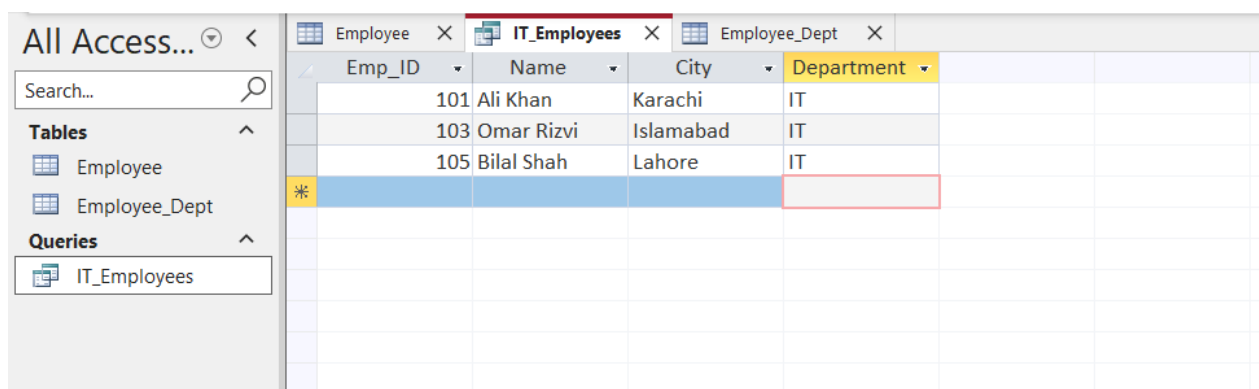
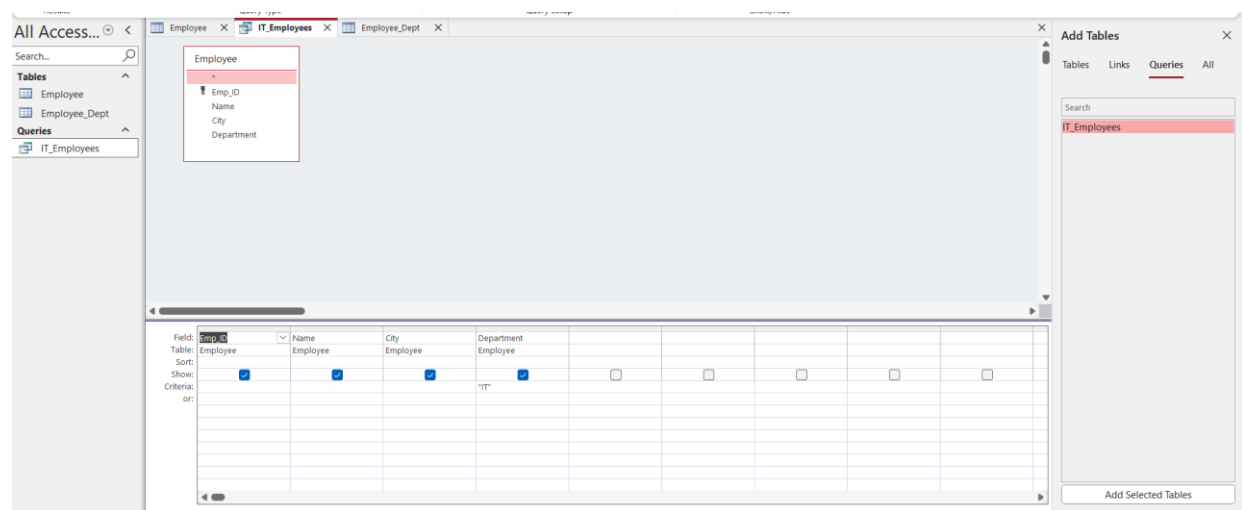
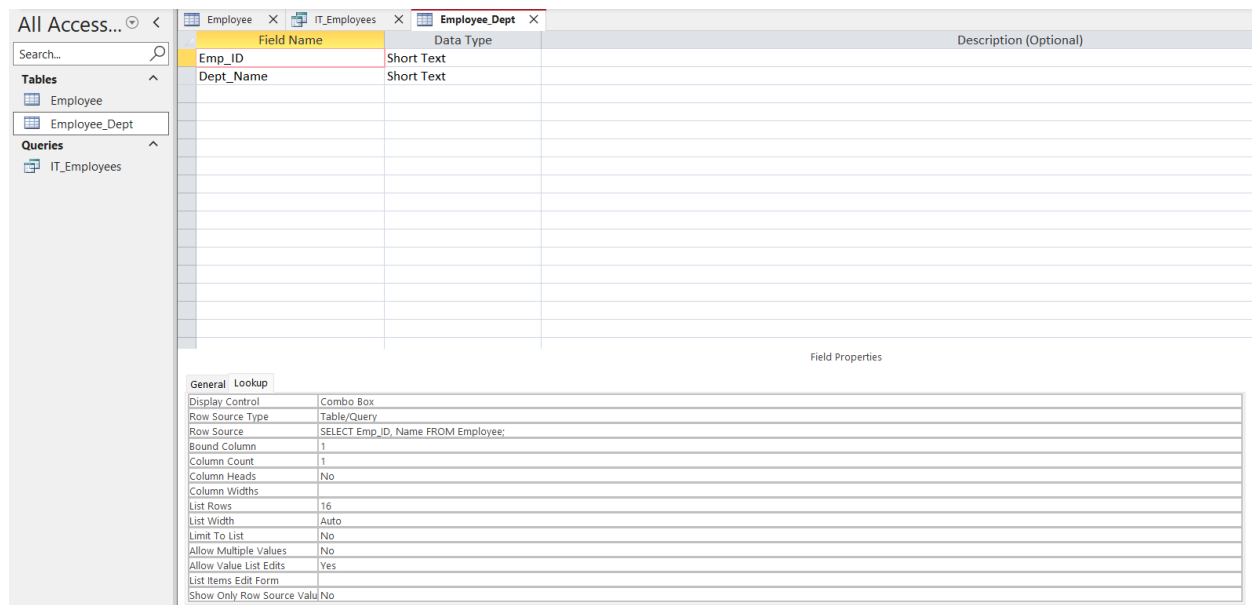
- Create a Query to extract information about Emp_ID, Name, City and Department of an employee where department is "IT".
- For Department create a new table Employee_Dept of two columns: Emp_ID and Dept_Name.
- "Emp_ID" column values are being taken from the EMPLOYEE table using Lookup Column technique.



Emp_ID	Name	City	Department	Click to Add
101	Ali Khan	Karachi	IT	
102	Sara Ahmed	Lahore	HR	
103	Omar Rizvi	Islamabad	IT	
104	Aisha Khan	Karachi	Finance	
105	Bilal Shah	Lahore	IT	



Emp_ID	Dept_Name
101	
102	
103	
104	
105	



LAB SESSION 13

CEA

CLO Covered

CLO #: 4

CPA-# 1

CPA-# 2

CPA-# 4

Problem Statement

Generation of multi tables queries relationships and report of database.

Guidelines and Evaluation Criteria Overview

Submission Details:

Criteria	Supreme (2)	Average (1)	Not Achieved (0)	Marks
	Able to correctly identify the problem(s) and all required data (input and output)	Able to correctly identify the problem(s) and all required data (input and output) on average	Not able to correctly identify the problem(s) and all required data (input and output).	
Problem Solving Skills	Uses an effective strategy to solve the problem(s) and clearly identifies the steps	Uses an effective strategy to solve the problem(s) and clearly identifies the steps on average	Uses an ineffective strategy to solve the problem(s) and completely misses the steps	
Implementation & Debugging	Efficient implementation and excellent debugging skills. Solution works without any error	Efficient implementation and excellent debugging skills. Solution works without any error on average	Incomplete implementation and ineffective debugging skills. Major errors in the implementation.	
Deliverable(s)	Generates correct and complete output.	Generates correct and complete output on average	Cannot generate the desired output.	
Depth of Knowledge	Thorough grasp of the concepts covered in the lab and can apply it in real-life situations	Thorough grasp of the concepts covered in the lab and can apply it in real-life situations on average	Does not grasp the concepts covered in the lab and cannot identify any real-life application	
Weighted CLO Score				
Remarks and Signature				

LAB SESSION 14

OBJECTIVE:

Creating Forms

Creating **forms** for your database can make entering data more convenient. When you create a form, you can design it in a way that works with your database and that makes sense to you.

In this lesson, you will learn how to **create** and **modify** forms. You'll also learn how to use form options like **design controls** and **form properties** to make sure your form works exactly the way you want.

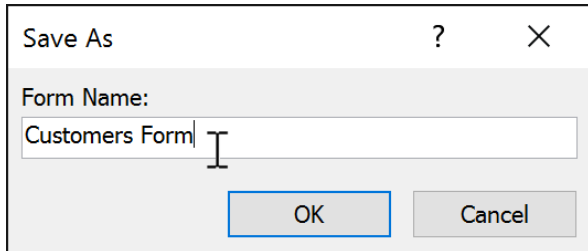
To create a form:

Access makes it easy to create a form from any table in your database. Any form you create from a table will let you view the data that's already in that table and add new data. Once you've created a form, you can modify it by adding additional fields and design controls like combo boxes.

1. In the Navigation pane, select the table you want to use to create a form. You do not need to open the table.
2. Select the **Create** tab, locate the **Forms** group, and click the **Form** command.
3. Your form will be created and opened in **Layout view**.

ID	Paid	Pre Order	Notes	Pickup Date
38	Yes	No		12/24/13
(New)	No	No		

4. To **save** the form, click the **Save** command on the **Quick Access Toolbar**. When prompted, type a **name** for the form, then click **OK**.

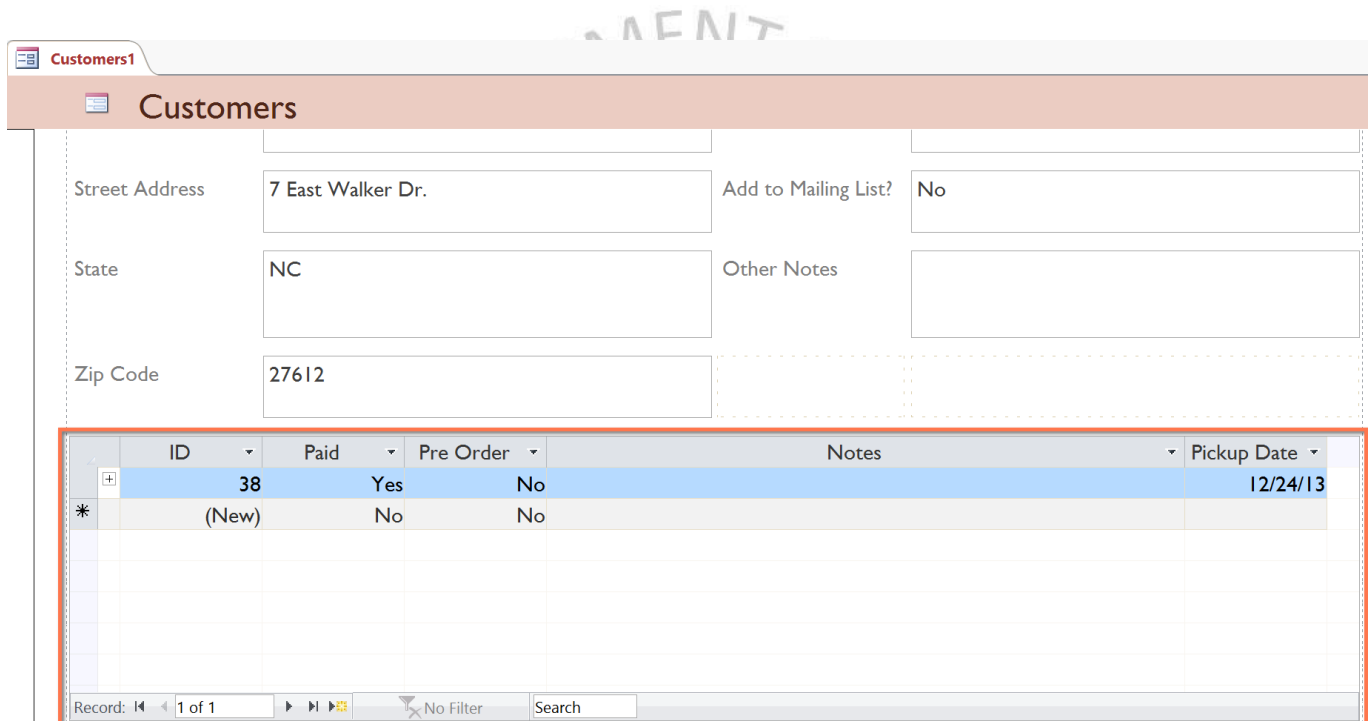


A 'Save As' dialog box with a title bar containing a question mark and a close button. The 'Form Name:' label is above a text box containing 'Customers Form'. Below the text box are 'OK' and 'Cancel' buttons.

About subforms

If you created a form from a table whose records are linked to another table, your form probably includes a **subform**. A subform is a **datasheet form** that displays linked records in a table-like format. For instance, the subform included in the **Customers** form we just created displays linked customer **orders**.

We probably don't need to include this subform because we just want to use the Customers form to enter and review contact information. If you find that you don't need a subform, you can easily **delete** it. To do this, simply click it and press the **Delete** key on your keyboard.



The 'Customers' form is shown with a tab labeled 'Customers1'. It contains fields for 'Street Address' (7 East Walker Dr.), 'State' (NC), 'Zip Code' (27612), 'Add to Mailing List?' (No), and 'Other Notes'. Below these fields is a subform displaying a table of customer orders. The table has columns for ID, Paid, Pre Order, Notes, and Pickup Date. The first row is highlighted in blue and shows ID 38, Paid Yes, Pre Order No, and Pickup Date 12/24/13. A second row is marked with an asterisk and shows '(New)', No, No, and an empty date field. The table is enclosed in a red dashed border.

ID	Paid	Pre Order	Notes	Pickup Date
38	Yes	No		12/24/13
*(New)	No	No		

However, subforms are often quite helpful. Depending on the content and source of your form, you might find that the subform contains useful information, like in the example below. In our **Orders** form, the subform contains the name, quantity, and price of each item contained in the order, which is all useful information.

New Order

Customer
Whitman
Order #
5
Pickup Date
12/4/13

Notes

☐ Pre Order
☒ Paid

Add Item

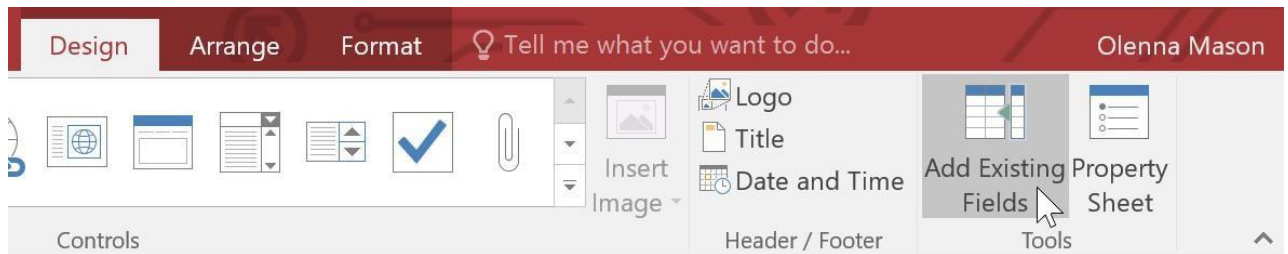
Category ▾	Product ▾	Quantity ▾	"Unit" ▾	Price ▴	Subtotal ▾
Cookies	Chocolate Chip	2	Single	\$1.50	\$3.00
Cookies	Fudge Brownie	1	Single	\$2.00	\$2.00
Cookies	Ginger Shortbread	1	Half-Dozen	\$10.50	\$10.50
Pastries	Brownies	1	One Dozen	\$19.00	\$19.00
Cakes	Black Forest	5	Single	\$22.00	\$110.00
Cakes	Coconut	2	Single	\$22.00	\$44.00
Cakes	Carrot Cake	1	Single	\$22.00	\$22.00

Adding additional fields to a form

When you use the **Form** command on an existing table, all of the fields from the table are included in the form. However, if you later add additional fields to the table, these fields will **not** automatically show up in existing forms. In situations like this, you can **add** additional fields to a form.

To add a field to a form:

1. Select the **Design** tab, then locate the **Tools** group on the right side of the Ribbon.
2. Click the **Add Existing Fields** command.



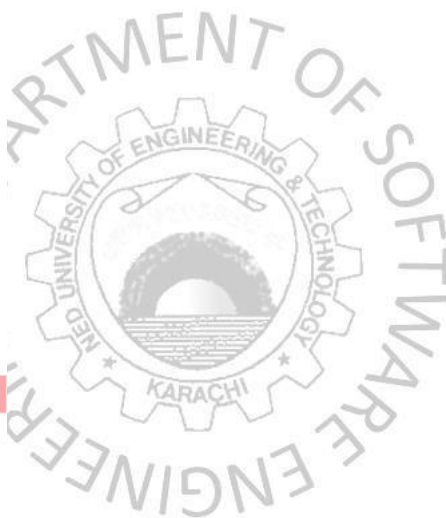
3. The **Field List** pane will appear. Double-click the desired field(s).

Field List

Show all tables

Fields available for this view:

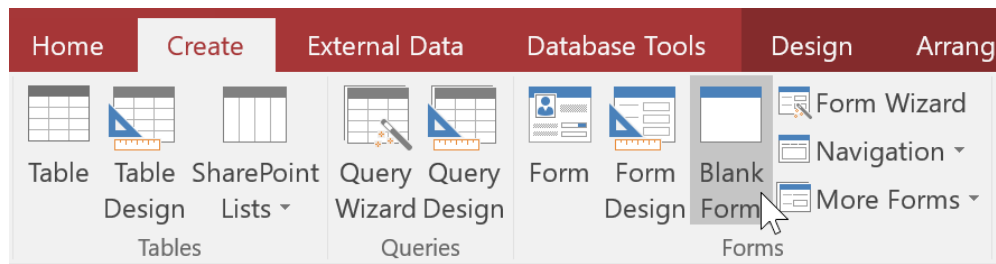
ID
First Name
Last Name
Street Address
State
Zip Code
Email
Phone Number
City
Add to Mailing List?
Other Notes



4. The field will be added.

Customers			
ID	I	Email	beck@email.com
Other Notes			
First Name	Tracey	Phone Number	919-555-2314
Last Name	Beckham	City	Raleigh

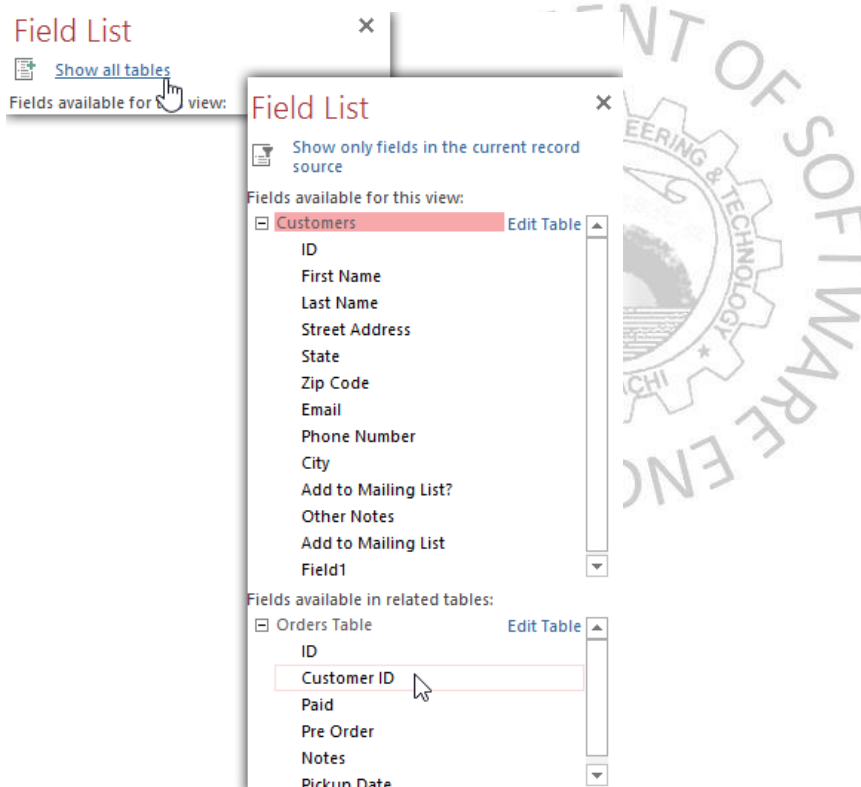
You can also use the above procedure to add fields to a totally blank form. Simply **create a form** by clicking the **Blank Form** command on the **Create** tab, then follow the above steps to add the desired fields.



To add a field from a different table:

You can also add fields from **different tables** in your database to the form.

1. From the **Field List** pane, click **Show All Tables**.
2. Click the plus sign + next to the table that contains the field you want to add, then double-click the desired field. The new field will be added.



Adding design controls

Design controls set restrictions on the fields in your forms. This helps you better control how the data is entered into your forms, which in turn helps keep the database consistent.

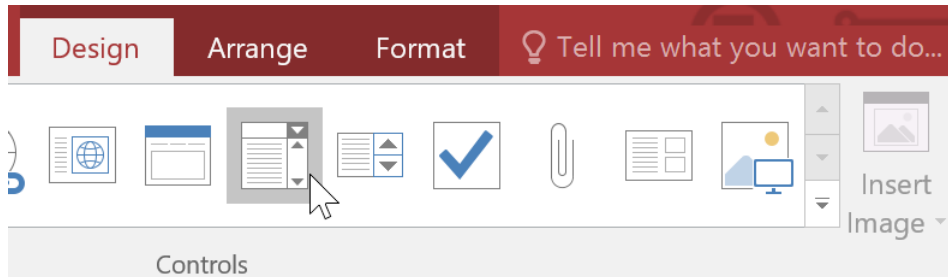
Combo boxes

A **combo box** is a drop-down list you can use in your form in place of a field. Combo boxes **limit** the information users can enter by forcing them to select only the **options** you have specified.

Combo boxes are useful for fields that have a limited number of possible valid responses. For instance, you might use a combo box to make sure people only enter a valid U.S. state while entering an address, or that they only choose products that already exist in your database while placing an order.

To create a combo box:

1. In **Form Layout** view, select the **Design** tab, then locate the **Controls** group.
2. Select the **Combo Box** command, which looks like a drop-down list.



3. Select the desired location for the combo box. A line will appear to indicate the location where your combo box will be created. In our example, we'll place it between the **City** field and the **Add to Mailing List?** fields.

Phone Number

919-555-2314

City

Raleigh

Add to Mailing List?

No



4. The **Combo Box Wizard** dialog box will appear. Select **I will type in the values that I want**, then click **Next**.

6. If necessary, **resize** the column so all of your text is visible. Once you are satisfied with your list, click **Next**.

Combo Box Wizard

What values do you want to see in your combo box? Enter the number of columns you want in the list, and then type the values you want in each cell.

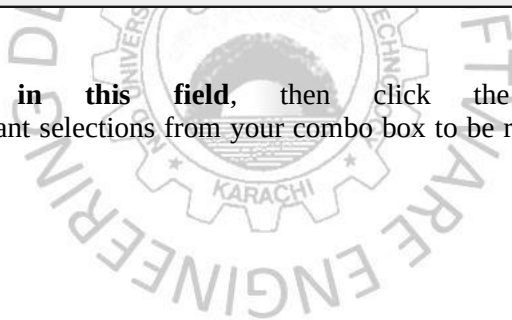
To adjust the width of a column, drag its right edge to the width you want, or double-click the right edge of the column heading to get the best fit.

Number of columns:

	Col1			
	No			
	Yes - Weekly			
	Yes - Special Events and Offers			
*				

Cancel < Back Next > Finish

7. Select **Store that value in this field**, then click the drop-down arrow and **select** the **field** where you want selections from your combo box to be recorded. After making your selection, click **Next**.



Combo Box Wizard



Microsoft Access can store the selected value from your combo box in your data base, or remember the value so you can use it later to perform a task. When you select a value in your combo box, what do you want Microsoft Access to do?

Q Remember the value for later use.

@ Store that value in this field:

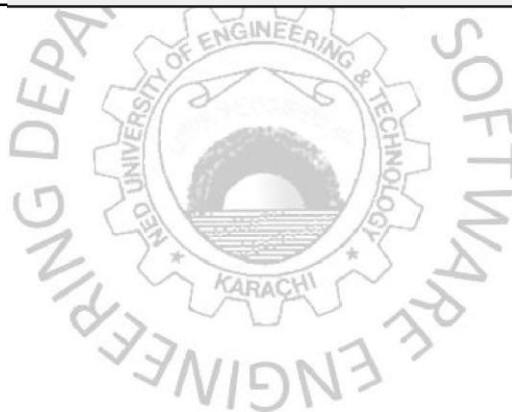
ID
First Name
Last Name
Street Address
State
Zip Code
Phone Number
City
Add to Mailing List?

Cancel

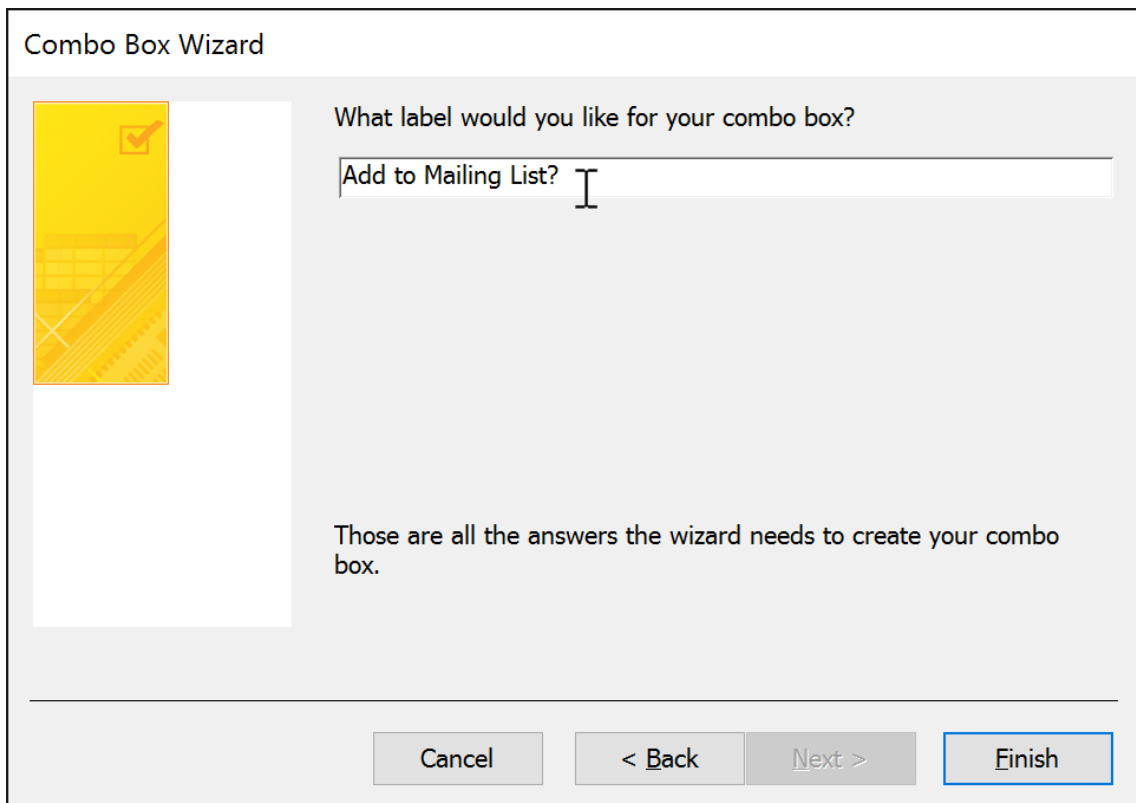
Back

Next >

Finish



8. Enter the **label**—or **name**—that will appear next to your combo box. Generally, it's a good idea to use the name of the field you chose in the previous step.



Combo Box Wizard

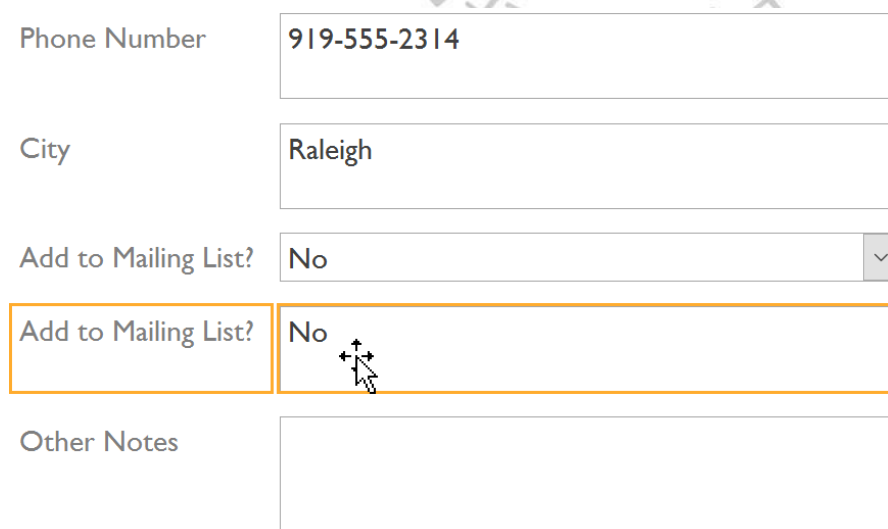
What label would you like for your combo box?

Add to Mailing List?

Those are all the answers the wizard needs to create your combo box.

Cancel < Back Next > Finish

9. Click **Finish**. Your combo box will appear on the form. If you created your combo box to **replace** an existing field, you should **delete** the first field. In our example, you might notice that we now have two fields with the same name. These two fields send information to the same place, so we don't need them both. We'll **delete** the one without the combo box.



Phone Number 919-555-2314

City Raleigh

Add to Mailing List? No

Add to Mailing List? No

Other Notes

10. Switch to **Form** view to **test** your combo box. Simply click the drop-down arrow and verify that the list contains the correct choices. The combo box can now be used to enter data.

Phone Number 919-555-2314

City Raleigh

Add to Mailing List? No

Other Notes

Yes - Weekly

Yes - Special Events and Offers

If you want to include a drop-down list with a long list of options and don't want to type all of them out, create a combo box and choose the first option in the Combo Box Wizard: **I want to get the values from another table or query.** This will allow you to create a drop-down list from a table field.

Combo Box Wizard

This wizard creates a combo box, which displays a list of values you can choose from. How do you want your combo box to get its values?

☒ I want the combo box to get the values from another table or query.

☐ I will type in the values that I want.

☐ Find a record on my form based on the value I selected in my combo box.

Search

Other Notes

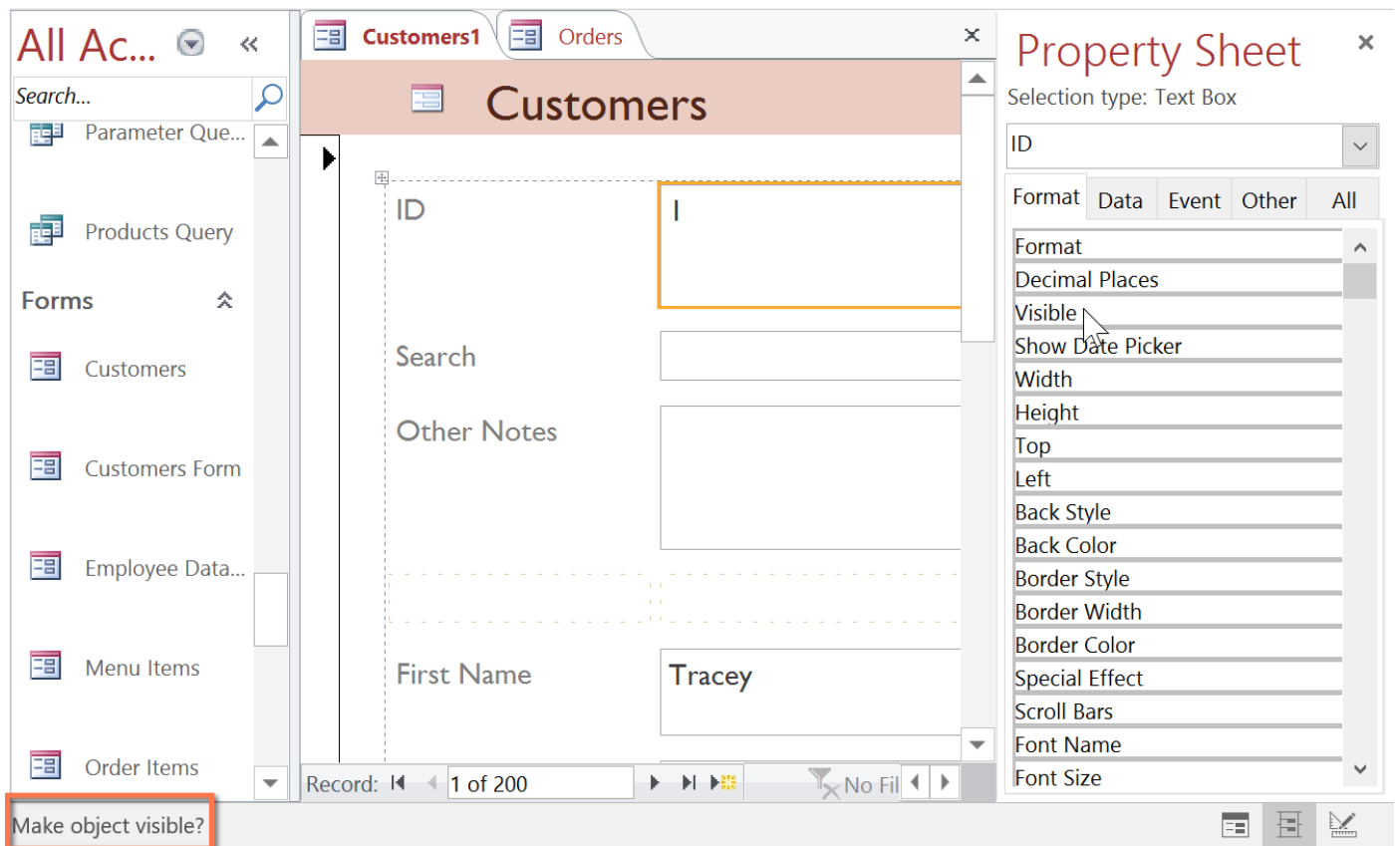
First Name

Tracey	Beckham	919-555-2314
Lucinda	George	919-555-4534
Jerrold	Smith	919-555-4564
Brett	Newkirk	919-555-7653
Chloe	Jones	919-555-8658
Quinton	Boyd	919-555-5112
Alex	Hinton	919-555-5460
Nisha	Hall	919-555-5753

Some users report that Access malfunctions while working with forms. If you have a problem performing any of these tasks in Layout view, try switching to Design view.

Customizing form settings with the Property Sheet

The **Property Sheet** is a pane containing **detailed information** on your form and each of its components. From the Property Sheet, you can make changes to every part of your form, both in terms of function and appearance. The best way to familiarize yourself with the Property Sheet is to **open** it and **select** various options. When you select an option, Access will display a brief description of the option on the **bottom-left border** of the program window.



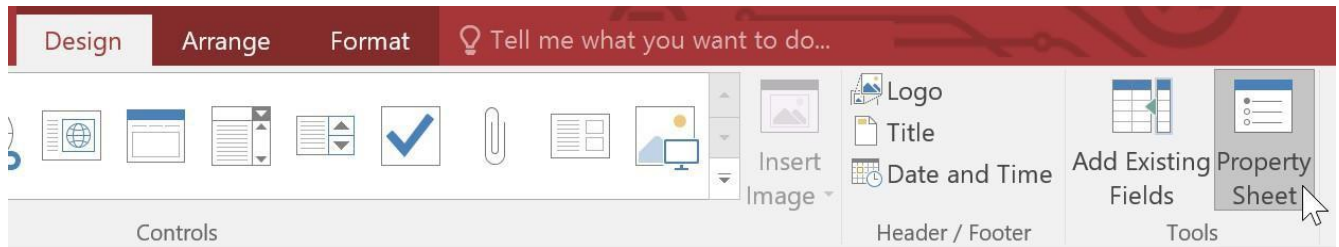
Pay close attention as you modify your form and its fields. It's possible to make subtle changes with the Property Sheet. Because there are so many options, it can sometimes be difficult to remember which one you used to modify each aspect of your form.

Modifying form settings

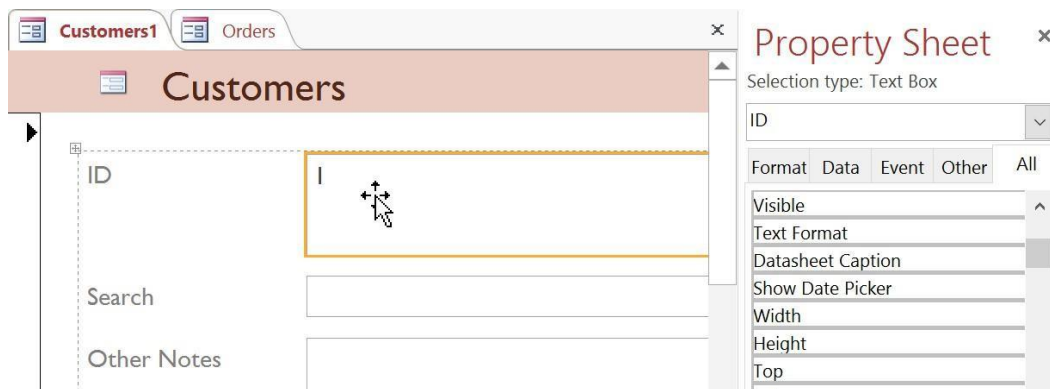
There are far too many options in the Property Sheet to discuss them all in detail. We'll review two useful ones here: **hiding** fields and setting fields with **dates** to **automatically fill in the current date**. Practicing these procedures should also give you a sense of how to work with other Property Sheet settings.

To hide a field:

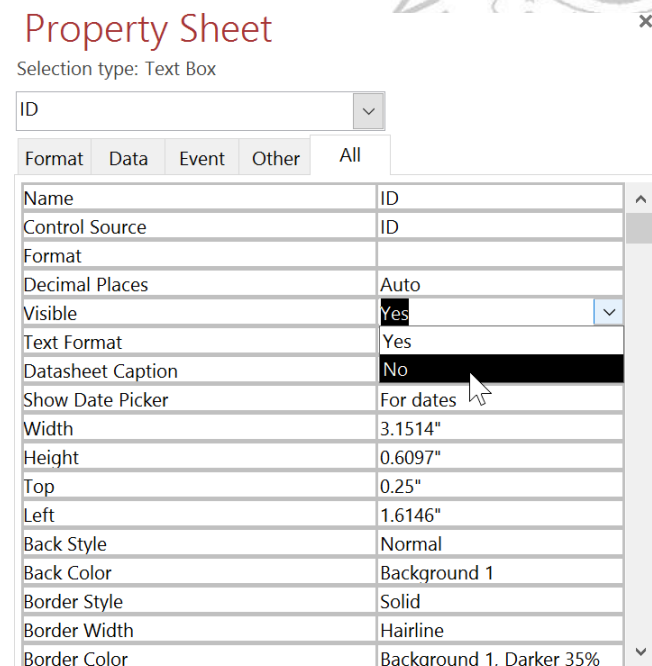
1. In either **Layout** or **Design** view, select the **Design** tab, then locate the **Tools** group. Click the **Property Sheet** command.



2. The **Property Sheet** will appear in a pane on the right. On the form, **select** the field you want to hide. In our example, we'll hide the **Customer ID** field because we don't want any of our users to try to edit it.



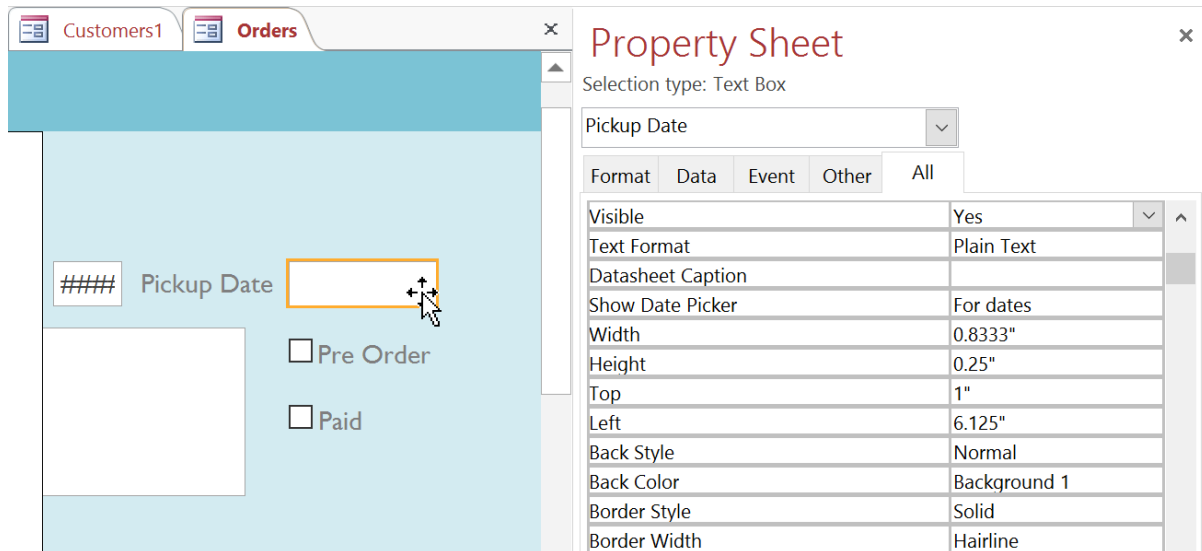
3. In the **Property Sheet**, click the **All** tab, then locate the **Visible** option on the fifth row.
4. Click the drop-down arrow in the column to the right, then select **No**.



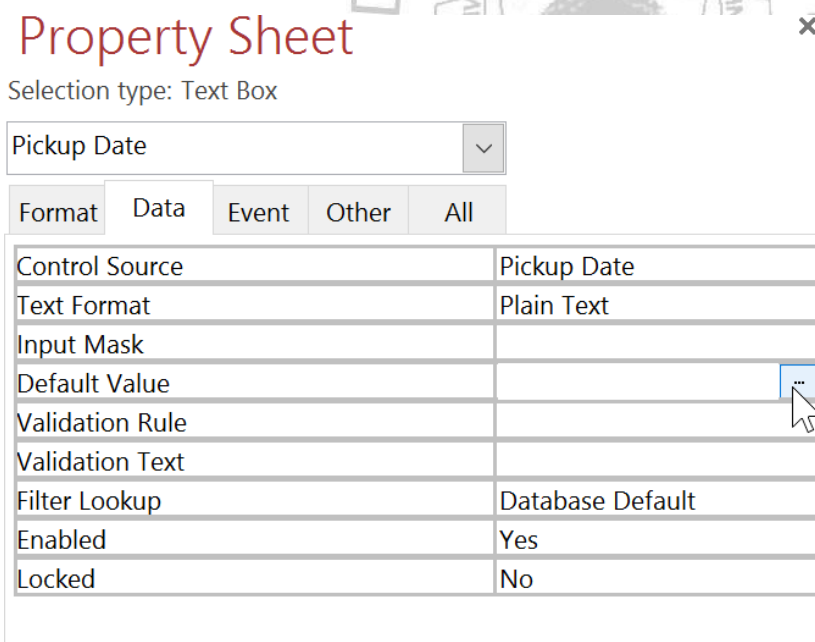
5. Switch to **Form** view to verify that the field is hidden.

To set a field to autofill with the current date:

1. In either **Layout** or **Design** view, select the **Design** tab, then locate the **Tools** group. Click the **Property Sheet** command.
2. The **Property Sheet** will appear in a pane on the right. On the form, **select** the field you want to automatically fill in the current date. This **must** be a field with the **date** data type. For our example, we'll select the **Pickup Date** field on our **Orders** form.

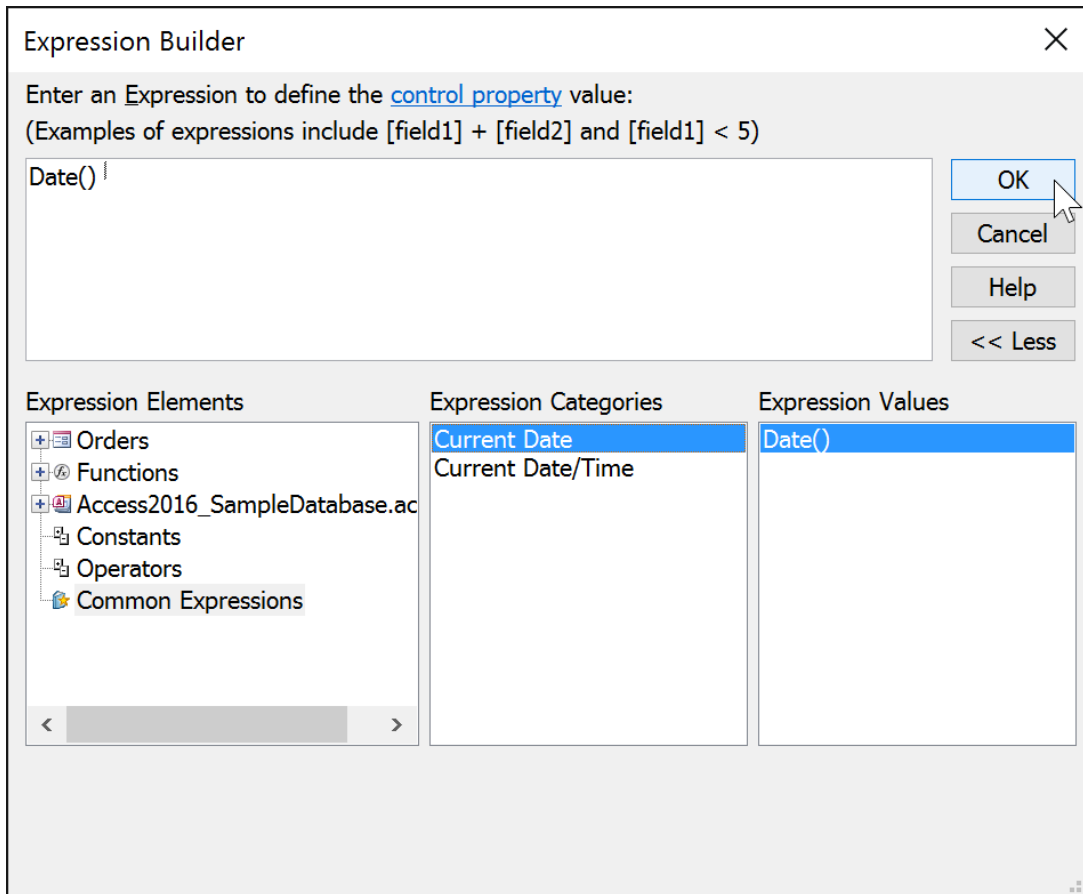


3. In the **Property Sheet**, click the **Data** tab, then select the **Default Value** field in the fourth row. Click the **Expression Builder** button that appears in the column to the right.

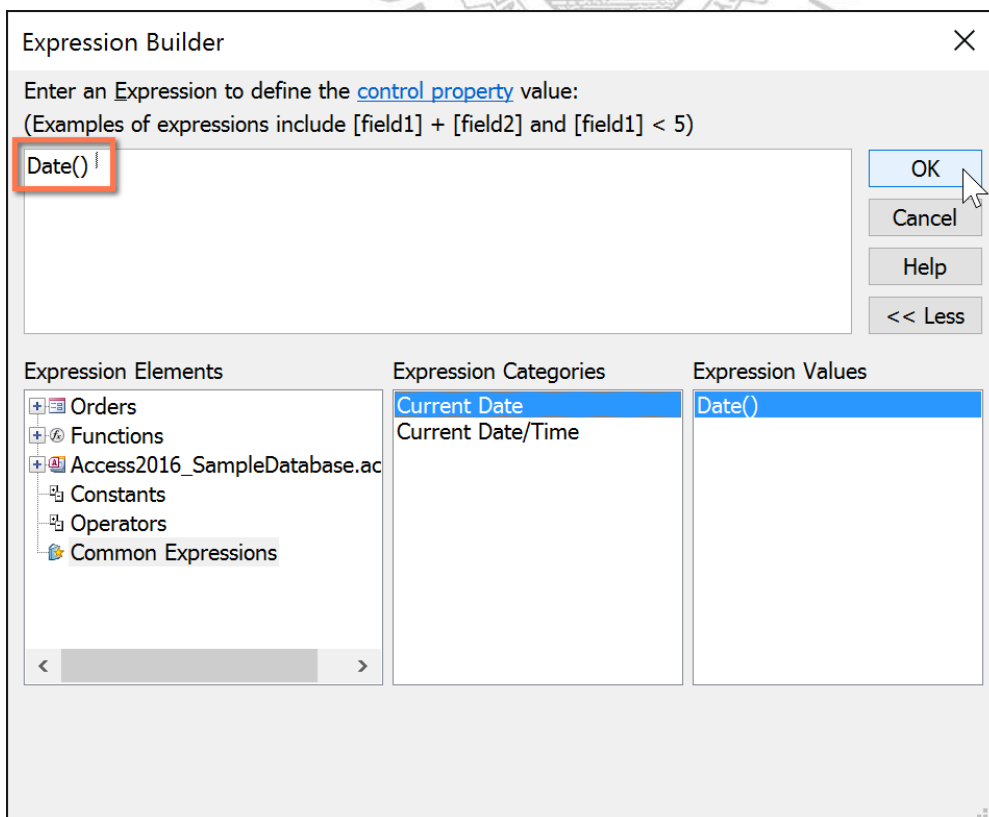


4. The **Expression Builder** dialog box will appear. In the **Expression Elements** list, click the words **Common Expressions**.

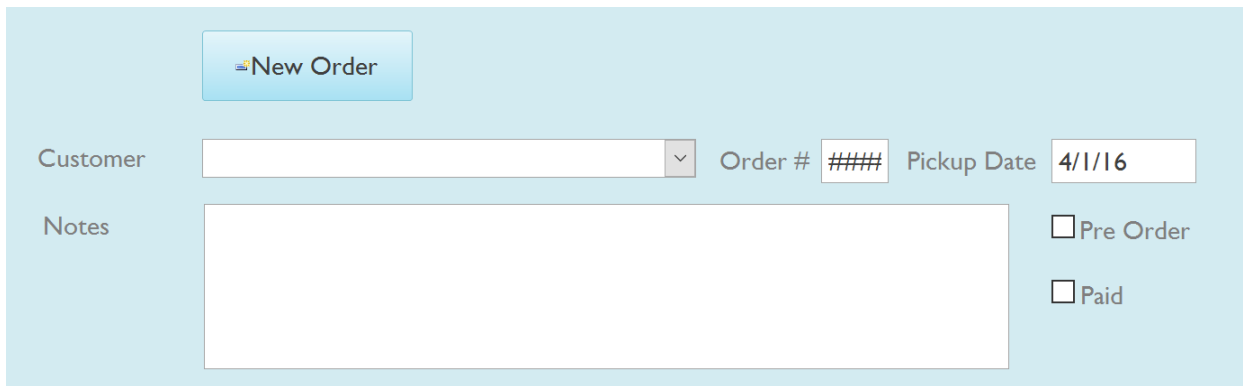
5. In the **Expression Categories** list, double-click **Current Date**.



6. The expression for **Current Date** will be added. Click **OK**.



7. Switch to **Form** view to verify that the expression works. When you create a **new record** with that form, the date field you modified will automatically fill in the current date.



The screenshot shows a Microsoft Access form with a light blue background. At the top center is a button labeled 'New Order' with a small icon. Below this, there are three fields: 'Customer' with a dropdown arrow, 'Order #' with a text box containing '####', and 'Pickup Date' with a text box containing '4/1/16'. Below the 'Customer' field is a large text box labeled 'Notes'. To the right of the 'Notes' box are two checkboxes: 'Pre Order' and 'Paid', both of which are currently unchecked.

Formatting Forms

After creating a form, you might want to modify its appearance. **Formatting** your forms can help make your database look consistent and professional. Some formatting changes can even make your forms easier to use. With the formatting tools in Access, you can customize your forms to look exactly the way you want.

In this lesson, you will learn how to **add command buttons**, **modify form layouts**, add **logos** and **other images**, and change form **colors** and **fonts**.

Access offers several options that let you make your forms look exactly the way you want. While some of these options—like **command buttons**—are unique to forms, others may be familiar to you.

Command buttons

If you want to create a way for users of your form to quickly perform specific actions and tasks, consider adding **command buttons**. When you create a command button, you specify an action for it to carry out when clicked. By including commands for common tasks right in your form, you're making the form easier to use.

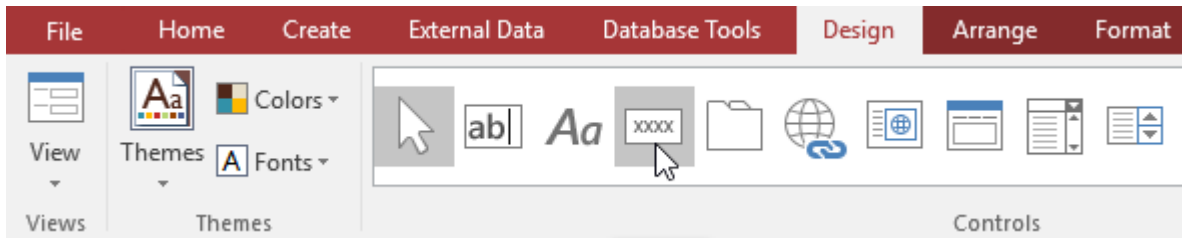
Access offers many different types of command buttons, but they can be divided into a few main categories:

- **Record Navigation** command buttons, which allow users to move among the records in your database
- **Record Operation** command buttons, which let users do things like save and print a record
- **Form Operation** command buttons, which allow users to quickly open or close a form, print the current form, and perform other actions
- **Report Operation** command buttons, which offer users a quick way to do things like preview or mail a report from the current record

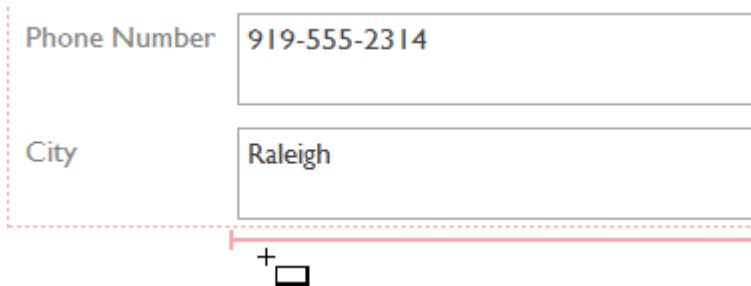
To add a command button to a form:

1. In **Form Layout** view, select the **Design** tab, then locate the **Controls** group.

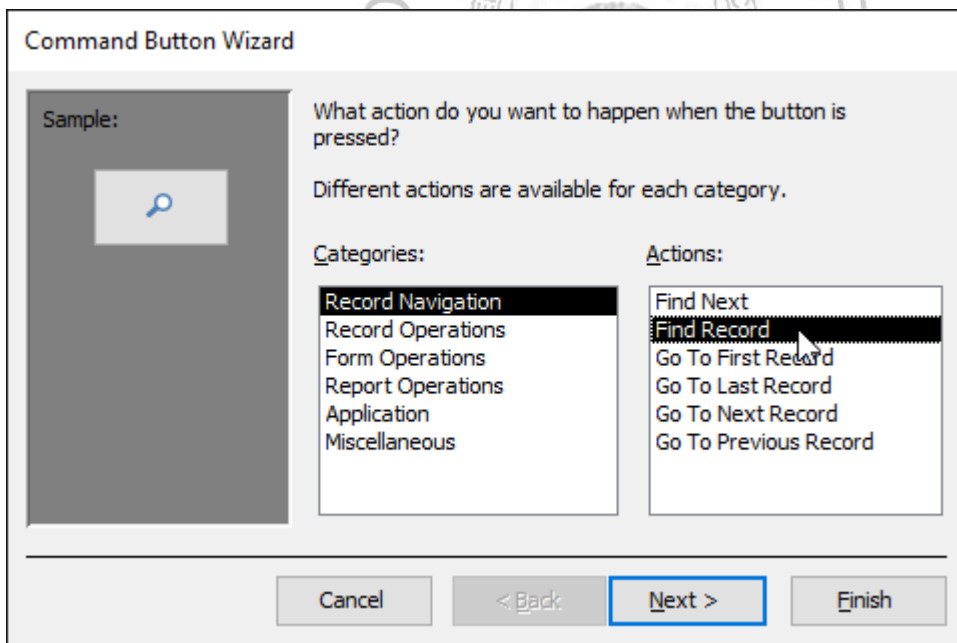
2. Click the **Button** command.



3. Choose the desired location for the command button, then click the mouse.



4. The **Command Button Wizard** will appear. In the **Categories** pane, select the category of button you want to add. We want to find a way to move more quickly to specific records, so we'll choose the **Record Navigation** category.
5. The list in the **Actions** pane will update to reflect your chosen category. Select the action you want the button to perform, then click **Next**. In our example, we'll choose **Find Record**.



6. You can now decide whether you want your button to include **text** or a **picture**. A live preview of your button appears on the left.

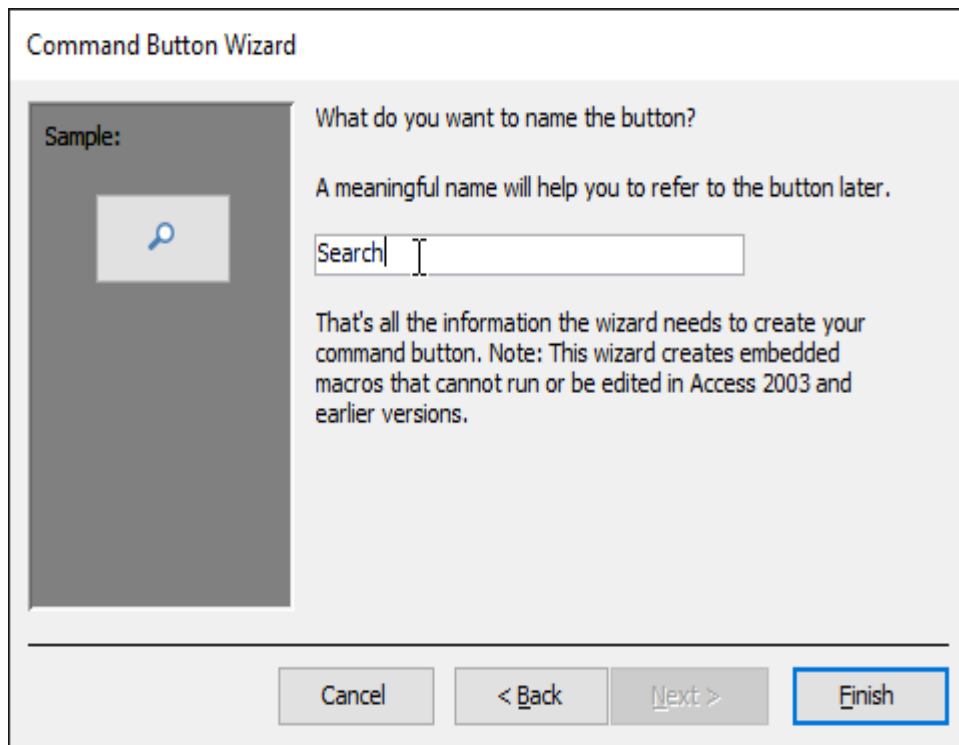
7. To include **text**, select the **Text** option, then type the desired word or phrase into the text box.

The screenshot shows the 'Command Button Wizard' dialog box. On the left, a 'Sample:' area displays a button with the text 'Search'. The main area asks 'Do you want text or a picture on the button?'. The 'Text' radio button is selected. Below it, a text box contains the word 'Search'. The 'Picture' radio button is unselected. Below it, a list box shows 'Binoculars (Find)' as the selected item, with a 'Browse...' button to its right. At the bottom, there is a checkbox for 'Show All Pictures' which is unchecked. The bottom of the dialog has four buttons: 'Cancel', '< Back', 'Next >', and 'Finish'. The 'Next >' button is highlighted with a blue border.

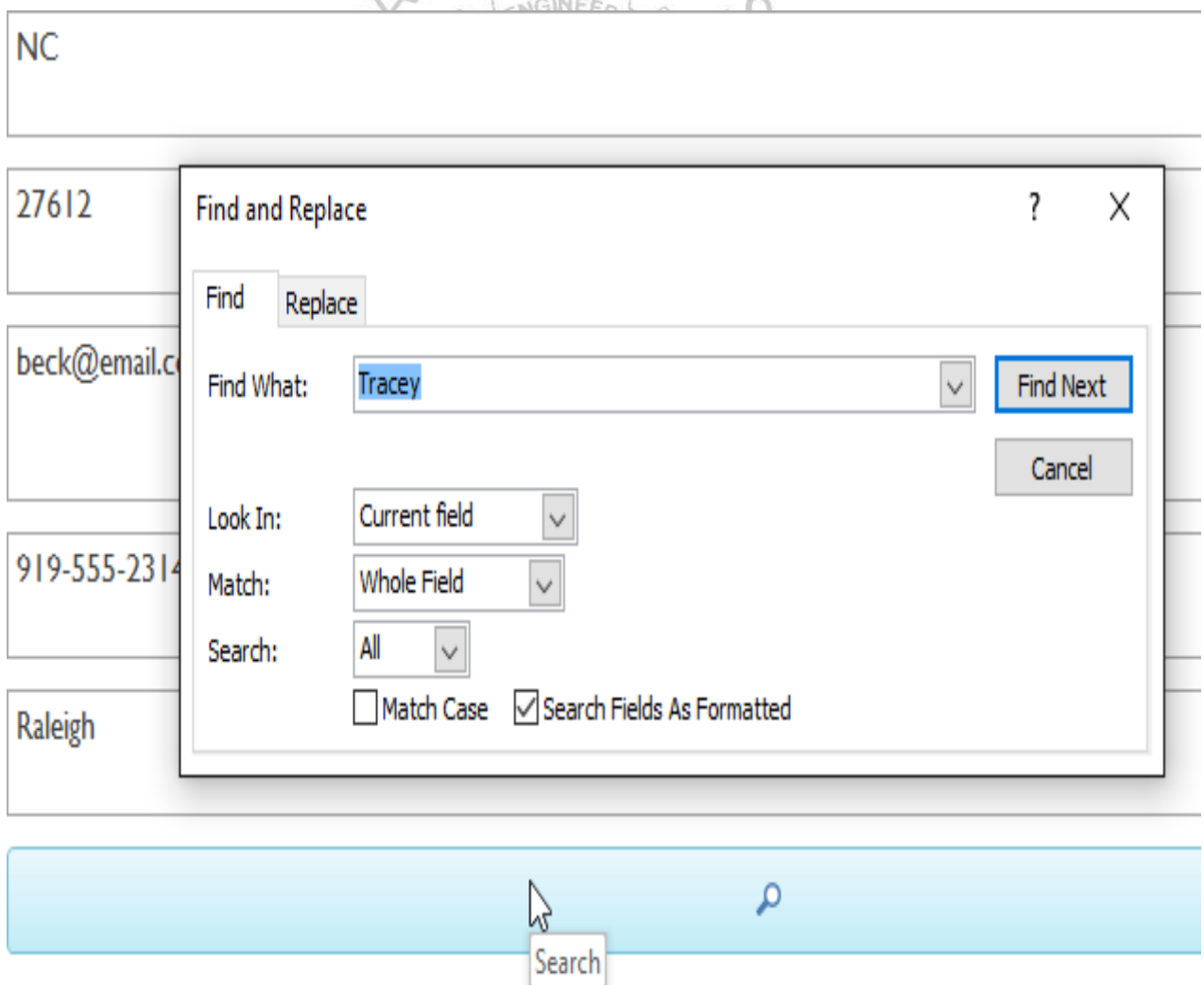
8. To include a **picture**, select the **Picture** option. You can decide to keep the default picture for that command button or select another picture. Click **Show All Pictures** to choose from another command button icon or **Browse** to choose a picture from your computer.

This screenshot shows the 'Command Button Wizard' dialog box with the 'Picture' radio button selected. The 'Sample:' area now shows a button with a magnifying glass icon. The 'Text' radio button is unselected, and its text box contains the word 'Search'. The 'Picture' radio button is selected, and the list box below it still shows 'Binoculars (Find)' as the selected item. The 'Browse...' button is still present. The 'Show All Pictures' checkbox remains unchecked. The bottom buttons are 'Cancel', '< Back', 'Next >', and 'Finish'. The 'Next >' button is highlighted with a blue border.

9. When you are satisfied with the appearance of your command button, click **Next**.
10. Type a **name** for the button. This name won't appear on the button, but knowing the name will help you quickly identify the button if you ever want to **modify** it with the **Property Sheet**. After adding the button name, click **Finish**.



11. Switch to **Form** view to test the new button. Our **Search** button opens the **Find and Replace** dialog box.



Some users report that Access malfunctions while formatting forms. If you have a problem performing any of these tasks in Layout view, try switching to Design view.

EXERCISE:

Manage the Form Design for Employee in an Organization database.

The screenshot shows the Microsoft Access interface with the 'Employee_form' open in Design view. The left pane displays the 'All Access Objects' task pane with 'Tables' and 'Forms' sections. The 'Forms' section shows 'Employee_form' selected. The main area displays the form design with five instances of the Employee table, each with fields for Emp_ID, Name, City, and Department. The data for the instances is as follows:

Emp_ID	Name	City	Department
101	Ali Khan	Karachi	IT
102	Sara Ahmed	Lahore	HR
103	Omar Rizvi	Islamabad	IT
104	Aisha Khan	Karachi	Finance
105	Bilal Khan	Lahore	IT